Discrete Math:-

Chapter No 1:-
the foundations:-
Logic and Proofs:-

Definations:-
i) Propositions:-

A Proposition is an declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

For Example: All these following declarative sentence are proposition

(i) Toronto is the capital of canada.

(ii) $1+1=2$ (iii) $2+1=4$

∴ Here proposition (i) and (iii) are false and (ii) is true.

For Example:- All these sentence are not proposition

(i) what time is it? (ii) Read this carefully (iii) $x+1=?$ iv) $x+y=2$

∴ we use letters to denote propositional variable.

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Convential letters used for proposital variables are p, q, r, s..... the truth value of a position is true denoted by T if it is true proposition, and truth value of a propostion is false, denoted by F, if it is a false proposition.

ii) Proposition calculas (propositional logic)

the area of logic that deals with the propoaition is called Propositional calculas or propositional logic.

It was first developed systemmticdlly by Greek philosopher Aristotle more than 2300 years ago.

iii) Compound composition:-

Many methomatical Statement are construct by combinig one or more proposition. New Proposition; called compound proposition are formed from existing propositional using logical operators.

iv) Negation:-

Let p be a propositional the negation of p is denoted by ¬p (also denoted by $\bar{p}$) is the statement. It is not the case that p.

- the proposition ¬p is read not p the truth value of negation of P. ¬p is opposite of the truth value of P.

For Example:-

Let we have a proposition michael's pc runs Linux:-

- Its negation is It is not the case that Michael's Pc runs Linux.
- this negation is more simplified. as Michael's pc does not run Linux.
- the truth table for negation of propositional p is given as.

P	¬P
T	F
F	T

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* the logical operation that are used to form new proposition from two or more exiting propositions are called connectives

Conjuction :.

Let p and q be propositions the conjunction of q and p denoted as $p \wedge q$, is the proposition p and q. the cunjuction $p \wedge q$ is true when both p and q are true and is false otherwise.

For Example:-

Let proposition p is "Rebecca" Pc has more than 16 GiB Free hard disk space and q propostion is the processes in Rebeca's Pc runs faster the /GiH2 the conjunction of these proposition. $p \wedge q$ is the proposition "Rebeca's Pc has more than 16 GiB free hard disk space and processor in Rebecca's pc runs faster than 1 GiH2". the truth table of conjuction of p and q $p \wedge q$ is given as:-

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjuction:-

Let p and q be the proposition the disjuction of p and q is denoted as $p \vee q$, is the proposition "p or q". the disjunction $p \vee q$ is false if bath 'p' and 'q' are false and is true. other wise.

For Example:-

the disjunction of p and q $p \vee q$ as given in above example is the proposition Rebecca's Pc has atleast 16 GiB free hard disk space, or the Processor in Rebecca's Pc runs faster than 1 GiH2.

the truth table of disjuction of p and q $p \vee q$ is given as:-

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

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Exclusive or:-

Let P and q be the propositional the exclusive or of P and q, denoted by $P \oplus q$ is the proposition that is true when exactly one of P and q is true and is false otherwise.

For Example:-

Students who have taken calculas or computer science, but not both can enroll inthis class.

the truth tables of exclusive or is given as:-

P	q	$P \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Conditional Statement:-

Let P and q be the propositions the conditional statement $P \rightarrow q$ is the proposition if P, then q: the conditional statement $p \rightarrow q$ is false when P is true and q is false, and is true otherwise. In conditional statement $P \rightarrow q$. P is called the hypothesis (or antecedent or Premise) and q is called the conclusion (or consequence) the statement. $p \rightarrow q$ is called conditional statement because $p \rightarrow q$ assents that q is true q on the condition that P holds. A conditional statement is also called implication.

for Example:- Example No 07:-

Let P be statement "Maria learns discrete Mathemetics and q is the statement Maria will find a good job; the conditional statement $P \rightarrow q$ is the statement: If Maria learns discrete Mathematic than she will find a good Job; the truth table of conditional statemer of P and q $P \rightarrow q$ is given belows.

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

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Example No 08:-

Find value of x
if $2+2=4$ then $x = x+1$
If $x=0$ before this statement is encountered (the symbol = stad for assignment).
the statement $x = x+1$ means the assignment of the value of $x+1$ to x)

Solutions:-

Because $2+2=4$ is true the assignment statement $x := x+1$ is excecuted statement Hence x has the value $0+1=1$ after this statement is encountered.

Converse:-

Let we have a conditional proposition $p \to q$, the proposition $q \to p$ is called converse of proposition $p \to q$.
truth table of converse is given as.

p	q	$q \to p$
T	T	T
T	F	T
F	T	F
F	F	T

Contrapostive:-

the propostion $\neg q \to \neg p$ is called contrapositive of proposition $p \to q$.

p	q	$\neg p$	$\neg q$	$\neg q \to \neg p$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

Inverse:-

the proposition $\neg p \to \neg q$ is called inverse of proposition $p \to q$.

p	q	$\neg p$	$\neg q$	$\neg p \to \neg q$
T	T	F	F	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

Biconditional statement:-

Let p and q be proposition the biconditional statement $p \leftrightarrow q$ is the proposition p if and only if q; the biconditional statement $p \leftrightarrow$ is true when p and q have same truth values, and is false otherwise. Biconditional state are also called bi-implications.

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For example:-

Let p be the statement "you can take flight" and q be statement "you buy a ticket". then $p \leftrightarrow q$ is the statement "you can take the flight of and only if you buy a ticket":

P	q	$P \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example No 11:-

Construct truth table of compound proposition.

$(P \vee \neg q) \rightarrow (p \wedge q)$

P	q	$\neg q$	$p \vee q$	$p \wedge q$	$(P \vee \neg q) \rightarrow p \wedge q$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Procedence of logical operators.

operators	precedence
$\sim$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Bit:-

A bit is a symbole with two possible values, namely 0 (zero) and 1 (one).

→ this meaning of word bit comes from binary digit because zeros and ones are digits used in binary representation of numbers.

Truth value	Bit
T	1
F	0

→ A variable is called a Boolan variable if its value is either true or false consequently, a Boolian variable can be represented using a bit.

Table for Bit operators OR, AND, and XOR.

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

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Bit String:-

A bit string is a sequence of zero or more bits. the length of his string is the number of bits in the string.

Exercise

Question No 1:-

a) Boston is the capital of Masschsetts
It is a proposition.
It is a proposition.

b) Miami is the capital of Florida:-
It is a proposition
It is a proposition

c) $2+3=5$
this is a proposition
Its truth value is true

d) $5+7=10$
It ie also a proposition. Its buth value. is false

e) $x+2=11$
It is not a proposition.

f) Answer this questions.
It not a propositkn

Question No 2:-

a) Do not pass go
It is not a proposition.

b) What time is it?
Its is a not proposition.

c) there are ino black flies in maine.
It's a proposition It's truth value is false. Because anyone has been to make knows,

d) $4+x=5$ It is not a proposition

e) the moon is made of green cheese.
It is a proposition. Its truth value is false.

f) $2^n \geq 100$
It is note a proposition Its truth depends upon. n.

Question No. 3:-
find negation:-

a) Mei has an Mps. players.
Mei does not have an Mp3 players.

b) there is no pollution in New Jersey.
It is note the case that there is no pollution in new Jersey.

c) $2+1=3$
It is not the case that $2+1=3$

d) the summer is maine is hot and sunny.
the summer is Maine is not hot and sunny.

Question No 4:-

a) Jennifer and Teja are friends.
Jennifer and Teja are not friends.

b) there are 13 items in a bakers dozen.
there are no 13 item in a bakers dozen.

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c) Abby sent more than 100 text messages every day.

Abby did not send more than 100 text message every day.

d) 121 is a perfect square.

Question No 5:-

find negations:-

b) Zech block e-mails and text from Jennifer.

It is not the case that Zach block e-mail and text from Jennifer.

c) $7 \cdot 11 \cdot 13 = 999$

It is not the case that.

$7 \cdot 11 \cdot 13 = 999$

OR.

$7 \cdot 11 \cdot 13 \neq 999$

d) Diene rode her bicycle 100 miles on Sunday.

Diene rode her did not bicycle 100 miles on sunday.

Question No 6:-

a) this proposition has true truth value.

b) this proposition has true truth tables value.

c) this proposition has false truth value.

d) Its truth value is false.

e) Its truth value is false.

Question No 7:-

a) the truth value of this proposition is false.

c) the truth value of this preposition is true.

e) Truth value of this proposition is true.

Question No 8:-

P: I bought a lottery tickets this week.

q: I won the million dollers Jack pot.

a) $\sim p$: I did not buy a lottery ticket this week.

b) $p \vee q$: I brought a lottery ticket this week or I won the million dollar jack pot.

c) $\sim p \vee q$: I did not buy a lottery ticket ticket these week I won million dollar Jack pot.

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e) $p \leftrightarrow q$

I bought a lottery ticket this week if and only if I won the million dollars jackpot.

g) $\sim p \wedge \sim q$

I did not buy a lottery ticket this week and I did not win the million dollars jackpot.

h) $\sim p \vee (p \wedge q)$

I did not buy a lottery ticket this week or I bought a lottery ticket this week and I won million dollar jackpot.

Question No 9:-

P: "Swimming at New Jersey shore is allowed"

q: "Sharks have been spotted near the shore."

a) $\sim q$)

Sharks have not been spotted near the Shore.

e) $\sim q \rightarrow p$

If sharks have not been spotted near the shore then swimming at new Jesey shore is allowed.

l) $p \leftrightarrow \sim q$

simming at new Jersey shore is allowed if and nly if sharks have not been spotted hear sharks.

h) $\sim p \wedge (p \vee \sim q)$

Swimming at new Jersey shore is not allowed and swimming at new Jersey shore is allowed or Sharks have not been spotted near the Shore.

Question No 10:-

P: the election is decided.

q: the votes have been counted.

b) $p \vee q$

the election is decided or the votes have been counted.

d) $q \rightarrow p$

If the votes have been counted then the election is decided.

f) $\sim q \rightarrow \sim p$

If the votes have not been counted then election is not decided.

g) $p \leftrightarrow q$

the election is decided if and only if the votes have been coun

Question No 11:-

P: It is below freezing

q: It is Snowing.

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a)
It is below freezing and snowing
$p \wedge q$

b) It is below freezing but not snowing
$p \wedge \sim q$

e) If it is below freezing it is also snowing.
$p \rightarrow q$

g)
That is below freezing is necessary and suffient for it to be snowing.
$p \leftrightarrow q$.

Question No 12:-

p: You have the flu.
q: you miss final examination
r: You pass the course

a) $p \rightarrow q$.
If you have the flu then you miss final examination

c) $q \rightarrow \sim r$
If you miss final examination then you denot pass the course.

e) $(p \rightarrow \sim s) \vee (q \rightarrow \sim r)$.
It is either the case if you have flu then yai donst pass course or the case that if you mis final examination then you don't pass course

Question No 13:-

p: You drive over 65 miles per hour.
q: You get a speeding ticket.

a)
You do not drive over 65 miles per hour. $\sim p$.

c) You will get a speeding ticket if you drive overs 65 miles per hour. $p \rightarrow q$

e) Driving over 65 miles per hour is sufficient for getting a speeding ticket.
$p \rightarrow q$

g)
$q \rightarrow p$

Question No 14:-

p: you get an A on final exam
q: you do every exercise in this book
r: You get an A in this class.

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a) $P \wedge \sim q$ b) $(P \wedge q) \rightarrow r$

c) $r \rightarrow p$ f) $r \leftrightarrow (q \vee p)$

Question No 15:-

p: Grizzing bears have Seenin area

q: Hiking is safe on the trail

r: Berries are ripe along the trial.

a) $r \wedge \sim p$ b) $(\sim p \wedge q) \wedge r$

c) $r \rightarrow (q \leftrightarrow \sim p)$ d) $\sim q \wedge (\sim p \wedge r)$

e) $q \leftrightarrow (\sim r \wedge \sim p)$ f) $p \rightarrow (q \wedge r)$

Question No 16:-

a) $2+2=4$ if and only if $1+1=2$

Bicondition is true

b) $1+1=2$ if and only if $2+3=4$

false

c) $1+1=3$ if and only if Monkeys can fly

true

d) $0>1$ if and only if $2>1$

false

Question No 17:-

a) If $1+1=2$, then $2+2=5$

false.

b) If $1+1=3$, then $2+2=4$

conditional statement is true.

c) If $1+1=3$ then $2+2=5$

true.

d) If monkey can fly, then $1+1=3$

true.

Question No 18:-

a) If $1+1=3$, then dogs can fly

True

b) If $1+1=2$, then dogs can fly

false

c) If $2+2=4$, then $1+2=3$

true.

Question No 19:-

a) cofee or tea comes with dinner.

this sentence is exclusive or because only one coffee or tea can come with dinner but not both.

d) You can pay using V.S. dollars or euros.

this sentence is inclusive or because use can pay both using U.S dollars or euros.

Question No 20:-

a) Experience with C++ or Java is required.

this sentence is inclusive or because both experience with C++ or Java is required but not both.

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b) Lunch includes soup or salad:-
this sentence is exclusive or because only Lunch can included soup and (B) salad

d) Publish or Perish:
this sentence is inclusive or because only one of publish or perish be done but not both.

Question No 22:-

a) If you get promoted, you wash the boss's car.

b) If Winds are form the South, then there Will be a spring thaw.

d) If willy cheats, then he caught

g) If cannot on a beat, then She gets sea sick.

Question No 23:-

a) If the wind blews from northeast, then it snows.

d) If you get to top of Longs peak, then you walk 8 miles.

h) If water is not too cold then In will go swimming.

Question No 24:-

a) If am to remember to send you address, then you will have to send me an e-mail message.

c) If you keep your textbook then it will be a useful reference in your future course.

f) If there is astorm then beach erodes.

Question No 25:-

a) You buy an icecrem can if and only if it is hot outside.

e) I take trains if and only if they run late on exactly those days.

Question No 26:-

b) you read newspaper every day if and only if you will be informed.

c) If rains, if and only if it is a Weekend day.

d) You can See the Wizard if and only if he is not in.

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Question No 27:-

a)

Converse:- If I skies tomorrow then it will snow today.

Inverse:- If it does not snow today then i will not sky tomorrow.

Contrapositive:- If I does not sky tomorrow then it will not snow today.

Question No 28:-

b)

Converse:- Whenever I go to beach it is sunny summer day.

Inverse:- Whenever it is not a sunny day, I donot go to beach.

contrapositive:- Whenever I do not go to beach, it is not a sunny summer day.

Question No 29:- b) $(p \vee r) \wedge (q \vee s)$

p	q	r	s	¬r	¬s	p∨r	q∨s	(p∨r)∧(q∨s)
T	T	T	T	F	F	T	T	T
T	T	T	F	F	T	T	T	T
T	T	F	T	T	F	T	T	T
T	T	F	F	T	T	T	T	T
T	F	T	T	F	F	T	F	F
T	F	T	F	F	T	T	T	T
T	F	F	T	T	F	T	F	F
T	F	F	F	T	T	T	F	F
F	T	T	T	F	F	T	T	T
F	T	T	F	F	T	T	T	T
F	T	F	T	T	F	F	T	F
F	T	F	F	T	T	F	T	F
F	F	T	T	F	F	T	T	T
F	F	T	F	F	T	T	T	T
F	F	F	T	T	F	F	F	F
F	F	F	F	T	T	F	F	F

Question No 30:-

a) $(q \to \neg p) \vee (\neg p \to \neg q)$

p	q	¬p	¬q	q→¬p	¬p→¬q	(q→¬p)∨(¬p→¬q)
T	T	F	F	F	T	T
T	F	F	T	T	T	T
F	T	T	F	T	F	T
F	F	T	T	T	T	T

Numbers of rows $= 2^n = 2^2 = 4$ rows

b) $(p \vee t) \wedge (p \vee s)$

Number of rows $= 2^n = 2^3 = 8$ rows

Question No 31:-

a) $P \wedge \neg P$

P	¬P	P∧¬P
T	F	F
F	T	F

c) $(p \vee q) \to q$

P	q	¬q	p∨q	(p∨q)→q
T	T	F	T	T
T	F	T	T	T
F	T	F	F	T
F	F	T	T	F.

Question No 32:-

b) $(P \oplus q) \to (P \wedge q)$

P	q	P⊕q	P∧q	(P⊕q)→(P∧q)
T	T	F	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

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Question No 33:- f) $(P \oplus q) \rightarrow (P \oplus \neg q)$

P	q	¬q	P⊕q	P⊕¬q	(P⊕q)→(P⊕¬q)
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	F	T	T

Question No 34:-

d) $\neg P \oplus \neg q$

P	q	¬P	¬q	¬P⊕¬q
T	T	F	F	F
T	F	F	T	T
F	F	T	F	T
F	T	T	T	F

Question No 35:-

f) $(\neg P \leftrightarrow q) \leftrightarrow (p \leftrightarrow q)$

P	q	¬P	¬q	¬P↔q	p↔q	(¬P↔¬q)↔(p↔q)
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	F	F	T
F	F	T	T	T	T	T

Question No 37:-

C) $(P \rightarrow q) \vee (\neg P \rightarrow r)$

P	q	r	¬P	P→q	¬P→r	(P→q)∨(¬P→r)
T	T	T	F	T	T	T
T	T	F	F	T	T	T
T	F	T	F	F	T	T
T	F	F	F	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	F	T
F	F	T	T	T	T	T
F	F	F	T	T	F	T

Question No 38:-

$((P \rightarrow q) \rightarrow r) \rightarrow s$

P	q	r	s	P→q	(P→q)→r	((P→q)→r)→s
T	T	T	T	T	T	T
T	T	T	F	T	T	F
T	T	F	T	T	F	T
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	T	F	T	T	F
F	T	F	T	F	T	T
F	T	F	F	T	F	T

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Question No 40:-

Question No 41:-

Question No 42:-

a) if $x+2=3$ then $x := x+1$

Since condition is true, statement is executed, so x incremented and now has value 2 $x := 1+1 = 2$

c) if $(2x+3)$ AND $(3x+4=7)$ then $x := x+1$

Since condition is true, the statement is excuted, so x increment and now has valu 2.

e) if $x<2$ then $x := x+1$

Since condition statement is true, so statement is excuted, so x incremented and How has value 1.

Question No 43:-

a) Let $p = 10\,1110$, $q = 010\,0001$

P	q	OR	AND	XOR
1	0	1	0	1
0	1	1	0	1
1	0	1	0	1
1	0	1	0	1
1	0	1	0	1
1	0	1	0	1
0	1	1	0	1

Question No 44:-

a) Let $p = 11000$, $q = 01011$, $r = 11011$

11000	01011	11011	(01011 v 11011)	11000 ∧ (01011 v 11011)
1	0	1	1	1
1	1	1	1	1
0	0	0	0	0
0	1	1	1	0
0	1	1	1	0

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Question No 45:-

"Fred is not happy"; "John is not happy"

i) the truth value of "Fred is not happy" is given as. . .

$1 - 0.8 = 0.2$

ii) the truth value of "John is not happy" is given as

$1 - 0.4 = 0.6$.

Question No 46:-

i) the truth value of the statement "Fred and John are happy" is given as $\min(0.8, 0.4) = 0.4$

i) the truth value statement "Neither Fred nor John is happy" this statement can be written as Fred is not happy and John is t happy."

$\min(0.2 + 0.6) = 0.2$

Question No 47:-

the truth value of statement "Fred happy or John is happy" is

$\max(0.8, 0.4) = 0.4$

e truth value of statement "Fred is happy or John is not happy" is

$\max(0.2, 0.6) = 0.6$.

Question No 48:-

this cannot be a proposition because it cannot have a truth value this string of letters, while appearing to be a proposition, is in fact Meaning less.

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Exercise No 3:-

Question No 1:-

a) $P \wedge T \equiv P$

P	T	P∧T
T	T	T
F	F	F ≡ P

c) $P \wedge F \equiv F$

P	F	P∧F
T	T	T
F	F	F ≡ P

f) $P \wedge P \equiv P$

P	P	P∧P
T	T	T
F	F	F ≡ P

e) $P \vee P \equiv P$

P	P	P∨P
T	T	T
F	F	F ≡ P

Question No 2:-

~ (~P) and P are equivalent.

P	~P	~(~P)
T	F	T
F	T	F = P

So, ~(~P) and P are logically equivalent.

Question No 3:-

a) $P \vee q \equiv q \vee P$

P	q	L·H·S P∨q	P	q	R·H·S q∨P
T	T	T	T	T	T
T	F	T	T	F	T
F	T	T	F	T	T
F	F	F	F	F	F

So $P \vee q \equiv q \vee P$.

Question No 4:-

b) $(P \wedge q) \wedge r \equiv P \wedge (q \wedge r)$

L·H·S = $(P \wedge q) \wedge r$

P	q	r	(P∧q)	(P∧q)∧r
T	T	T	T	T
T	T	F	T	F
T	F	T	F	F
T	F	F	F	F
F	T	T	F	F
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

R·H·S = $P \wedge (q \wedge r)$

P	q	r	q∧r	P∧(q∧r)
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	T	F
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

L·H·S = $(P \wedge q) \wedge r$ = R·H·S = $P \wedge (q \wedge r)$

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Question No 5:-

$P \wedge (q \vee r) = (P \wedge q) \vee (P \wedge r)$

L·H·S = $P \wedge (q \vee r)$.

P	q	r	$q \vee r$	$P \wedge (q \vee r)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	T	F
F	T	F	T	F
F	F	T	T	F
F	F	F	F	F

R·H·S = $(P \wedge q) \vee (P \wedge r)$

P	q	r	$P \wedge q$	$P \wedge r$	$(P \wedge q) \vee (P \wedge r)$
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	F	F	F
F	T	F	F	F	F
F	F	T	F	F	F
F	F	F	F	F	F

So, L·H·S = R·H·S.

Question No 6:-

$\sim (P \wedge q) \equiv \sim P \vee \sim q$

L·H·S = $\sim(P \wedge q)$

P	q	$P \wedge q$	$\sim(P \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

R·H·S = $\sim P \vee \sim q$

P	q	$\sim P$	$\sim q$	$\sim P \vee \sim q$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

R·H·S = L·H·S.

Question No 7:-

a) John is rich and happy.

According to demorgan's Law the negation of this statement is "Jan is not rich or Jan is not happy"

b) carles will bicycle or run tomorrow.

According to demorgan's law the negation of this statement is it is not the case that carlos will bicycle and run tomorrow.

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d) Ibrahim is smart and hardworking:-
Ace. to Demorgan's law the negation is
"Ibrahim is not smart or he is not hardworking."

Question No 08:-

b) Yeshiko know Java and calculas.
Ace. to Demorgan's law negation is yeshiko does not know Java or he does not know calculas.

c) James is young and strong.
James into young or he is not strong.

Question No 9:-

a) $(P \wedge q) \to P$

P	q	P∧q	(P∧q)→P
T	T	T	T
F	F	T	
T	F	T	
F	F	T	

so it is tautology.

d) $(P \wedge q) \to (P \to q)$

P	q	P∧q	P→q	(P∧q)→(P→q)
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

So, it is tautology.

$\sim(P \to q) \to \sim q$

P	q	~q	P→q	~(P→q)	~(P→q)→~q
T	T	F	T	F	T
T	F	T	F	T	T
F	T	F	T	F	T
F	F	T	T	F	T

So it is tautology.

Question No 10:-

b) $[(q \to r) \wedge (P \to q)] \to (P \to r)$

P	q	r	P→q	q→r	(P→q)∧(q→r)	P→r	[(P→q)∧(q→r)]→(P→r)
T	T	T	T	T	T	T	T
T	T	F	F	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	F	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

So it is tautology.

d) $[(P \vee q) \wedge (P \to r) \wedge (q \to r)] \to r$

P	q	r	P∨q	P→r	q→r	(P∨q)∧(P→r)∧(q→r)	[(P∨q)∧(P→r)∧(q→r)]→r
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

So, it is tautology.

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Question NO 13:-

b) $P \wedge (P \wedge q) \equiv P$

P	q	$P \vee q$	$P \wedge (P \wedge q)$
T	T	T	T
T	F	T	T
F	T	T	F ≡ P.
F	F	F	F

Question NO 14:- $(\sim P \wedge (P \rightarrow q)) \rightarrow \sim q$

P	q	$\sim P$	$\sim q$	$P \rightarrow q$	$\sim P \wedge (P \rightarrow q)$	$(\sim P \wedge (P \rightarrow q)) \rightarrow \sim q$
T	T	F	F	T	F	T
T	F	F	T	F	F	T
F	T	T	F	T	T	F
F	F	T	T	T	T	T.

this not tautology. this is contigency.

Question No 16:- $P \leftrightarrow q$ and $(P \wedge q) \vee (\sim P \wedge \sim q)$ are equilent.

P	q	$P \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

these are equilent.

P	q	$P \wedge q$	$\sim P \wedge \sim q$	$(P \wedge q) \vee (\sim P \wedge \sim q)$
T	T	T	F	T
T	F	F	F	F
F	T	F	F	F
F	F	F	T	T

So, it is equivalent.

Question NO 19:-

$\sim P \leftrightarrow q$ and $P \leftrightarrow \sim q$ are equivalent.

P	q	$\sim P$	$\sim P \leftrightarrow q$
T	T	F	F
T	F	F	T
F	T	T	T
F	F	T	F

P	q	$\sim q$	$P \leftrightarrow \sim q$
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	F

So, they are equivelent.

Question NO 22:-

$(P \rightarrow q) \wedge (P \rightarrow r)$ and $P \rightarrow (q \wedge r)$

P	q	r	$P \rightarrow q$	$P \rightarrow r$	$(P \rightarrow q) \wedge (P \rightarrow r)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

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P	q	r	q∧r	P→(q∧r)
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	T	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

So it is equivelent.

Question No 25:- (P→r) ∨ (q→r) and (p∧q)→r

P	q	r	P→r	q→r	(P→r)∨(q→r)
T	T	T	T	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

So, they are equivelent.

P	q	p∧q	r	(p∧q)→r
T	T	T	T	T
T	T	T	F	F
T	F	F	T	T
T	F	F	F	T
F	T	F	T	T
F	T	F	F	T
F	F	F	T	T
F	F	F	F	T

Question No 28:- P↔q and ¬P↔¬q

P	q	P↔q	¬P	¬q	¬P↔¬q
T	T	T	F	F	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

So they are logically equivelent.

Question No 30:-

(P∨q) ∧ (¬P∨r) → q∨r

P	q	r	¬P	P∨q	¬P∨r	q∨r	(P∨q)∧(¬P∨r)	(P∨q)∧(¬P∨r) → q∨r
T	T	T	F	T	T	T	T	T
T	T	F	F	T	F	T	F	T
T	F	T	F	T	T	T	T	T
T	F	F	F	T	F	F	F	T
F	T	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T	T
F	F	T	T	F	T	T	F	T
F	F	F	T	F	T	F	F	T

tautology.

Question No32:- (P∧q)→r and (P→r)∧(q→r)

P	q	r	P∧q	(P∧q)→r	P→r	q→r	(P→r)∧(q→r)
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	T	T	T
T	F	F	F	T	F	T	F
F	T	T	F	T	T	T	T
F	T	F	F	T	T	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

So they not logically equivelent.

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Logical Equivalence involving:

Conditional statement	Biconditional statement
$p \to q \equiv \neg p \vee q$	$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
$p \to q \equiv \neg q \to \neg p$	$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \vee q \equiv \neg p \to q$	$p \leftrightarrow \equiv (q \wedge p) \vee (\neg p \wedge \neg q)$
$p \wedge q \equiv p \wedge \neg q$	$\neg (r \leftrightarrow q) \equiv p \leftrightarrow \neg q$
$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$	
$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$	
$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$	
$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$.	

Propostional satisfiablelity:

A compound proposition is satisfiable if there is an assignment of truth value to its variables that makes it true.

unsatisfiable:

When a compound proposition is false for all assignment of truth values to its variables, then compound proposition is unsatisfiable.

Question No 61:

a) $(p \vee q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$

1) In this question we discuss two cases. When we put p and q both are true then $(p \vee q)$ and $(\neg p \vee q)$ become but the factor $(\neg p \vee \neg q)$ becomes false.

2) When we put p is true and q is false then the $(p \vee q)$ and $(\neg p \vee q)$ becomes true but factor $(\neg p \vee \neg q)$ becomes false.

So, form above two cases, it is clear that there is no truth value that make it satisfied, therefore it is unsatisfiable proposition.

c) $(p \leftrightarrow q) \wedge (\neg p \leftrightarrow q)$

We know that this Proposition is only satisfiable if both factor ar true.

i) When we put p and q are true then $(p \leftrightarrow q)$ becomes true but $(\neg p \leftrightarrow q)$ becomes (false).

ii) When we put p is true and q is false then $(\neg p \leftrightarrow q)$ becomes true but $(p \leftrightarrow q)$ is false.

So from above discussion we come to know that there is no truth value that make it satisfiable. therefore it is unsatisfied proposition.

Question No 62:

a) $(p \vee q \vee \neg r) \wedge (p \vee \neg q \vee s) \wedge (p \vee \neg r \vee s) \wedge (\neg p \vee \neg q \vee s) \wedge (p \vee q \vee s)$

In this proposition there are case that make it satisfiable.

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When p is true and q is false and r, s are also true then all the proposition becomes true and hence the compound proposition is satisfiable

1.4:-
Predicates and Quantifiers:- (Preconditions)-

the statement that describe valid input are known as precondition and the condition that output should satisfy when the program has run are known as post conditions.

Quantifier:-

Quantification:-

the quantification expresses the extent to which a predicate is true over a range of element.

Predicate calculas:-

the area of logic that deals with predicates and quantifiers is called Predicates calculas.

universal Quantifier:-

the universal quantification of P(x) is the statement "P(x) for all value of x in domain".

the notation $\forall x\, P(x)$ denotes the universal quantifier. We read $\forall x\, P(x)$ as "for all x P(x)" or "for every x P(x)". An element for which P(x) is false is called a countere example of $\forall x\, P(x)$.

Quantifier:-

Statement	When true?	When false?
$\forall x\, P(x)$	P(x) is true for every x	there is an x for which P(x) is false.
$\exists x\, P(x)$	there is an x for which P(x) is true.	for every x.

Extential/Existential Quantifiers:-

the existential quantifier of P(x) is proposition "there exists an element x in the domain such that P(x)".

We use notation $\exists x P(x)$ for existential quantifier of P(x). Here $\exists$ is called existential quantifier.

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Exercise 1.4:-

Question 1:-

"$x \leq 4$"

a) P(0)	b) P(4)	c) P(6)
$0 \leq 4$	$4 \leq 4$	$6 \leq 4$
true	~~fals~~/true	false.

Question No 2:-

a) this is true because there a in range.
b) this false since there is ~~a~~no in lesson.
c) false since there is no a true in parents.
d) True, since is parents false in this.

Question No 3:-

$Q(x,y)$ = "x is capital of y"

d) Q (New york, New York)

the truth value of this statement is false.

Question No 4:- P(x) = "$x^2 > 1$"

a) $x=0$

No, the value of x will remain same as this condition is false.

c) $x=2$

As statement is true so, value of x executed and becomes 1.

Logical Equivalences involving Quantifiers:-

Statement involving predicates and quantifeirs are logically equivelent if and only if they have same truth value no matter which predicates are substituted into these statement and which domain of discourse is used for the variable in these propositional function. We use notation $S \equiv T$ to indicates that two statement S and T involving predicates and quantifier are logically equivelent.

Demorgan's Law for Quantifiers:-

Negation	Equivelent statement	When true?	When false?
$\neg \exists x\, P(x)$	$\forall x \neg P(x)$	For every x is false.	there is an p(x) is true for x
$\neg \forall x\, P(x)$	$\exists x \neg P(x)$	there is an x for which P(x) is false	P(x) is true for every x.

Question No 5:-

a) $\exists x\, P(x)$

there is exist a student that spends more than five hours every.

b) $\forall x\, P(x)$

Every student in the school spends more than five hour every weekday in class.

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c) $\exists x \neg P(x)$

Same student in the school that do not spends more than five hours every weekday.

d) $\forall x \neg P(x)$

All students do not spend more than five hours every weekday.

Question No 7:-

c) $-\exists x\, N(x)$

No students in the school has visited North Dakota.

e) $\neg \forall x\, N(x)$

It is not true that every students in the school has visted North Dakota.

Question No 8:-

a) $\forall x\, (C(x) \rightarrow F(x))$

If all people are comedian then they are funny. Alternatly. (All people are comdian and funny)

b) $\forall x\, (C(x) \wedge F(x))$

All people are comedian and funny.

c) $\exists x\, (C(x) \rightarrow F(x))$

there exist a man such that if he is comedian then he is funny.

d) $\exists x\, (C(x) \wedge F(x))$

there exist a man who is comedian and funny.

Question No 9:

a) $\exists x\, (P(x) \wedge Q(x))$

b) $\exists x\, (P(x) \wedge \neg Q(x))$

c) $\forall x (P(x) \vee Q(x))$

d) $\neg \exists x\, (P(x) \vee Q(x))$

Question No 10:

a) $\exists x\, (C(x) \wedge D(x) \wedge F(x))$

b) $\forall x\, (C(x) \vee D(x) \vee F(x))$

c) $\exists x\, (C(x) \wedge F(x) \wedge \neg D(x))$

d) $\neg \exists x\, (C(x) \wedge D(x) \wedge F(x))$

e) $(\exists x\, (C(x))) \wedge (\exists x\, (D(x))) \wedge (\exists x\, (F(x)))$

Question No 11:-

$x = x^{2}$

a) $P(0)$

$(0) = (0)^2 = 0$

so, it is true.

c) $P(2)$

$(2)^{2} = (2)^{2}$

$2 = 4$

so, it is false.

e) $\exists x\, P(x)$

this is true because when we put $P(0)$ it is true.

f) $\forall x\, (P(x))$

this is false because when we put $P(2)$ it is false

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Question No 12:-

"$x+1>2x$"

a) $Q(0)$ | c) $P(1)$ | d) $\exists x\, Q(x)$

$0+1>2(0)$ | $1+1>2(1)$ | this is true because the exist an x that make $Q(n)$ true.

$1>2$ | $2>2$

so, false | false.

f) $\forall x > Q(x)$

the exist at least one x that make $Q(x)$ true so, it is true.

Question No 13:-

a) $\forall x\ (n+1>n)$

the truth value of this statement is true because all value of x in domain satisfy this equality and for all integers its value is true.

b) $\exists n\ (2n=3n)$

this statement is true because there exist at least one n that make it true.

Question No 14:-

a) $\exists x\ (x^3=-1)$

since $(-1)=-1$, true

b) $\exists x\ (x^4<x^2)$

since $(\frac{1}{2})^4<(\frac{1}{2})^2$, true.

c) $\forall x\ ((-x^2)=x^2)$

since $(-x)^2=((-1)x)^2=-1^2 \cdot x^2=x^2$

so $\forall x.((-x)^2=x^2)$ is true.

d) $\forall x\ (2x>x)$

there exist at least one x that makes it false as $2(-2)>-2$ $-4>-2$ so it is false.

Question No 15:-

a) $\forall x\ (n^2 \geq n)$

this is true for value of x in domains. so it is true.

d) $\exists n\ (n^2<0)$

this is false for all integers. so it is false.

Question No 16:-

c) $\forall x\ (x^2+2 \geq 1)$

True (left hand side is always at least 2)

d) $\forall x\ (x^2 \neq x)$

false. not true for $x=1$ or $x=0$

Question No 17:-

$D=0,1,2,3,4$

a) $\exists x P(x)$

$P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$

b) $\forall x\ P(x)$

$P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$

c) $\exists x \neg P(x)$

$(\neg P(0)) \vee (\neg P(1)) \vee (\neg P(2)) \vee (\neg P(3)) \vee (\neg P(4))$

d) $\forall x \neg P(x)$

$(\neg P(0)) \wedge (\neg P(1)) \wedge (\neg P(2)) \wedge (\neg P(3)) \wedge (\neg P(4))$

e) $\neg \exists x\, P(x)$

$\neg (P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4))$

f) $\neg \forall x\, P(x)$

$\neg (P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4))$

Question No 19:-

$D=1,2,3,4,5$

e) $\forall x\ ((x \pm 3) \rightarrow P(x)) \vee \exists x \neg P(x))$

$[((1 \pm 3) \rightarrow P(1)) \wedge ((2 \pm 3) \rightarrow P(2)) \wedge ((3 \pm 3) \rightarrow P(3)) \wedge ((4 \pm 3) \rightarrow P(4)) \wedge ((5 \pm 3) \rightarrow P(5))] \vee [(\neg P(1)) \vee (\neg P(2)) \vee (\neg P(3)) \vee (\neg P(4)) \vee (\neg P(5))]$.

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Question No 20:-

D = -5, -3, -1, 1, 3, 5

d) $\exists x\,((x \geq 0) \wedge P(x))$

$(L-5 \geq 0) \wedge P(-5)) \vee ((-3 \geq 0) \wedge P(-3)) \vee ((-1 \geq 0 \wedge P(-1))$

$\vee ((1 \geq 0) \wedge P(1)) \vee ((3 \geq 0) \wedge P(3)) \vee ((5 \geq 0) \wedge P(5)))$

e) $\exists x\,(\neg P(x)) \wedge \forall x\,((x<0) \rightarrow P(x))$

$[(\neg P(-5)) \vee (\neg P(-3)) \vee (\neg P(-1)) \vee (\neg P(1)) \vee (\neg P(3)) \vee (\neg P(5))$

$\wedge [P(5) \wedge P(-3) \wedge P(-1)]$

Question No 21:-

a) Everyone is studing dicrete Mathematics:- the Domain consist of some students that are studing discrete mathematics is true and the domain consisting all students is false.

b) Every one is older than 21 Year. this statement is true for those domain that consist of that people which are older than 21 Year. And is false for the domain that consist all people in the world.

Question No 24:-

a) Everyone in your class has callular phone.

Let C(x) be the propositional function x is in your class.

Let P(x) be "x has a callular phone" then we have $\forall x\,P(x)$ the first way, or $\forall x(C(x) \rightarrow P(x))$ the second way.

c) there is a Person in your class who cannot swim?

Let S(x) be the proposition "x can swim" and C(x) be the proposition "x in your class"

i) we can write this statement as $\exists x\,(\neg S(x))$ the first way:

ii) the second way is $\exists x\,(C(x) \wedge \neg S(x))$

Question No 25:-

a) No one is Perfect:

Let P(x) be "x is a perfect" the domain in people of inverse. this means that every one has a property of not being Profect So, we can write as $\forall x\,[P(x)]$ or $\neg \exists x\,P(x)$.

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b) Not everyone is perfect.
Let $p(x)$ be "x is perfect".
So, if we can write as
$\forall x\, p(x)$ "everyone is perfect"
then not everyone is perfect
is written as $\neg \forall x\, p(x)$.

c) All your friends are perfect.
Let $p(x)$ be "x are perfect"
and $F(x)$ be "x is your friends"
so, we can write this statement
as: if someone is your friend, then he
is perfect
$\forall x\, (F(x) \rightarrow p(x))$.

Question No 27:-
Logical expression=?

Let
$c(x)$ is the proportion function.
$c(x)$, is your school or class.

a) A student in your school have lived in veitnam:-
Let
$p(x)$ "x lived in vietman"
Existential quantifier.
$\exists x\, p(x)$.

d) Everyone in your class enjoys thai food:-
$M(x)$ "x enjoys thai food
Universal quantifier.
$\forall x\, M(x)$

e) Someone in your class does not play hockey.
$L(x)$ "x plays hockey"
Existential quantifier.
$\exists x \sim L(x)$
x does not play hockey Now,
the logically expression is given
as $c(x)$ "students in the
class" $\exists x\, (c(x) \wedge \sim c(x))$

Question No 29:-
$T(x)$ "x is tatuology"
$c(x)$ "x is contradiction"

(a)
Some Propositions are tautologic:
Existential quantifier
$\exists x\, T(x)$

(c) the disjunction of two contengence can be tautology.
Existential quantifier.
Let $p(x)$ and $p(y)$ be the.

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Two Contegencies.

Disjunction of two contingencies

$= P(x,y)$

$\exists x \exists y (\neg T(x) \wedge \neg P(x) \wedge \neg T(y) \wedge \neg P(y) \wedge P(x,y))$

F ∧ F ∧ F∧F.

F ∧ F₁ Wrong - question.

d) the conjunction of two totadogy can be tatology:-

Conjunction of two totaulogies $= T(x) \wedge T(x)$

$\forall x \forall y (T(x) \wedge T(y)) \rightarrow T(x \wedge y)$

$T \wedge T \rightarrow T \quad T \rightarrow T \Rightarrow \boxed{T}$

universal quantifier.

Question No 31:-

$Q(x,y,z)$

$x = 0,1,2$

$y = 0,1$

$z = 0,1$

Conjunction and Disjunction = ?

(a) $\forall y\, Q(0,y,0)$

Existential quantifier = ∨

universal quantifier = ∧

(a) $\forall y\, Q(0,y,0)$

universal quantifier =

$Q(0,y,0)$

$y = (0,1)$

$Q(0,0,0) \wedge Q(0,1,0)$

d) $\exists x \neg Q(x,0,1)$

Existential quantifier.

$\neg Q(x,0,1)$

$x = 0,1,2.$

$\neg Q(0,0,1) \vee \neg Q(1,0,1) \vee \neg Q(2,0,1)$

Question No 33:-

Logically express = ?

Simple english = ?

Negation of expression = ?

a)

Some old dogs can learn new tricks.

$T(x)$ is the propositional functions

"x can learn new tricks"

"Some old dogs can learn new tricks"

$\exists x\, t(x)$

(Existential Quantifier)

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→ **Negation:-**

$\sim \exists x\, T(x)$

"Every old dog does not learn new tricks."

→Note:-
No negation is to the left of the quantifier.

So, $\sim \exists(x)\, T(x) = \forall x \sim T(x)$

"No old old can learn new trickes"

(d) there is a no dog that can talk.

Let $P(x)$ is a propositional function

"x that can talk"

Existential Quantifer.

$\sim \exists x\, P(x)$

"there is a no dog that can talk"

Negation:-

$\exists x\, P(x)$

"there, is a dog that can talk"

Questional No 35:-

Counterexample:-

$\forall x\ (x^2 \geq x)$

As, $\forall x\ (x^2 \geq x)$

Let $x = 1$

$P(1)$ $\forall 2\ (1)^2 \geq 2)$

$4 \geq 2$

Hence, this Proportion is true. And has no counter example.

(C)

$\forall x\ (x = 1)$

As, $\forall x\ (x = 1)$

Universal Quantifier.

$\forall = 0, \pm 1, \pm 2, \ldots\ldots$

$P(2) = 2 \neq 1$

So,

this proposition is not true. If provided a counter example.

Question No 39:-

F(P) "printer P is out of service"

B(P) "printer P is busy"

L(j) "Print Job j is lost."

Q(j) "Print Job j is queued"

(a)

$$\exists P\,(F(P) \wedge B(P)) \rightarrow \exists j\, L(j)$$

$\exists \rightarrow$ If there is a Printer.

$\wedge \rightarrow$ Both

$\rightarrow$ then.

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Now,
"If there is a printer that is who both out of service and busy, then some job has been lost"

(d) $(\forall P B(P) \wedge \forall j Q(j)) \rightarrow \exists j L(j)$

If every printer is busy and every print job is queued then some print job is lost.

Question No 41:-

Predicates, quantifer and logical connective.

b)

When there is an active alert, all queued messages are transmitted.

the proposition are given as,

$A(x)$ "x is active alert"

$Q(x)$ "x is squeued messages"

$T(x)$ "x is the messages transmitted"

the logical expression is given as,

there is $\exists x\, A(x) \rightarrow \forall x\, Q(x) \rightarrow T(x)$

(there is shows existantial quantifier)".

(d).

Each Participant = universal quantifier.

Did not put on a special list $= \neg P(x)$

Was billed $= M(x)$

$\forall x (\neg P(x) \rightarrow M(x))$

Question No 43:-

$\forall x\, (P(x) \rightarrow Q(x))$

→ Equivalent.

$\forall x\, P(x) \rightarrow \forall x\, Q(x)$

Universal quantifier is used Means Proposition May be true or false.

Let. $P(x)$ = this proposition sometime true and also sometime false.

$Q(x)$ = Let this proposition is always false.

(i)

If $P(x)$ is T and $Q(x)$ is F then

$\forall x\, (P(x) \rightarrow Q(x))$

$T \rightarrow F \Rightarrow \boxed{F}$

(ii)

If $P(x)$ is F and $Q(x)$ is false F.

$\forall x P(x) \rightarrow \forall x\, Q(x)$.

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$F \to F$ [T]

Hence, $\forall x\,(P(x) \to Q(x))$ and $\forall x\,P(x) \to \forall x\,Q(x)$ are not logically equivalent.

Question No 45:-

$\exists x\,(P(x) \vee Q(x))$

→ **Equivalent**

$\exists x\,P(x) \vee \exists x\,Q(x)$

Existential quantifier is used. Means. It may be T or F but not both.

$P(x)$ = this proposition may be true or false but not both.

$Q(x)$ = T or F but not both.

(i)

$P(x)$ is T and $Q(x)$ is F.

$\exists x\,(P(x) \vee Q(x))$

$T \vee F = [T]$

(ii)

When $Q(x)$ is T and $P(x)$ is F

$\exists x\,P(x) \vee \exists x\,Q(x)$

$F \vee T = [T]$

Both statement are true when at least one $p(x)$ and $Q(x)$ is true

Hence, they are equivalent.

Question No 47:-

x is constant

A is a variable

↓

May be T or F.

(a)

$(\forall x\,P(x) \wedge A \equiv \forall x\,(P(x) \wedge A)$

If We take A is true then.

L.H.S:-

$(\forall x\,P(x)) \wedge A$

$= \forall x\,P(x) \longrightarrow (A)$

Because, the conjunction of any proposition with a true proposition has the same truth value as the proposition.

R.H.S:-

$\forall x\,(P(x) \wedge A)$

When A is true.

$= \forall x\,(P(x) \wedge T)$

$= \forall x\,P(x) \longrightarrow (B)$

Hence,

the two proposition are logically equivalent.

Question No 49:-

$\forall x\,(P(x) \to A) \equiv \exists x\,P(x) \to A$

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Case I:

→ If A is true.

L.H.S =

$\forall x\,(p(x) \to A)$ $\quad p(x) \to T$

$= p(x) \to A$ is true.

R.H.S =

$\exists x\, p(x) \to A$

$p(x) \to A$ is also true.

L.H.S = R.H.S

Case II:-

→ If A is false.

Let the proposition $p(x)$ is false then.

L.H.S

$= \forall x(p(x) \to A)$

$= F \to F \quad = \boxed{T} \to (A)$

R.H.S

$\exists x\,(p(x) \to A)$

$\exists x\, p(x) \to A$

$= F \to F \quad = \boxed{T} \to (B)$

So, $\forall x\,(p(x) \to A) \equiv \exists x\, p(x) \to A$

→ Let the proposition $p(x)$ is true

L.H.S:-

$\forall x(p(x) \to A)$

$= T \to F \quad \boxed{F} \to A$

R.H.S:

$\exists x\, p(x) \to A$

$= T \to F \quad \boxed{= F}$

$\forall x\,(p(x) \to A) \equiv \exists x\, p(x) \to A$

Hence, both the statement are equivalent.

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Exercise No 1-7:

Question No 1:
(Direct Proof)
Show sum of two odd integers is even:

Let m and n are two odd integers.

Indirect proof we assume that first proposition is true and final we show that second is also true

By defination of odd integers.

$m = 2K+1$, Where K is integers of some.

$n = 2t+1$, Where t is some integers.

Now we find their sum

$$m+n = (2K+1)+(2t+1)$$
$$= 2K+1+2t+1 \Rightarrow 2K+2t+2$$
$$2(K+t+1) \quad \because K+t+1 = \mathbb{Z}$$
$$m+n = 2(\mathbb{Z})$$

So, by diffination sum is twice of of an integer and its provided that the second proposition that, their sum is even is true.

Question No 3: Square of an even number is an even number:

Left a be a number By using direct proof rule let assume that P(x): a or b even number is true. So by defination of an even number.

$a = (2K) \rightarrow (i) \quad \because K$ is an integers.

$b = (2t) \quad \because t$ is an integers

Now we have to find or proof that their square is also even squaring on both side of eq (i)

$$a^2 = (2K)^2$$
$$a^2 = 4K^2$$
$$a^2 = 2(2K^2)$$

As its square is multiple of two, so by defination it is also even number.

Question No 05:
Answer:

We can give a direct proof.

Suppose $m+n$ is even so, by defination

$m+n = 2K \quad \because 2K$ integers.

and $n+p$ is also even so, by defination

$n+p = 2t \quad = t$ integers.

their sum is given as.

$$m+n+n+p = 2K+2t$$
$$m+p+2n = 2(K+t)$$
$$m+p = 2K+2t-2n \quad \Rightarrow m+p = 2(K+t-n)$$

so it is also even numbers.

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Question No 07:-

Every odd integer is difference of two squares:-

Let a is an odd integers and expressed as,

$$a = 2k+1 \longrightarrow ①$$

Let two integers are f and g which are expressed as

$$f = k+1$$
$$g = k$$

the diffirence of their squares as

$$f^2 - g^2 = (k+1)^2 - (k)^2$$
$$= k^2 + 2k + 1 - k^2$$
$$f^2 - g^2 = 2k + 1 = a$$

So, the statement is proved.

Question No 09:-

the proposition to be provide here is, If r is rational and i is an irrational when $s = r+i$ is an irrational number.

By proving with contradiction let we sppose that r is rational, i is rational and s is rational umber. as,

$$s = a/b \qquad b \neq 0 \qquad r = c/d$$

$$s + (-r) = a/b - c/d$$
$$= \frac{ad - bc}{bd}$$

this shows that $s+(-r)$ is rational. But $s+(-r) = r+i-r = i$ this is forcing us to conclude that i is rational but in general i is irrational. So, our assumption that s is rational is incorrect and generally s is irrational So it is proved.

Question No 11:-

To disprove this statement Let a and b are two Irrational numbers. as.

$$a = \sqrt{2}$$
$$b = \sqrt{2}$$

their product

$$a \cdot b = \sqrt{2}\sqrt{2}$$

$\therefore$ 2 which is rational number.

Question No 13:-

Let p(x) is: x is irrational and Q(x): $1/x$ is irrational We prove it by contraposition Let.

$$\sim Q(x) \longrightarrow \sim p(x)$$

Now, We apply direct rule on it Now, We assume that.

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$\frac{1}{x}$ is rational and can be written as.

$1/x = p/q$, $q \neq 0$

$1/(1/x) = 1/(p/q)$

$x = p/q$ $\quad p \neq 0$

So $\sim p(x)$ is also true, it means x is also rational numbers.

Question No 15:-

if $x+y \geq 2$ then $x \geq 1$ or $y \geq 1$

Let $p(x)$: $x+y \geq 2$

$Q(x) = x \geq 1$, $y \geq 1$.

the contrapositive of this is given as

$\sim Q(x) \longrightarrow \sim p(x)$ in english

"If it is not true that $x \geq 1$ or $y \geq 1$, then it is not true that $x+y \geq 2$".

Now are direct method on it as

assume that.

If it is not true that $x \geq 1$ or $y \geq 1$. then by demargons Law $x < 1$ or $y < 1$.

Adding these inequalities then

$x+y < 2$. this is negation of $x+y \geq 2$ and our proof is complete.

Question No 17:-

If n is an integers and n^3+5 is odd, then n is even.

a) By using proof of contrapositions:-

In this question we prove its contrapositive. If n is odd, then n^3+5 is even. By using direct proof on this contrapositive.

Assume n is odd by deffination of odd number.

$n = 2k+1$

$n^3+5 = (2k+1)^3+5$

$= 8k^3+12k^2+6k+6$

$= 2(4k^3+6k^2+3k+3)$

thus (n^3+15) is two time some integers.

So, it is even.

b) By using proof of contradiction:-

Suppose that n^3+5 is odd and n. is odd.

If n is odd then its product is also odd as,

$n = 2k+1$

$n^3 = (2k+1)^3$

$= 8k^3+12k^2+6k+1$

$n^3 = 2(4k^3+6k^2+3k)+1$

So, n^3 is odd.

But when 5 added in it then it becomes even.

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as $n^3+5 = 8k^3+12k^2+6k+1+5$

$n^3+5 = 2(4k^3+6k^2+3k+3)$

As n^3+5 is twice of some integer So, n^3+5 is even. So, we can say that our assumption is false and n^3+5 is odd.

Question No 19:-

the Proposition we are trying to prove is If a is a positive integers greater than 1. then $(a)^2>a$. our proof is vacous one since. Hypothesis is false theconditional statement is automatically true.

Question No 21:-

the proposition we are trying to Prove is given as "If a and b are positive real number then $(a+b)' \geq a'+b'$" our proof is direct proof.

$$(a+b)' = a+b = a'+b'$$

Finally, by addition rule, We can concluded form $(a+b)' = a'+b'$ that

$$(a+b)' \geq a'+b'$$

Question No 25:- Show that at Least ten of any 64 days chosen must fall on the some day, of Weak.

We give proof by contradiction. If there were nine. of fewer days on each day of Week, this would acceding for at most $9 \cdot 7 = 63$ days. But We chosen 64days. this contradiction shows that at least ten of the days must be on the same day of the Week.

Question No 27:-

If n is a positive integers, then n is odd if and only if 5n+6 is odd.

We must prove two conditional statements.

first We assume that n is odd and 5n+6 is also odd. We use direct Proof on it.

Let n is odd be true now by defination $n = 2k+1$.

$$5n+6 = 5(2k+1)+6$$
$$= 10k+5+6$$
$$= 10k+11$$
$$= 2(5k+5)+1.$$

So, 5n+6 is also odd second We proof it by contraposition. Let.

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n is even odd 5n+6 is also even.

By defination.

$n = 2K$

$5n+6 = 5(2K)+6$

$= 10K+6$

$= 2(5K+3)$

So, 5n+6 is also even.

Here we have proved the statement.

Question No 29:-

this is a true proposition We give a proof by contradiction.

Suppose that m is neither 1 nor -1 then mn has a fector (namely |m|) larger than 1.

on the other hand mn=1, 1 has no such factor.

So, We can conclude that

$m = 1$ or $m = -1$ and $n = 1$ or $n = -1$

therefore. $mn = 1$

So, it has proved.

Question No 31:-

In this question We take x as even and prove all these parts

By defination.

$x = 2K$ ∵ K is an integer.

i) $3x+2$ is even

$= 3(2K)+2$

$= 6K+2$

$= 2(3K+1)$

Here, 3x+2 is two multiple of some integer and thus 3x+2 is even.

ii) x+5 is odd

Put $x = 2K$ ∵ x is even

$= 2K+5$

$= 2K+4+1$

$= 2(K+2)+4$

So, by defination x+5 is odd.

iii) x^2 is even.

Put $x = 2K$

$= (2K)^2$

$= 4K^2$

$= 2(2K^2)$

So, x^2 is also even. therefore this statement is proved.

Question No 35:-

$\sqrt{x+3} = 3-x$ it is correct.

i) $\sqrt{x+3} = 3-x$ given.

ii) $x+3 = 9-6x+x^2$ by saquaring on b/s

iii) $0 = x^2-7x+6$ by sattracting (x+3) on b/s

iv) $0 = (x-1)(x-6)$ by factrization.

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v) $x=1$ or $x=6$ as $(a-b)=0 \Rightarrow a=a, b=a$ all the securing step are not reversible So all step are true but the last step (v) is not valid because when we put $x=6$ then.

$$\sqrt{6+3} = 3-6$$

$$\sqrt{9} = 3 \implies 3=-3$$

that is false.

Questions No 37:-

Suppose that we have proved $P_1 \to P_4 \to P_2 \to P_5 \to P_3 \to P_1$. Imagin these statement arranged around a circle. then to prove that each one of these proposition (say p_i) implies each of others (say p_i), We just have to fallow the circle starting at p_i, until we come to p_i using hepothetical syllogism repeatedly.

Question No 39:-

We can give a very satisfying proof by contradiction here. Suppose that all of number $a_1, a_2, \ldots, a_n$ are less than their average, which we can denoted by A.

In symbols we have $a_i < A$ for all if we add these n inequalities, We see that.

$$a_1 + a_2 + \cdots a_n < nA.$$

$$A = \frac{a_1 + a_2 + \cdots + a_n}{n}$$

the above two formulae contradicts each others. however they say that $nA < nA$. thus our assumption is false and we conclude that at least one of numbers a_i is greater than or equal to their average.

Question No 41:-

We can prove that there four statement are equivelent in a circular way.

(i) $\to$ (ii) $\to$ (iii) $\to$ (iv) $\to$ (v).

First we prove

(i) $\to$ (ii) in wording.

If n is even the n+1 is odd.

Assume n is even as.

$$n = 2K$$

$$n+1 = 2K+1.$$

Hence $n+1$ is odd.

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Second: We prove (ii)→(iii)

suppose (n+1) is odd.

$n+1 = 2K+1$.

$3n+1 = 2n + (2K+1)$

$= 2n + 2K + 1$

hence, it is odd.

third: We prove (iii)→(iv)

suppose,

$3n+1$ = is odd say as

$3n+1 = 2K+1$

then $3n = (3n+1) - 1$

$= 2k+1-1$

$3n = 2K$

So, $3n$ is even.

Finally:

We have to prove that if $3n$ is even then n is even.

By using - contraposition.

Suppose that n is odd.

$n = 2K+1$

$3n = 3(2K+1)$

$= 6K+3$

$= (6K+2)+1$

$= 2(3K+1)+1$

So, we can say that $3n$ is odd. So, proof is completed by contraposition.

1.8: Proof Methods and Strategy:

Exercise No 1.8:-

Question No 1:-

Prove that $n^2+1 \geq 2^n$ when in is a positive integer with $1 \leq n \leq 4$.

In prove $n^2+1 \geq 2^n$ where n is 1, 2, 3, 4, We use Proof of exhaustion.

i) When $n = 1$.

$(1)^2 + 1 \geq 2^1$

$2 \geq 2$

ii) for $n = 2$

$(2)^2 + 1 \geq (2)^2$

$4+1 \geq 4$

iii) fo $n = 3$

$(3)^3 + 1 \geq 2^3$

$9+1 \geq 8$

$10 \geq 8$

iv) For $n = 4$

$(4)^2 + 1 = 2^4$

$16+1 = 16$

$17 \geq 16$

Hence, $n^2+1 \geq 2^n$ is proof that.

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Question No 02:-

the cubes that might go into the sum are 1, 8, 27, 64, 125, 216, 343, 512, 729. We must show that no two of these sum to a number on this list. If We try the 45 combination $(1+1, 1+8, \ldots 1+729), (8+8, 8+27, \ldots 8+729, \ldots 729+729)$ We see that none of them Works. Having exheusted the possibilities, We conclude that no cube less than 1000 is sum of two cubes.

Question No 03:-

By using hint, We consider two cases determined by relative Size of x and y.

Case I: If $x \geq y$.

By defination.

$\max(x,y) = x$ and.

$\min(x,y) = y$

So, In this case

$\max(x,y) + \min(x+y) = x+y$

as desired

Case II: If $x < y$

$\max(x,y) = y$

$\min(x,y) = x$

So, in this case

$\max(x+y) + \min(x,y) = x+y$

$= x+y$

as desired

So, it is proved.

Question No 4:-

In this question We use three case to prove this statement by case:

Case I: When a is smallest it means that $a < \min(b,c)$

So,

$\min(a, \min(b,c)) = \min(\min(a,b),c)$ $\because a<b<c$

$a = a$ $\because a<b$

So L.H.S = R.H.S.

case II:

When b is smallest.

L.H.S = $\min(a, \min(b,c))$

$\because (b,c) < a$

$\because b < a$

R.H.S = $\min(\min(a,b),c)$

$\min(a, \min(b,c)) = \min(\min(a,b),c)$

$b = b$

So L.H.S = R.H.S.

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Case III: When c is smallest

L.H.S = min (a, min (b, c))

= (b, c) < 0

∵ c < b

= c

R.H.S = min (min (a, b), c)

= (a, b) > c

= c

So L.H.S = R.H.S.

thus, by these three cases the statement is proved.

Question No 5:-

Because $|x-y| = |y-x|$ the values of x and y can be interchange. therefore, by using without less of generality, We can assume that $x \geq y$

$|x-y| = x-y$.

In this case the first expression will be as, $\min(x,y) = \frac{x+y-|x-y|}{2}$

$= \frac{x+y-(x-y)}{2} \Rightarrow \frac{x+y-x+y}{2}$

$= \frac{2y}{2} = y$

So, y is the minimum.

the second expression is given as.

$\max(x,y) = \frac{x+y+|x-y|}{2}$

$= \frac{(x+y)+(x-y)}{2}$

$= \frac{x+y+x-y}{2} = 2x/2$

$= x$

So, Maximum Value is x.

Question No 7:- $|x|+|y| \geq |x+y|$

there are several case to prove it.

1) We assume that x and y both are positive real number then.

$|x|+|y| = x+y = |x+y|$

2) If x and y both are negative then:

$|x|+|y| = (-x)+(-y) = |x|+|y|$

3) In third case we assume that if one of x and y is positive and other is negative. In this case we use without loss of generally and. Let x is positive and y is negative

i.e $x \geq 0$, $y < 0$

this case has two subcases.

i) When $x \geq -y$ then $x+y \geq 0$.

therefore, $|x+y| = x+y$ and we use into as $|x+y| \geq 0$ on the other hand $|x|+|y| = x+|y|$ is positive larger than x as :-

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$|x| + |y| > x$

So, We can write as.

$x+y < x < |x|+|y|$, as desired.

(ii) Let $x < -y$ then

$|x+y| = -(x+y) = (-x)+(-y)$

So, this number is less than or equal to $-y$ on their hand.

$|x|+|y| = |x|+(-y)$ is a positive and grander than $-y$. so, nequality is.

$$|x+y| \leq -y \leq |x|+|y|$$

Question No 9:-

We want to find consecutive squares that are for apart. If n is large enough, then $(n+1)^2$ will be much greater than n^2. Let $n=100$ then $100)^2 = 10000$ and $(100+1)^2 = 10201$ So, 201 consecutive numbers 10001, 10002 0200 are not Perfect square. the first oo of these ~~100~~ will satisfy the equirement of this question. our proof ; constructive because we exhibited umber.

Question No 12:-

We try some small numbers and discover that only a pair of consecutive number that is 8 and 9 in which one is perfect square and other is Perfect cube. as:-

$8 = 2^3$ — Perfect cube.

$9 = 3^2$ — perfect square.

so, there or it is a consecutive Proof.

Question No 13:-

this is a non-constructive Proof. Let $x = 2$ (rational) and $y = \sqrt{2}$ (irrational) if $x^y = (2)^{\sqrt{2}}$ is irrational So,.

We are done.

But if not, then Let.

$x = 2^{\sqrt{2}}$ and $y = \sqrt{2}/4$

x is rational as assumption and y is irrational. So,

$x^y = \left(2^{\sqrt{2}}\right)^{\sqrt{2}/4}$.

$= 2^{\sqrt{2}\cdot\sqrt{2}/4}$

$= 2^{2/4} = 2^{1/2} = \sqrt{2}$

which is an irrational.

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Question No 15:-

a) $\exists x \forall y \, (P(y) \leftrightarrow x=y)$

this statement asserts the existance of x with a certain property if y=x then P(x) is true. But if y is anything other than x, then P(y) is not true.

thus x is the unique element that makes P true.

b) $\exists x \, P(x) \wedge \forall x \forall y \, (P(x) \wedge P(y) \rightarrow x=y)$

the first clause here says that there is an element that makes P true. the second clause says that whenever two element both makes P true, they are infact same element. To gether this says that P is satisfied by exactly one element.

Question No 17:-

$|a-c| = |b-c|$

$|a-c| = |b-c| \approx a-c = b-c$ or $a-c = -b+c$

the first of these is equivalent to $x=b$, which contradicts the assumption made in this problem. so, the original equation is equivelent to

$$a-c = -b+c.$$

$$a-c = -b+c$$

$$a+b = 2c$$

$$c = \frac{a+b}{2}$$

thus there is a unique solution Further more, this c is an integer, because sum of odd integer a and b is even.

Question No 19:-

We are being to solve

$$n = (k-2) + (k+3) \text{ for } k.$$

this equation is equivalent to

$$n = k-2+k+3$$

$$n = 2k+1$$

$$k = \frac{n-1}{2}$$

this is one only value of k that makes our equation true. Since n is odd, n-1 is even So, k is an integer.

Question No 21:-

there are two case in this equation:

Case I:

If x is itself an integer, then we can take n=x and so, No other

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Solution is possible in this case. Since if integer n is (negative) greater than x. then is at least x+1, which would make ≥1.

Case II:-

If x is not an integer, then round it up to next integer and call that integer n. We let $\varepsilon = n - x$ clearly $0 \leq \varepsilon < 1$ this is only ε that will work with this n and n cannot be any large >, since ε is constrained to be less than 1.

Question No 22:-

Use forward reasoning

$$\left(\frac{x-1}{x}\right)^2 \geq 0$$

using hint.

$$\left(\frac{x}{x} - \frac{1}{x}\right)^2 \geq 0$$

$$1 + \frac{1}{x^2} \geq 2$$

$$\left(1 - \frac{1}{x}\right)^2 \geq 0$$

$$\frac{x^2+1}{x^2} \geq 2$$

$$\left(1 + \frac{1}{x^2} - 2\right) \geq 0$$

So, proved.

Adding 2 on b/s,

Question No 34:-

Prove that $\sqrt[3]{2}$ is irrational

We can:- proved it by contradiction.

Let $2^{4/3}$ is rational. So by defination

$$2^{4/3} = p/q \quad \because p, q \text{ are integer.}$$

Cubing on b/s.

$$2 = p^3/q^3 \Rightarrow p^3 = 2q^3$$

So, p^3 is even and hence p is even and written as $p = 2s$

$$(2s)^3 = 2q^3$$

$$8s^3 = 2q^3$$

$$4s^3 = q^3$$

So, q^3 is also even and hence q is even. By conclusion we say that both p and q are even have common factor that they are divided by 2.

this contradicts our choice of p/q

So. our assumption is false that $\therefore \sqrt[3]{2}$ is rational and original statement that $\sqrt[3]{2}$ is true, so it is proved.

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Question No 36:-

the average of different number is certainly always b/w two number Furthermore, average of a rational number and irrational number must be irrational Because the equation.

$$a = \frac{x+y}{2}$$ leads to

$$y = 2a - x$$

that which would be rational if were rational.

Exercise No 4.1 :-

Question No 1:

Does 17 divide each of these numbers.

a) 68

We can write this as $a|b$. therefore $a=17$ $b=68$

$68/17$ So,

By diffination.

$68 = 17 \times 4$ or

$\frac{68}{17} = 4$ $\boxed{c=4}$

So it divides 68

b) 84

We can write this as

$17|84$ or

$84|17$

$17\overline{)84}$ (4, 68, 16)

No, this do not divide 84 and its remainder's 16.

c) 357

We can write as $17|357$

$357/17 = 21$

So $c=21$ it mean $357 = 17 \cdot 21$

So it divides

d) 1001

We can write this as

$17|1001$ (or) $1001|17$

$17\overline{)1001}$ (18, 996, 15)

No. it camot divide 1001. Because remainder is 15.

Question No 3:-

Prove Part (i) of theorem 1 is true.

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Prove if a|b, then a|bc ; for all integers c:-

Find we take that.

a|b

By defination

b=aS ∵ S = some integer.

the next step is that.

bc = as(c)

= a(sc)

as bc is multiple of a and some constant So, by defination

∴ a|bc.

So, it is proved.

Question No 4:-

Prove that (iii) of theorem 1.

if a|b and b|c, then a|c.

By defination we know that if a|b then.

b = at (∵ t is integers)

and if b|c then.

c = bk (∵ k= is integer)

So,

c = bk

c = atk

c = a(tk)

as, c is multiple of a and some integer so,

also, it is proved.

Question No 5:-

Show that if a|b and b|a, where a and b are integers then

a = b or a = -b.

the given condition imply that there are integers s and t such that. a|b and b|a.

b = at a = bs.

combining these, we obtain

a = bs

a = ats; since a≠0, we conclude that st = 1.

Now the only way for this to happen is for s = t = 1 or

s = t = -1

therefore either a = b or a = -b.

Question No 6:-

If a, b, c and d are integers, where a≠0 such that a|c and b|d then ab|cd.

If a|c and b|d

We know that

c = at d = bs.

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then;

$cd = (at)(bs)$

$cd = ab(ts)$

therefore either. so it is proved that $ab|cd$.

Question No 7:-

the given condition means that $ac|bc$.

By defination

$bc = (ac)t$ ∵ t is integer.

since $c \neq 0$, we can divide by c on b/s to obtain

$$\frac{bc}{c} = \frac{ac}{c}t$$

$$b = at$$

this is the definition of $a|b$ as desired.

Question No 8:-

if $a|bc$, where a, b and c are positive integers and $a \neq 0$, then $a|b$ or $a|c$.

if $a|bc$ by deffination.

$bc = at$

$b = a\left(\frac{t}{c}\right)$ So, $a|b$.

or

$c = a\left(\frac{t}{b}\right)$ So, it is proved.

So, $a|c$

Question No 9:-

a) 19 is divided by 7

19 in term of

$a = dq + r$

$19 = 7 \cdot 2 + 5$

So, $\boxed{q=2}$ and $\boxed{r=5}$

c) 789 is divided by 23

$a = 789$, $d = 23$

in term of

$a = dq + r$

$789 = 23 \cdot 34 + 7$

So, $\boxed{q=34}$, $\boxed{r=7}$

e) 0 is divided by 19.

In term of $a=0$, $d=19$

$a = dq + r$

$0 = 19 \cdot 0 + 0$

So, $\boxed{q=0}$, $\boxed{r=0}$

g) -1 is divided by 3.

$a = -1$ $d = 3$

in term of

$a = dq + r$.

$-1 = 3 \cdot (-1) + 2$

So, $\boxed{q=-1}$ $\boxed{r=2}$

Question No 10:

a) 44 is divided 8

$a = 44$, $d = 8$

$a = dq + r$ in term of

$44 = 85 + 4$

$\boxed{q=5}$ $\boxed{r=4}$

d) -1 is divided by 23

$a = -1$, $d = 23$

in term of

$a = dq + r$

$-1 = 23(-1) + 22$

So, $\boxed{q=-1}$, $\boxed{r=22}$

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Question No 11:-

What time does a 12-hour clock read.

a) 80 hours after it reads 11:00?

We are doing arithmetic module 12 For this exercise.

Because $11+80$ mode $12=7$, the clock reads 7:00

b) 40 hourse before it reads 12:00?

Becaus $m=12$

$12-40 \bmod 12 = -28 \bmod 12 = -28+36$ mode $12=8$, the clock reads 8:00

c) 100 hours after it reads 6:00?

mod $= m = 12$

Because.

$6+100 \bmod 12 = 10$

the clock reads 10:00.

Question No 13:-

a) $= 11 \pmod{13}$

b) $= 9 \pmod{13}$

Find integer c with $0 \le c \le 12$ such that a) $c \equiv 9a \pmod{13}$

$9 \cdot 11 \bmod 13 = 36 \bmod 13 = 10 \longrightarrow$ Remainder.

c) $c \equiv a+b \pmod{13}$

$4+9 \pmod{13} = 13 \pmod{13} = 0$

Question No 14:-

Suppose that a and b are integer $a=11 \pmod{19}$, and $b=3 \pmod{19}$. Find the integer c is $0 \le c \le 18$ such that.

a) $c \equiv 13a \pmod{19}$

$c = 13a \pmod{19}$

$c = 13 \cdot 11 \pmod{19}$

$c = 143 \pmod{19}$

$c = 143 \bmod 19 = 10 \rightarrow$ Remainder.

b) $c \equiv 8b \pmod{19}$

$c = 8b \pmod{19}$

$= 8 \cdot 3 \bmod 19$

$= 24 \bmod 19$

$24 \bmod 19 = 5$ — Remainder.

(c) $c \equiv a-b \pmod{19}$

$c = a-b \pmod{19}$

$= 11-3 \pmod{19}$

$= 8 \bmod 19$

$8 \bmod 19 = +4$ - Remainder.

(d) $c \equiv 7a+3b \pmod{19}$

$c = 7a+3b \pmod{19}$

$= 7 \cdot 11 + 3 \cdot 3 \pmod{19} = 77+9 \pmod{19}$

$= 86 \pmod{19}$

$= 86 \bmod 19 = 10$ - Remainder.

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(e) $c \equiv 2a^2 + 3b^2 \pmod{19}$

$c = 2a^2 + 3b^2 \pmod{19}$

$= 2(11)^2 + 3(3)^2 \pmod{19} = 242 + 27 \pmod{19}$

$= 269 \mod 19$

$= 269 \mod 19 = 3$ — Remainder

(f) $c \equiv a^3 + 4b^3 \pmod{19}$

$c = a^3 + 4b^3 \pmod{19}$

$= (11)^3 + 4(3)^3 \pmod{19}$

$= 1331 + 108 \mod 19$

$= 1439 \mod 19$

$= 1439 \mod 19 = 14$ — Remainder

Question No 15:-

Let m be a positive integer. Show that $a \equiv b \pmod{m}$ if $a \bmod m = b \bmod m$.

the given condition that $a \bmod m = b \bmod m$. a and b have same remainder when divided by m.

$a = q_1 m + r$ → (i) and $b = q_2 m + r$ → (ii)

some integer q_1 and q_2 and r.

Subtracting by eq (i) and (ii)

$a - b = q_1 m - q_2 m + r - r$

$= q_1 m - q_2 m$

$a - b = (q_1 - q_2) m$

m divided by a-b.

$a \equiv b \pmod{m}$.

Question No 16:-

Let m be a positive integer. Show that $a \bmod m = b \bmod m$ if $a \equiv b \pmod{m}$

the given condition that $a \equiv b \bmod m$ a and b have same remainder when divided by m.

Respectively,

$a = q_1 m + r$ $\quad b = q_2 m + r$

compare a and b.

$q_1 m + r = q_2 m + r$. • Prove wrong

same remainder. this question

$q_1 m + r - r = q_2 m$ and available to

$q_1 m - q_2 m = 0$ prove right next

$a - b = (q_1 - q_2) m$. page

$a \bmod m = b \bmod m$.

Question No 17:-

Show that if n and k are positive integer then $\lceil n/k \rceil = \lfloor (n-1)/k \rfloor + 1$.

the quotient n/k lies b/w two consective integer $b-1$ and b. there exist a positive integer $b-1 < n/k \leq b$.

$\lceil n/k \rceil = b$

$n/k > b-1$

$n > k(b-1)$

$n-1 \geq k(b-1)$

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$(n-1)/k \geq (b-1)$

$\lceil (n-1)/k \rceil \geq (b-1)$ ..

on the other hand:

$\lfloor (n-1)/k \rfloor \leq (n-1)/k < n/k \leq b$

$\lfloor (n-1)/k \rfloor < b.$

$\lfloor (n-1)/k \rfloor = b-1.$

Question No 23:-

Find a divided m and a modm when

a) $a = 228$ $m = 119.$

Because $228 = a$ $m = 119.$

$228 = 1 \cdot 119 + 109$, we have

228 divided 119 = 1.

228 mod 119 = 109.

b) $a = 9009$, $m = 223.$

Because $a = 9009.$ $m = 223$

$9009 = 40 \cdot 223 + 89$ we have

9009 divided 223 = 40

9009 mod 223 = 89

c) $a = -10101$, $m = 333$

Because $a = -10101$ $m = 333$

$-10101 = -31 \cdot 333 + 222$ we have.

-10101 divided 333 = -31

-10101 mod 333 = 222.

d) $a = -765432$ $m = 38271.$

Because $a = -765432$ $m = 38271.$

$-765432 = -21 \cdot 38271 + 38259$ we have

-765432 divided 38271 = -21.

-765432 mod 38271 = 38259.

Question No 25:-

Find the integer a such that

a) $a \equiv -15 \pmod{27}$ and $-26 \leq a \leq 0$

$a \equiv -15 \pmod{27}$

Because -15 already satisfied the equality, the answer is -15.

b) $a \equiv 24 \pmod{31}$ and $-15 \leq a \leq 15$

$a \equiv +24 \pmod{31}$

Because 24 is too larger to satisfy the equality we substract 31 and obtain the answers -7.

c) $a = 99 \pmod{41}$ and $100 \leq a \leq 140$

$a = 99 \pmod{41}$

Because 99 is to small to satisfy the inequality we added 41 and obtain the answer is 140.

Question No 24:-

Find the integer a such that

a) $a = 43 \pmod{23}$ and $-22 \leq a \leq 0$

$a = 43 \pmod{23}$

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Because 43 is to large to satisfy inequality we substract to 22 we obtain answer +21.

b) $a = 17 \pmod{29}$ and $-14 \le a \le 14$.

$a = 17 \pmod{29}$

Because 17 is to small satisfy to inequality we subtract 29 and obtain the answer -14.

c) $a = -11 \pmod{21}$ and $90 \le a \le 110$

$a = -11 \pmod{21}$

Because -11 already satisfied the equality the answer is -15.

Question No 16:

Let m be a positive integer. show that $a \bmod m = b \bmod m$ if $a \equiv b \pmod{m}$

Proof:

[Note: "because this theorem is a bioconditional, We must prove it in both direction:"]

First assume

$a \equiv b \pmod{m}$

then $m \mid (a-b)$, so there is $k \in Z$ such that.

$a - b = mk$

then according to the division algorithm

there is $q \in Z$ such that

$a = mq + r$, $0 \le r < m$.

using

$a = mq + r$ to replace a in $a - b = mk$

$mq + r - b = mk$

So,

$mq - mk + r = b$

$m(q - k) + r = b$

this shows that r is remainder when b is divided by m. So $b \bmod m = (r = a \bmod m)$

$b \bmod m = a \bmod m$

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Exercise No. 1.5 :-

Question No 1:-

a) $\forall x \exists y \ (x<y)$

For every real number x there exist a real number y such that x is less than y.

b) $\forall x \forall y \ [((x \geq 0) \wedge (y \geq 0)) \rightarrow (xy \geq 0)]$

For every real number x and real number y, if x and y both are non-negative then their product it is non-negative.

c) $\forall x \forall y \ \exists z \ (xy = z)$

For Every real number x and real number y their exist a real number z such that $xy = z$

Question No 3:-

a) $\exists x \exists y \ Q(x,y)$

there exist a some student in your class who has sent a message to some student in your class.

c) $\forall x \exists y \ Q(x,y)$

Every student in your class has not a message to atleast one students in (your) class.

e) $\forall y \exists x \ Q(x,y)$

Every student in your class has been sent a massage From at least one student in your class.

Question No 7:-

a) $\neg T$ (Abdallah Hussain, Japanese)

Abdallah Hassain, does not like Taponese.

b) $\exists x \ T(x, \text{Korean}) \wedge \forall x \ T(x, \text{Mexican})$

Some student at your school Like Korean, cuisine, and every student in your Like mexican cuisine.

(c)

there is some cuisine that either Monique Aresnault or Jay Johnson Like.

e) $\exists x \exists z \forall y \ (T(x,y) \leftrightarrow T(z,y))$

there are two students at your school who have exactly the some tast.

Question No 9:-

a) Every body Leaves Jerry.

$\forall x \ (x, \text{Jerry})$

b) there is somebody whom every body leaves.

$\exists x \forall x \ (x,y)$

e) there is somebody whom no one loves.

$\exists y \forall x \neg < (x,y)$.

i) Every loves himself or herself.

$\forall x < (x,y)$

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Question No 11:

a) Loise has asked professor Michals a question.

A (Lois, Professor, Michael)

d) Some students has not asked any faculty member a question.

$\exists x \left(S(x) \wedge \forall y \left(f(y) \rightarrow \neg A(x,y)\right)\right)$

f) Some student has asked every faculty member a question.

$\exists x \left(S(x) \wedge \forall y \left(f(y) \rightarrow A(x,y)\right)\right)$

Question No 13:

(a)

Let $N(x,y)$: "x needs a course in y" the domain of x is computer science students and domain of y is academic subject So,

$\forall x\ N(x, \text{discrete Math})$.

(c)

Let $P(x,y)$: "x has taken y." the domain of x is students in this class and domain for y is computer Science courses.

$\forall x \exists y\ P(x,y)$.

(f)

Let $P(z,y)$ is "z is in y" and $Q(x,z)$ is "x has been in z" x range over student in class, y ranges over buildings on campus, z ranges over room

$\exists x \exists y \forall z \left(P(z,y) \rightarrow Q(x,z)\right)$

Question No 17:

a) Every user has access to exactly one mail box.

Let $A(u,m)$ means that user u has access to mailbox m $(\forall u \forall s)$

$\forall u \exists m \left(A(u,m)\right)$

(c)

Let $E(s,x)$ mean that Website s has extension x. and $A(x,s)$ mean: that user u can access Website S. So,

$\forall u \forall s \left(E(s,edu) \rightarrow A(u,s)\right)$

Question No 20:

a) Product of two negative integers is positive.

$\forall x \forall y \left((x<0) \wedge (y<0) \rightarrow (xy>0)\right)$

b) Average of two positive integers is positive.

$\forall x \forall y \left((x>0) \wedge (y<0) \rightarrow (xy>0)\right)$

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c) $\neg \forall x \forall y \left((x<0) \wedge (y<0) \rightarrow (x-y<0)\right)$

d) $\forall x \forall y \left(|x+y| \leq |x| + |y|\right)$

Question No 21:

$\forall x \exists a, \exists b, \exists c, \exists d \left((x>0) \rightarrow x = a^2+b^2+c^2+d^2\right)$

Where domain consist of all integers.

Question No 24:

a) $\exists x \forall y (x+y=y)$

there exist a additive identity — a number that when added to every number does not change its value.

(c) $\exists x \exists y ((x \leq 0) \wedge (y \leq 0) \wedge (x-y>0))$

the difference of two non-positive number is not necessarily non positive.

Question No 27:

a) $\forall x \exists m (n^2 < m)$

the import of this statement is that no matter how large n might be, we can always an integer m bigger than n^2. So, this is surely true. For e.g. we could always take $m = n^2+1$

c) $\forall n \exists m (n+m=0)$

Here m depends on n since we can take $m = -n$. So, this statement is true.

(i) $\forall n \forall m \exists p (p = (m+n)/2)$

this statement says that average of two integers is always an integer.

Let we take $m=1$ and $a=2$ and

$p = \frac{m+n}{2} = \frac{1+2}{3} = \frac{3}{2} = 1\frac{1}{2}$ which is not integer So, this statement is false.

Question No 29:

(a) $\forall x \forall x (x=y)$

$P(1,1) \wedge P(1,2) \wedge P(1,3) \wedge P(2,1) \wedge P(2,2) \wedge P(2,3) \wedge P(3,1) \wedge P(3,2) \wedge P(3,3)$.

Question No 30:

a) $\neg \exists y \exists x P(x,y)$

$\forall y \forall x \neg P(x,y)$

Question No 32:

a) $\exists z \forall y \forall x \neg T(x,y,z)$

using Demorgans Law.

$= \neg \exists z \forall y \forall x \; T(x,y,z)$

$= \forall z \neg \forall y \forall x \; T(x,y,z)$

$= \forall z \exists y \neg \forall x T \; (x,y,z)$

$= \forall z \exists y \exists x \neg T(x,y,z)$

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(c) $\exists x \exists y \ (Q(x,y) \leftrightarrow Q(y,x))$

$= \neg \exists x \exists y \ (Q(x,y) \leftrightarrow Q(y,x))$

$= \forall x \neg \exists y \ (Q(x,y) \leftrightarrow Q(x,y))$

$= \forall x \forall y \ \neg (Q(x,y) \leftrightarrow Q(y,x))$

$= \forall x \forall y \ (\neg Q(x,y) \leftrightarrow Q(y,x))$

Question NO 34:-

$\forall x \forall y \ ((x \neq y) \rightarrow \forall z \ ((z = x) \vee (z = y)))$

the domain which consist of one or two member of united state make it true. And domain consist of more than two member make it false.

Question No 36:-

a) No one has lost more than one thousand dollars playing lottery.

Let $L(x,y)$ mean that person x has lost y dollars playing lottery, the original statement is than

$\neg \exists x \exists y \ (y > 1000 \wedge L(x,y))$

its negation.

$\exists x \exists y \ (y > 1000 \wedge L(x,y))$

in english :- someone has lost more than thousand dollars playing lottery.

(c) No student in this class has sent e-mail to exactly two other students in this class.

Let $P(x,y)$ mean that person x has sent e-mail to person y. the original statement is given as.

$\neg \exists x \exists y \exists z \ (y \neq z \wedge x \neq y \wedge x \neq z \wedge \forall w \ (w \neq x) \rightarrow E(x,w) \leftrightarrow (w = y \vee w = z))$

the negation is same but in positive reaction. In English.

Some students in this class has sent e-mail to exactly two other students in this class.

Question No 39:-

a) $\forall x \forall y \ (x^2 = y^2 \rightarrow x = y)$

Since Square of a number and its additive inverse are some, so, we have many counter example such as $x = 2$, $y = -2$

b) $\forall x \exists y \ (y^2 = x)$

this statement is saying that every number has a square root. If x is negative ($\because x = y$) x is not perfect square than equation $y^2 = x$ is false.

c) $\forall x \forall y \ (xy \geq x)$

this statement says that (every number has a square root)

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Product of two integers is greater than the first integers since Product of a negative integer and a positive integer is not positive. So, Let we have

$n = 17$ and $y = -1$ their product is not equal to $+17$ but equal to (-17). so, this is counter example.

Question No 41:

$\forall x \forall y \forall z \quad ((x-y)z = x \cdot (y \cdot z))$

Question No 43:

the distributive Is: given as:

$$x(y+z) = xy + xz$$

By using quantifiers.

$\forall x \forall y \forall z \quad (x(y+z) = xy + xz)$

Question No 45: $\forall x \exists y \ (xy = 1)$

a) nonzero real numbers.

In universe of nonzero real number this is surely true as $y = 1/x$.

b) non zero integers.

this is false because Let $x = 3$. $y = ?$

So, $xy = 3 \times 1 = 3 \neq 1$

Previous Exercise 4.1:

Question No 8:

Prove or disprove that if a/bc, where a, b, and c are positive and integers $a \neq 0$, then a/b or a/c.

Solution:

If a/bc then By defination.

$bc = a(m) \longrightarrow (1)$ m is an integers

therefore,

$cb/c = a(m/c)$ Divided by c both side.

$b = a(m/c)$

as b is multiple of a and integers (m/c) then By defination a/b

From (1)

$bc = a(m)$

$c = a(m/b)$ divided by b both side.

as c is multiple of a and integers m/b then By defination a/c

$b = \frac{a}{c}(m)$ ∴ ∴ ∴

$a/b \longrightarrow$ not possible as $\frac{a}{c}/\frac{m}{c}$ is not an integers

$c = \frac{a}{b}(m)$ ∴

$a/c \longrightarrow$ not possible as $\frac{a}{b}$ or $\frac{m}{b}$ is not an integer.

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Question No 2:-

Prove that if a is an integer other than 0, then

(a) 1 divided a. $a \neq 0$

Solution:

$= 1/a$

If 1 divides a then it gives an integer.

Let m be an integer. So, By defination $\frac{a}{1} = m$

$a = 1(m)$

a is multiples of 1 and integer m.

(b) a divided 0.

$a/0$

a divide 0 then it is an integers.

If m is an integers then by defination of divisibility. $a/0 = m$

$0/a = m \Rightarrow 0 = a(m)$

0 is multiply a and integers m.

Question No 4:-

Prove that part (iii) of theorem 1 is true

Solution:-

If a/b and b/c then by defination of division it is an integer that are m and n.

$b = a(m)$, and $c = b(n)$

therefore,

$c = b(n)$

$= a(m)(n)$

$c = a(mn)$

as c is multiple of a and integers $(m.n)$ So a/c by defination.

Question No 6:-

Show that if a, b, c and d are integers, where $a \neq 0$, such that a/c and b/d then ab/cd.

Solution:-

If a/c and b/d then by defination it is integers that are m, n

$c = (m)a$, $d = b(n)$

therefore,

$cd = a(m).b(n)$

$cd = ab\ (m.n)$

as cd is multiple of ab and integers mn then. So, by defination

ab/cd.

Question No 10:-

What are the quotient and remainder when.

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(a) 44 is divided by 8?

this question is already solve in previous page !

Question No 12:

What time does a 24-hour clock read,

(a) 100 hours after it reads 2:00 ?

Solution:

$28 + 100 \mod 24 = 128 \mod 24$

$24 \overline{)128}$ (4), 96, 32, 24, 8 ← Remainder

$= 8$

8 clock read but reading to remainder in 6 clock.

(b) 45 hours before it reads 12:00?

Solution:

$12 - 45 \mod 24 = -33$

$24 \overline{)-33}$ (2), 48, 15 ← Remainder

$\mod 24 = -33 + 48$

$\mod 24 = 15$

Clock reads 15 o'clock.

(c) 168 hours after it read 19:00?

Solution:

$19 + 168 \mod 24 = 187 \mod 24$

$24 \overline{)187}$ (7), 168, 19 ← Remainder

$= 19$

Clock reads 19 o'clock.

Question No 26:

List five integers that are congruent to 4 modulo 12.

Solution:

We just need to short from 4 and repeatedly added 12 unitile 5 digits.

So, the five digits that are congruent to 4 modulo 12 are 4, 16, 28, 40, 52.

Question No 27:

List all integers blw -100 and 100 that are congruent to -1 modulo 25.

Solution:

We just need start at -1 and repeatedly subtract or add 25 until we exceed the desired range.

-1, -26, -51, -76, 24, 49, 74, 99

Question No 28:

Decide whether each of these integers in congruent to 3 modulo 7.

Solution:

For these problems, we need to divide 7 and see whether the remainder equal to 3.

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Remember that, quotient can be negative, but remainder must satisfy

$0 \leq r < 7$.

(a) 37

$37 = 7 \cdot 5 + 2$

as remainder is not equal to 3.

So, $37 \not\equiv 3 \pmod 7$

(b) 66

$66 = 7 \cdot 9 + 3$

as remainder is equal to 3

So, $66 \equiv 3 \pmod 7$

(c) -17

$-17 = 7(-3) + 4$

as remainder is not equal to 3

So, $-17 \not\equiv 3 \pmod 7$

(d) -67

$-67 = 7 \cdot (-10) + 3$

as remainder is equal to 3.

$-67 \equiv 3 \pmod 7$

Question No 29:

Decide whether each of these integer is congruent to 5 modulo 17.

(a) 80

$80 = 17 \cdot 4 + 12$

as remainder is not equal to 5

$80 \not\equiv 5 \pmod{17}$

(b) 103

$103 = 17 \cdot 6 + 1$

as required is not equal 5.

$103 \not\equiv 5 \pmod{17}$

(c) -29

$-29 = 17 \cdot (-2) + 5$

As required to equal 5. Remainder 5.

$-29 \equiv 5 \pmod{17}$

(d) -122

$-122 = 17(-8) + 14$

$r \neq 5$ So, required is not equal to 5.

$-122 \not\equiv 5 \pmod{17}$

Question No 30:-

Find each of these value.

(a) $(177 \bmod 31 + 270 \cdot \bmod 31) \bmod 31$.

$(177 \bmod 31 + 270 \bmod 31) \bmod 31$.

Working modulo 31. We have

$177 + 270 = 447$

$447 \bmod 31 \equiv 13$

So answer is 13.

(b) $(177 \bmod 31 \cdot 260 \bmod 31) \bmod 31$

Working modulo 31. We have.

$177 \cdot 260 \bmod 31$.

$46020 \bmod 31 \equiv 9$

So answer is 9.

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Question No 31:

Find each of these Value.

(a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 31$.

$(-133 \bmod 23 + 261 \bmod 23) \bmod 31$

Working modulo 23. We have

$-133 + 261 = 128$

$128 \bmod 23 = 13$

So, Answer is 13.

(b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

$= (457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

Working modulo 23. We have

$457 \cdot 182 = 83174$

$83174 \bmod 23 = 6$

$83174 = 23 \cdot 3616 + 6$

So, Answer is 6.

Question No 33:

Find the each of these Value.

(a) $(99^2 \bmod 32)^3 \bmod 15$

$= (99^2 \bmod 32)^3 \bmod 15$

$= (3^2 \bmod 32)^3 \bmod 15$ $\because 99 \bmod 32 = 3$

$= (9 \bmod 32)^3 \bmod 15$

$= 9^3 \bmod 15$

$729 \bmod 15 = 9$

$= 9$ Ans

(b) $(3^{41} \bmod 17)^2 \bmod 11$

$= (3^{41} \bmod 17)^2 \bmod 11$

$= (81 \bmod 11)^2 \bmod 11$

$= (13)^2 \bmod 11$

$= 169 \bmod 11$ $(\because 169 = 11 \cdot 15 + 4)$

$= 4$. Ans:

(c) $(19^3 \bmod 23)^2 \bmod 31$.

$= (19^3 \bmod 23)^2 \bmod 31$.

$= (-4)^3 \bmod 23)^2 \bmod 31$.

$= (-64 \bmod 23)^2 \bmod 31$.

$= 5^2 \bmod 31$.

$= 25 \bmod 31$.

$= 25$.:

(d) $(89^3 \bmod 79)^4 \bmod 26$.

$= (89^3 \bmod 79)^4 \bmod 26$

$= (10^3 \bmod 79)^4 \bmod 26$

$= (1000 \bmod 79)^4 \bmod 26$

$= 52^4 \bmod 26$

$0^4 \bmod 26 = 0$

$= \text{Ans } 0$

Question No 34:

Show that if $a \equiv b \pmod m$ and $c \equiv d \pmod m$. Where a, b, c, d and m are integers with $m \geq 2$, than $a - c \equiv b - d \pmod m$.

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Solution:-

Condition are

if $a = b \pmod m$ and

$c = d \pmod m$

Show:-

$a - c = b - d \pmod m$

Proof:

By using conditions.

$a \equiv b \bmod m$ $\quad$ $c \equiv d \bmod m$.

$a = b + km$ — (i) $\quad$ $c = d + K'm$ — (ii)

Subtracting eq (i) and (ii)

$a - c = (b + km) - (d + k'm)$

$a - c = b - km - d - k'm$

$a - c = b - d + m(k - k')$ — (iii)

By comparing eq (iii) with equation.

$a = b + mk$

So by defination

$a - c = b - d \pmod m$

Question NO 35:-

~~Show that if $a \equiv b \pmod m$ and $c \equiv d \pmod m$, then have~~

~~$a \pm b \equiv c \pm d$~~

Show that if $n \mid m$ where n and m are integer greater than 1. and if $a \equiv b \pmod m$, Where a and b are integers. than $a \equiv b \pmod n$.

Solution:-

For $n \mid m$ we know that it means that $m = nk$ — (i)

Second condition is

$a \equiv b \pmod m$ it means that

$a = b + k'm$ — (ii)

Put value of m in ii,

$a = b + k'(nk)$

$a = b + n(k'k)$ → (iii)

So by defination.

$a \equiv b \pmod n$

Question NO 45:-

Write out the addition and multiplication table for Z_5. (where by addition and multiplication we mean +5 and ·5)

Solution:-

We will use + and · For these operation to save space and improve the appearance of table.

0+0=0	0+1=1	0+2=2	0+3=3	0+4=4
1+0=1	1+1=2	1+2=3	1+3=4	1+4=0
2+0=1	2+1=3	2+2=4	2+3=0	2+4=1
3+0=3	3+1=4	3+2=0	3+3=1	3+4=2
4+0=4	4+1=0	4+2=1	4+3=2	4+4=3

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$0\cdot0=0$	$0\cdot1=0$	$0\cdot2=0$	$0\cdot3=0$	$0\cdot4=0$
$1\cdot0=0$	$1\cdot1=1$	$1\cdot2=2$	$1\cdot3=3$	$1\cdot4=4$
$2\cdot0=0$	$2\cdot1=2$	$2\cdot2=4$	$2\cdot3=1$	$2\cdot4=3$
$3\cdot0=0$	$3\cdot1=3$	$3\cdot2=1$	$3\cdot3=4$	$3\cdot4=2$
$4\cdot0=0$	$4\cdot1=4$	$4\cdot2=3$	$4\cdot3=2$	$4\cdot4=1$

Question No 47:-

Determine whether each of the functions $f(a)=a \bmod d$ and $g(a)=a \bmod d$, where d is a fixed positive integers, from the set of integer to the set of integers is one-to-one. and determine whether each of these function is onto.

Solution:

If $d=1$, then $f(a)=a$ and $g(a)=a$. therefore f is clearly one-to-one and onto. and g is neither. If $d>1$, then f is still onto, because $f(db)=b$ for any desired $b\in Z$, but is clearly not one-to-one. because $f(0)=f(1)=0$.

Further more, g is clearly not onto, because its range is just $\{0,1,2,\ldots\ldots,d-1\}$, and it is not one-to-one because $g(0)=g(d)=0$.

Exercise No 4.3

Example No 2:

the prime factorization of 100, 641, 999 and 1024 are given,

Solution

$100=2\cdot2\cdot5\cdot5=2^2\cdot5^2$

$641=641$

$999=3\cdot3\cdot3\cdot37=3^3\cdot37$

$1024=2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot2=2^{10}$

Example No 3:-

Show that 101 is prime.

Solution:

the only Prime not exceeding $\sqrt{101}$ are 2, 3, 5 and 7. Because 101 is not divisible by 2, 3, 5 and 7. (the quotient of 101 and each of these integer is not an integer), it fellow that 101 is prime.

Example No 4:-

Find the factorization of 7007.

Solution:

To find the prime factorization of 7007.

$7007=7\cdot7\cdot11\cdot13$

$=7^2\cdot11\cdot13$

7	7007
7	1001
11	143
13	13
	1

the prime factorization of 7007 is $7\cdot7\cdot11\cdot13=7^2\cdot11\cdot13$.

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theorem 2:

If n is composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

Proof:

If n is composite integer.

By defination of composite integer. we know that factor a

$1 < a < n$.

By diffination of factor positive.

$n = ab$,

b is positive integer greater 1.

$a \leq \sqrt{n}$ or $b \leq \sqrt{n}$

$b > \sqrt{n}$ or $a > \sqrt{n}$

then $ab > \sqrt{n} \cdot \sqrt{n} = n$.

When is a contradiction, consequently

$a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

In, either case, n has a prime divisor Less than or equal to $\sqrt{n}$.

Example 5:

the numbers

$2^2 - 1 = 3$,

$2^3 - 1 = 7$

$2^5 - 1 = 31$

$2^7 - 1 = 127$

$2^{11} - 1 = 2047$

2047 is not a Mersenne prime because $2047 = 23 \cdot 89$.

Example 8:

$n^2 + 1$,

where n is a positive integer.

For example.

$5 = 2^2 + 1$, $17 = 4^2 + 1$, $37 = 6^2 + 1$

that $n^2 + 1$ is prime or the product of at most two primes.

Greatest Common Divisors and Least Common Multiples:

the largest integer that divides both of two integers is called the greatest common divisor of these integer.

Definination 2:

Let a and b be integer, not both zero. the larges integers (is called greatest) d such that d/a and d/b is called. the greatest common divisor of a and b. the greatest common divisor of a and b is denoted by gcd (a,b)

Example 10:

What is greatest common divisor of 24 and 36?

Solution

the positive common divisor of 24 and 36 are, 1, 2, 3, 4, 6 and 12. Hence,

gcd (24, 36) = 12

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Example No 11:

What is the greatest common divisor of 17 and 22?

Solution:

the integer 17 and 22 have no positive common other than 1. so that

$\gcd(17,22) = 1$.

Definition 3:

the integer a and b are relatively prime if their greatest common divisor 1.

Example 14:

Because the prime factorization of 120 and 500 are

$120 = 2^3 3 \cdot 5$

$500 = 2^2 \cdot 5^3$.

$\gcd(120,500) = 2^{\min(3,2)}, 3^{\min(1,0)}, 5^{\min(1,3)}$

$= 2^2, 3^0, 5^1 = 20$

Example No 15:

What is the least common multiple of $2^3 3^5 7^2$ and $2^4 3^3$?

Solution:

We have $\text{lcm}\ 2^3 3^5 7^2, 2^4 3^3$

$= 2^{\max(3,4)}, 3^{\max(5,3)}, 7^{\max(2,0)}$

$= 2^4, 3^5, 7^2$

Lemma 1:

Let $a = bq + r$, where a, b, q and r are integer. this $\gcd(a,b) = \gcd(b,r)$

Proof:

Likewise, suppose that d divide both b and r. the d also divide $bq + r = a$. When we successively apply. Hence, any common divisor of b and r is also common divisor of a and b.

Suppose that a and b are positive integer with $a \geq b$. Let $r_0 = a$ and $r_1 = b$. when we successively apply the division algorithm, we obtain.

$r_0 = r_1 q_1 + r_2 \quad 0 \leq r_2 < r_1$

$r_1 = r_2 q_2 + r_3 \quad 0 \leq r_3 < r_2$.

.

.

$r_{n-2} = r_{n-1} q_{n-1} + r_n \quad 0 \leq r_n < r_{n-1}$

$r_{n-1} = r_n q_n$.

Remainder of zero occurs in this sequence of Successiv division. $a = r_0 > r_1 > r_2 > \cdots \geq 0$

$\gcd(a,b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \cdots =$

$\gcd(r_{n-2}, r_{n-1}) = \gcd(r_{n-1}, r_n)$

$= \gcd(r_n, 0) = r_n$

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Example No 16:-

Find the greatest common divisor of 414 and 662. using the Euclidean algorithm

Solution:

Successive uses of the divison algorithm give:

$662 = 414 \cdot 1 + 248$

$414 = 248 \cdot 1 + 166$

$248 = 166 \cdot 1 + 82$

$166 = 82 \cdot 2 + 2$

$82 = 2 \cdot 41$.

Hen gcd $(414, 662) = 2$.

Because 2 is non zero remainder.

Q.No 5:-

How many prime factorization of 10!

Solution:-

$10! = 2 \cdot 3 \cdot 4 \cdot 5, 6 \cdot 7 \cdot 8, 9 \cdot 10.$

$= 2 \cdot 3 \cdot (2 \cdot 2) \cdot 5 \cdot (2 \cdot 3) \cdot 7 (2 \cdot 2 \cdot 2) \cdot 3 \cdot 3 \cdot 2 \cdot 5$

$= 2 \cdot 3 \cdot 2^4 \cdot 5 \cdot 2 \cdot 3 \cdot 7 \cdot 2^3 \cdot 3^2 \cdot 2 \cdot 5$

$= 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$ Ans.

Q.No 6:-

How many zeros are there at the end 100!?

Solution:-

To Find number of zeros at the end of 100! We use the formula:

We divided 100 by 5 and 5^2 and so, on until the numerator is greater than denominator.

So,

$100! \rightarrow \frac{100}{5} + \frac{100}{5^2}$

for $\frac{100}{5^3} = \frac{100}{125}$ which is the incorrect So,

Number of zero. $= \frac{100}{5} + \frac{100}{5^2} \Rightarrow 20 + \frac{100}{25}$

$= 20 + 4$

$= 24$ zeros

Q.No 7:

Express in pseudocode the trial division algorith for determine whether an integer is prime?

Solution:

We given a proof by contradiction Let suppose.

$\log_2 3$ is a rational number

By definition it can be written by.

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$\log_2 3 = p/q$

$2^{p/q} = 3$

$3^q = 2^{p/q \times q}$

$3^q = 2^p$

This is violation of fundmental theorem of Arithmetic.

Since it gives two different prime factorizations.

Same number Hence our assumption (that $\log_2 3$ is rational) is wrong and we conclude that $\log_2 3$ is irrational.

Q.No 13:-

Prove or disprove that there are three consecutive odd positive integers that are primes that is, odd prime of the form p, p+2, and p+4,

Solution:

Here in this question we have three odd prime of the form p, p+2 and p+4, For first prime 2 which is even so, we don't put in it but we put prime 3 and get.

$p = 3$

$p+2 = 3+2 = 5$

$p+4 = 3+4 = 7$.

So, 3, 5 and 7 are consecutive Primes.

Q.No 15:-

Which positive integers Less than 30 are relatively prime to 30?

Solution:-

in this question we have find integers Less than 30. that are relatively prime to each other.

We Knows that the integers that have (gcd) is 1 are relatively Primes So, only Primes number Less than 30. fulfill this condition as. the prime number Less then 30, are 1, 2, 3, 5, 7, 11, 13, 17, 19, 21, 29.

gcd (2,30) = 1 gcd (13,30) = 1

gcd (3,30) = 1 gcd (17,30) = 1

gcd (5,30) = 1 gcd (19,30) = 1

gcd (7,30) = 1 gcd (21,30) = 1

gcd (11,30) = 1 gcd (29,30) = 1.

So,

2, 3, 5, 7, 11, 13, 17, 19, 23, 29 are relativly to 30.

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Q. No 17:-

Determine whether integers are relatively prime

(a) ~~$2^4 3^1 5^5$~~, 11, 15, 19

Solution:

since $\gcd(11, 15, 19) = 1$

$\gcd(11, 19) = 1$.

$\gcd(15, 19) = 1$. So, these are relatively primes.

(b) 14, 15, 22,

$\gcd(14, 15) = 1$, $\gcd(15, 22) = 3$

which is greater than 1 So, these three number are not pairwise relatively prime.

(c) 12, 17, 31, 37,

$\gcd(12, 17) = 1$, $\gcd(12, 31) = 1$, $\gcd(12, 37) = 1$.

$\gcd(17, 31) = 1$, $\gcd(17, 37) = 1$, $\gcd(31, 31) = 1$

So, these four number are pairwise relatively prime.

Q. No 18:

We call a positive integer perfect if it equal the sum of its positive divisors other than itself.

(a) Show that 6 and 28 are perfect.

Solution:

As given is condition

the divisors of 6 are 1, 2, 3, 6

and divisors of 28 are 1, 2, 4, 7, 14, 28

So, $6 = 1 + 2 + 3$.

$28 = 1 + 2 + 4 + 7 + 14$.

So 6 and 28 both are perfect.

Q. No 19:-

Show that if $2^n - 1$ is prime. then n is prime. [Hint: use the identity $2^{ab} - 1 = (2^a - 1) \cdot 2^{a(b-1)} + 2^{a(b-2)} + \cdots + 2^a + 1$]

Solution:

We have condition that if $2^n - 1$ is prime, this n is prime. We prove this by contrapositive

Let

n is not prime According to definition.

$n = ab$ where a and b both are greater than 1.

According to condition

$2^n - 1$

put $n = a$ put $n = b$

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$2^a - 1 > 1$ and $2^b - 1 > 1$

As both factors are greater than 1 So,

$2^n - 1 = 2^{ab} - 1$ is product of two integers each greater than 1. So, it is not prime.

Q. No 20:-

Determine whether each of these integers is prime, Verifying some of Mersenne's claims.

(a) $2^7 - 1$

$2^7 - 1 = 127$

It is prime and it is verifying the Mearsennes. claims.

(b) $2^9 - 1$.

$2^9 - 1 = 511$.

It is not prime to verifying.

(c) $2^{11} - 1$.

$2^{11} - 1 = 2047$.

which is prime and hence it is Verifying Mersenn's claims.

Q. No 21:-

Find these value of the Euler ϕ-function

Solutions.

We compute ϕ here by enumerating the set of positive integers less than n that are relatively prime to n.

(a) $\phi(1)$

$\phi(1) = |\{1\}| = 2$.

(b) $\phi(10)$

$\phi(10) = |\{1, 3, 5, 7, 9\}| = 4$

Q. No 25:-

Find greatest common divisors

(a) $3^7 \cdot 5^3 \cdot 7^3$. $2^{11}, 3^5 \cdot 5^9$.

Let

$a = 3^7 \cdot 5^3 \cdot 7^B$

$b = 2^{11} \cdot 3^5 \cdot 5^9$

So, to find gcd(a,b) we use second formula as:

$\gcd(a,b) = 3^{\min(7,5)} 5^{\min(3,9)}$

So, $\gcd(a,b) = 3^5 \cdot 5^3$ Ans.

(c) $a = 3^{13}, 5^{17}$ $b = 2^{12} \cdot 7^{12}$.

Here there is no common factors or common prime divisor So,

gcd (a,b) = 1.

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(f) 1111, 0
the gcd of any positive integers and 0 is that integers, so answer is 1111.

(b) $11.13.17$, $2^9, 3^7, 5^5, 7^3$. (27)
these number have no common prime factors so, the Lcm is their product.
Lcm is $2^9.3^7.5^5.7^3.11.13.17$.

(a) $3^{13}, 5^{17}, 2^{12}.7^{21}$ (27)
these number have no common prime factor. So, Lcm is their product
Lcm is $3^{13}.5^{17}, 2^{12}, 7^{21}$

Q.NO 27:-
Find least common multiple
(a) $3^7, 5^3, 7^3$, $2^{11}, 3^5, 5^9$.
To find lcm of two number whose Prime factorization are given, we just need to take larger expenent for each point.
In this part, Let.
$a = 3^7, 5^3, 7^3$
$b = 2^{11}, 3^5, 5^9$.
In 'b' there is no factor of 2 so, we can put 2^0 in 'a' as.

$a = 2^0, 3^7, 5^3, 7^3$
In 'b' there is no factor of 7 so, we can 7^0 in 'b' as.
$b = 2^{11}, 3^5, 5^9, 7^0$
By using formula
$\text{Lcm}(a,b) = 2^{\max(0,11)}, 3^{\max(7,5)}, 5^{\max(3,9)}, 7^{\max(3,0)}$
$= 2^{11}, 3^7, 5^9, 7^3$ Ans.

Q.No 29:-
Find gcd (92928, 123552). and Lcm (92928, 123552), and Verify that gcd(92928, 123552). Lcm (92928, 123552) = 92928 · 123552.

Solution:
As given is hint we first find their Prime factorization.
$92928 = 2^8.3.11^2$
$123552 = 2^5.3^3.11.13$
For gcd (92928, 123552)
$\text{gcd}(92928, 123552) = 2^{\min(8,5)}, 3^{\min(3,1)}, 11^{\min(2,1)}$
$= 2^5.3.11 = 1056$

For Lcm (92928, 123552)
$\text{Lcm}(92928, 123552) = 2^{\max(8,5)}, 3^{\max(3,1)}, 11^{\max(1,2)}$
$= 2^8, 3^3, 11^2.13$
$= 10872576$

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Product of gcd and Lcm.

gcd (92928, 123552). Lcm (92928, 123552)

$= 1056 \times 10872576$

$= 11,481,440,256$ → (i)

And their product their selves.

$92928 \times 123552 = 11,481,440,256$, — (ii)

So, From eq (i) and (ii) it is verified.

Q. No 30:

ab is product of two integers is $2^7 3^8 5^2 7^{11}$ and their great common divisor is $2^3 3^4 5$. What their least common multiple?

product of two integer $2^7 3^8 5^2 7^{11}$

their gcd $= 2^3 3^4 5$

Let we have to integers a and b. We know that:

gcd (a,b). Lcm(a,b)

$2^7 \cdot 3^8 \cdot 5^2 \cdot 7^{11} = 2^3 3^4 5\, 7^0$ Lcm (a,b)

Ucm (a,b) $= 2^{7-3} \cdot 3^{8-4}, 5^{2-1} 7^{11-0}$

$= 2^4, 3^4, 5, 7^{11}$ Ans

Q. No 31:-

Show that if a and b are positive integers, then ab = gcd (a,b) · Lcm (a.b).

Solution:

the important observation to make here is that smaller of any two number plus greater of any two number is always equal to sum of two number

In gcd (a,b) :

exponent of prime p is smallest of exponent a and b. as

gcd (a,b) = min (a,b).

I Lcm (a,b)

exponent of prime p is greater of exponent a and b. as.

Lcm (a,b) = max (a,b)

So, at their product.

gcd (a,b) . Lcm (a,b) the exponent. (Smaller and greater) are added. And as given in observation, it equal to sum of these value; so it is proved that gcd (a,b). Lcm (a,b). = ab.

Q. No 33:-

use the Euclidean algorithm to find

(a) gcd (12,18)

By using Eudidean algorithm

$18 = 12 \cdot 1 + 6$

$12 = 6 \cdot 2 + 0$

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We know that

gcd (12,18) = gcd(6,12) = gcd (6,0)

As gcd (6,0) = 6 So,

gcd (12,18) = 6.

(c) gcd (1001, 1331)

$1331 = 1001 \cdot 1 + 330$

$1001 = 330 \cdot 3 + 11$

$330 = 11 \cdot 30 + 0$

$11 = 0 \cdot 11 + 0$

So,

gcd (1001, 1331) = gcd (1001, 330) = gcd (330, 11) = gcd (11, 0) = 11

(e) gcd (1000, 5040)

$5040 = 1000 \cdot 5 + 40$

$1000 = 40 \cdot 25 + 0$

$40 = 0 \cdot 40 + 0$

So, gcd (1000, 5040) = gcd (1000, 40)

gcd (40, 0) = 40

Q No 35:

How many division are required to find gcd (34, 55) by using Euclden Algorithm.

(i) $55 = 34(1) + 21$

(ii) $34 = 21(1) + 13$

(iii) $21 = 13(1) + 8$

(iv) $13 = 8(1) + 5$

(v) $8 = 5(1) + 3$

(vi) $5 = 3(1) + 2$

(vii) $3 = 2(1) + 1$

(viii) $2 = 1(1) + 1$

(xi) $1 = 1(0) + 0$

this find to 34 & 55 nine division steps are required gcd (34, 55) = 1.

Q.No 39:-

Find the gcd be linear combination of Euclidean algorithm.

(a) (10, 11)

Two show gcd by linear combination we first use Euclidean Algorithm.

$11 = 10 \cdot 1 + 1$

$10 = 1 \cdot 10 + 0$.

As the gcd (10, 11) = 1 as,

$1 = 11 - 10 \cdot 1$

(b) (21, 44)

By using Euclidean algorithm.

$44 = 21(2) + 2$

$21 = 2(10) + 1$

$2 = 1 \cdot 2 + 0$

the least remainder is 1.

gcd (21, 44) = 1

Now we take last steps

$1 = 21 - 10 \cdot 2$

$1 = 21 - 10(44 - 21 \cdot 2)$

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thus we have expressed 1 as a linear combination of 21 and 44, namely gcd (21,44) = 21·21 + (-10)44

(c) (36, 48)

By using Euclidean Algorithm.

$48 = 36 \cdot 1 + 12$

$36 = 12 \cdot (3) + 0$

the gcd (36,48) is 12, so we can express it as linear combination.

$12 = +48 - 1 \cdot 36.$

Q.No 40 (a)

Find the s and t are used to Euclidean algorithm. (9.11)

We first use Euclidean Algorithm.

$11 = 9 \cdot 1 + 2$

$9 = 2(4) + 1.$

$2 = 1 \cdot 2 + 0.$

So last remainder is 1. and gcd (11,9) = 1.

$1 = 9 - 2 \cdot 4$

$1 = 9 - 4(11 - 9 \cdot 1)$

$= -44 + 9 \cdot 5$

$= 5 \cdot 9 - 44.$

So, we have solved the question s and t are 5 and -4 respectively.

Q.No 42:

Use the extend Euclidean algorithm to express gcd (252, 356) as a linear combination of 252 and 356.

We use Euclidean algorithm.

$356 = 252(1) + 104$

$252 = 104(2) + 44.$

$104 = 2 \cdot (44) + 16$

$44 = 2 \cdot (16) + 12$

$16 = 12 \cdot (1) + 4$

$12 = 3 \cdot 4 + 0.$

q_i represent the quotients.

$q_1 = 1$, $q_2 = 2$ $q_3 = 2$ $q_4 = 2$, $q_5 = 2$

Initially, extended Eclean algorithm.

Sets $s_0 = 1$, $t_0 = 0$, $s_1 = 0$ t_1

By using extend Euclidean algorithm.

$s_2 = s_0 - q_1 s_1 = 1 - 1 \cdot 0 = 1.$

$t_2 = t_0 - q_1 t_1 = 0 - 1 \cdot 1 = -1.$

$s_3 = s_1 - q_2 s_2 = 0 - 2 \cdot 1 = -2$

$t_3 = t_1 - q_2 t_2 = -1 - 2(-1) = 3.$

$s_4 = s_2 - q_3 s_3 = -1 - 2(-2) = 5$

$t_4 = t_2 - q_3 t_3 = -1 - 2 \cdot 3 = -7.$

$s_5 = s_3 - q_4 s_4 = -2 - 2 \cdot 5 = -12.$

$t_5 = t_3 - q_4 t_4 = 3 - 2(-7) = 14.$

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Question No 45:-

How many division are required to find gcd (34, 55) using Euclidean algorithm?

Solution:

$55 = 34(1) + 21$

$34 = 21(1) + 13$

$21 = 13(1) + 8$

$13 = 8 \cdot 1 + 5$

$8 = 5 \cdot (1) + 3$

$5 = 3(1) + 2$

$3 = 2(1) + 1$

$2 = 1 \cdot 2 + 0$

$1 = 0 \cdot 1 + 0$

So, the successively division are 55, 34, 21, 13, 8, 5, 3, 2, 1 which are nine in total.

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Exercise 4.4:

Question No 1:

Show that 15 is an inverse of 7 modulo 26:

We know that b is an inverse of a modulo m if

$$ab = 1 \pmod{m}$$

Here we have

$$a = 7, \quad b = 15, \quad m = 26.$$

$$a \cdot b = 7 \times 15 = 105$$

$$105 = 26 \times 4 + 1 = 1 \pmod{26}$$

So,

15 is inverse of 7 modulo 26.

Question No 2:

Show that 937 is an inverse of 13 modulo 2436.

We know that b is inverse of a modulo m if:

$$ab = 1 \pmod{m}$$

Here we have.

$$a = 13 \quad b = 937 \quad m = 2436$$

$$a \cdot b = 13 \times 937$$

$$12181 = 2436 \times 5 + 1 = (1 \pmod{2436})$$

So, 937 is an inverse of 13 mod 2436.

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Question No 04:-

By inspection. find an inverse of 2 modulo 17.

the inverse of a modulo m is an integer b for which $ab \equiv 1 \pmod{m}$

$a = 2$

$m = 17$.

First we a form Eusliden algorith.

$17 = 8 \cdot 2 + 1$.

$2 = 1 \cdot 2 + 0$

So, gcd $(a, m) = 1$.

we write the gcd $(a, m) > 1$ as multiple of m

$1 = 17 - 8 \cdot 2$

Here inverse is coeffient of a that is -8. So,

$-8 \bmod 17 = 9 \bmod 17$.

9 is inverse of modulo m.

Question No 05:-

Find inverse of a mod m when.

(b) $a = 19$, $m = 51$.

We first use EUslidean theorem

$51 = 19 \cdot 2 + 13$

$19 = 13 \cdot 1 = 6$.

$13 = 6 \cdot 2 + 1$.

$6 = 6 \cdot 1 = 0$

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So the the last non zero remainder is gcd of a and m as gcd $(a, m) = 1$.

we can write it as multiple

$1 = 13 - 6 \cdot 2$

$1 = 1 \cdot 13 - 2(19 - 13 \cdot 1)$

$1 = 3 \cdot 13 - 2 \cdot 19$.

$= 3(51 - 19 \cdot 2) - 19 \cdot 2$.

$1 = 3 \cdot 51 - 8 \cdot 19$.

As, we Know the inverse of a mod m is the - coefficient of a which is equal to -8.

So, its inverse is $\boxed{-8}$

Q. No 6: (C)

$a = 144$, $m = 233$ find inverse of b of a mod m.

First we Euclidan Algorithm to find gcd (a, m).

$233 = 144 + 89$

$144 = 89 \cdot 1 + 55$

$89 = 55 \cdot 1 + 34$

$55 = 34 \cdot 1 + 21$

$34 = 21 \cdot 1 + 13$

$21 = 13 \cdot 1 + 8$

$13 = 8 \cdot 1 + 5$

$8 = 5 \cdot 1 + 3$

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$5 = 3 \cdot 1 + 2$

$3 = 2 \cdot 1 + 1$

$2 = 1 \cdot 2 + 0$.

So, gcd is 1. which can be written as.

$1 = 3 \cdot 1 - 2 \cdot 1$.

$= 3 \cdot 1 - 1(5 - 3 \cdot 1)$

$= 2 \cdot 3 - 5 \cdot 1$.

$= 2(8 - 5 \cdot 1) - 5 \cdot 1$

$= 2 \cdot 8 - 3 \cdot 5$

$= 2 \cdot 8 - 3(13 - 8 \cdot 1)$

$= 5 \cdot 8 - 3 \cdot 13$

$= 5(21 - 13 \cdot 1) - 13 \cdot 3$

$= 5 \cdot 21 - 8 \cdot 13$

$= 5 \cdot 21 - 8(34 - 1 \cdot 21)$

$= 34 \cdot 55 - 21 \cdot 89$

$= 34(144 - 89 \cdot 1) - 21 \cdot 89$

$= 34 \cdot 144 - 55 \cdot 89$.

$= 34 \cdot 144 - 55(233 - 144 \cdot 1)$

$= 89 \cdot 144 - 55 \cdot 233$.

the inverse in then coefficient of b which is thus 89.

Q.No.8:-

Show that an inverse of a modulo m, where a is an integer and $m > 2$ is a positive integer, does not exist if gcd $(a, m) > 1$.

Given:-

a is an integer and $m > 2$ is a positive integer.

gcd $(a, m) > 1$.

An inverse of a mod m does not exist.

Solution:-

We prove this by proof of contradiction

Let we have inverse of a mod m is b.

By defination.

$ab = 1 \pmod m$

this means that m divides $ab - 1$ as

By definition :- if m divides $ab - 1$ the..

$ab - 1 = mc$

$ab - mc = 1$

Now we select the gcd (a, m). By definition we can say that if d is divides then it divides both a and m. as

$= d/a$

$= d/m$.

It can also divide ab and m

$= d/ab$

$= d/m$

As $ab - mc = 1$.

we can say that d divide. 1. as..

$= d/1$

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Now, a positive integer than can divide 1 can be only it self.
So, there is a contradaction that gcd (a,m) >1. So, our assumption that inverse of a mod m exist. is false.
Hence,
there does not exist an inverse of a modm.

Q. No 9:-

Solve congurance $4x \equiv 5 \pmod 9$ using inverse of 4 modulo 9 found in part (a) of Exercise 5:

We have already found the inverse of 4 modulo 9. in (a) part of exercise. that which is equal to 7.
To solve this equality we just multiply 7. on both side of.

$$4x \equiv 5 \pmod 9$$
$$7 \cdot 4x \equiv 7 \cdot 5 \pmod 9$$
$$28x \equiv 35 \pmod 9$$

So, we can write as $\because 28 \bmod 9 = 9 \cdot 3 + 1$.

$$1 \cdot x \equiv 8 \pmod 9 \quad \because 35 \bmod 9 = 9 \cdot 3 + 8$$
$$x \equiv 8 \pmod 9$$

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Q. No 10:-

Solve congurence $2x \equiv 7 \pmod{17}$ using inverse of 2 mod 17 found a part (a) of exercise 5.

We have already found the inverse of 2 mod 7 which is equal to -8. we just justify to multiply it on b/s of the congurence.

$$2x \equiv 7 \pmod{17}$$
$$-8 \cdot 2x \equiv -8 \cdot 7 \pmod{17}$$
$$-16x \equiv -56 \pmod{17}$$

So we can write as.

$$\because -16x \equiv -56 \pmod{17}$$
$$\because -16 \bmod 17 = 17 \cdot 1 + 1$$
$$1 \cdot x \equiv 12 \bmod 17 \quad \because -56 \bmod 17 = 17 \cdot 4 - 12$$
$$x \equiv 12 \bmod 17$$

Linear Congruence:

A congruence of the form

$$ax \equiv b \pmod m$$

where m is a positive integer, and and b are integer, and x is variable. is called Linear congruence. Such congruence rise throughout number theory and its application.

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theorem 1:

If a and m are relatively prime integer and $m > 1$, then an inverse of a modulo m exists. Furthermore, this inverse is unique modulo m. that is, a unique integer $\bar{a}$ less than m that is an integer of a modulo m and every other inverse of a modulo m is congruent to $\bar{a}$ modulo m.

Proof :-

By theorem 6 of section 4.3 because gcd $(a,m) = 1$. there are integer s and t such that:

$sa + tm = 1$

$\Rightarrow sa + tm = 1 \pmod m$.

Because $tm = 0 \pmod m$ it follows that:

$sa = 1 \pmod m$

Example No 2:-

Find an inverse of 101 modulo 4620.

$4620 = 101 \cdot 45 + 75$

$101 = 75 \cdot 1 + 26$

$75 = 26 \cdot 2 + 23$

$26 = 23 \cdot 1 + 3$

$23 = 7 \cdot 3 + 2$

$3 = 2 \cdot 1 + 1$.

$2 = 1 \cdot 2 + 0$.

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So gcd is (1). which can be written

$1 = 3 - 1 \cdot 2$

$= -1(23 - 7 \cdot 3) = -1 \cdot 23 + 8 \cdot 3$

$= 8 \cdot 3 - 1 \cdot 23 = 8(26 - 1 \cdot 23) = 8 \cdot 26 - 9 \cdot 23$

$= 8 \cdot 26 - 9(75 - 2 \cdot 26) = -9 \cdot 75 + 26 \cdot 26$

$= -9 \cdot 75 + 26(101 - 1 \cdot 75) = 26 \cdot 101 - 35 \cdot 75$

$= 26 \cdot 101 - 35(4620 - 45 \cdot 101)$

$= -35 \cdot 4620 + 1601 \cdot 101$

Example No 3:-

what are the solution of the linear congurence $3x = 4 \pmod 7$

Solution:

$3x = 4 \pmod 7$

-2 multyply both side

$-2 \cdot 3x = -2 \cdot 4 \pmod 7$

Because $-6 \equiv 1 \bmod 7$ $\quad -8 = 6 \bmod 7$

$x \equiv -8 \equiv 6 \pmod 7$

$3x \equiv 3 \cdot 6 \pmod 7$

$3x \equiv 18 \bmod 7$.

$3x \equiv 4 \pmod 7$

theorem 2:-

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the remainder Chinese theorem:

Let $m_1, m_2 \dots m_n$ be pairwise relatively prime positive integer greater than one $a_1, a_2 \dots a_n$ an arbitrary integer. then the system

$x \equiv a_1 \pmod{m_1}$

$x \equiv a_2 \pmod{m_2}$

.

.

$x = a_n \pmod{m_n}$

has a unique solution $m = m_1 m_2 \dots m_n$ (that is a solution x with $0 \le x \le m$. and all other solution are congurent modulo m to this solution)

Proof:

To construct a simultaneous solution, first let,

$M_k = m/m_k$.

We know that there is an integer y_k, an inverse of M_k modulo m_k, such than.

$M_k y_k = 1 \pmod{m_k}$.

To constract a simultaneous solution, form the sum.

$x = a_1 M_1 y_1 + a_2 M_2 y_2 + \dots + a_n M_n y_n$

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$M_k y_k \equiv 1 \pmod{m_k}$ we see that

$x \equiv a_k M_k y_k \equiv a_k \pmod{m_k}$

for $k = 1, 2 \dots n$. We shows that x is a simultaneous solution to the n congruence.

Example No 5:-

Let $m = 3, 5, 7 = 105$

$M_1 = m/3 = 105/3 = 35$

$M_2 = m/5 = 105/5 = 21$

$M_3 = m/7 = 105 = 15$

2 is an inverse of $M_1 = 35$ modulo 3,

$35 \cdot 2 \equiv 2 \cdot 2 \equiv (1 \bmod 3)$

1 is an inverse of $M_2 = 21 \bmod 5$.

$21 \cdot 1 \bmod 5$

$21 \equiv 1 \pmod 5$

1 is an inverse of $M_3 = 15 \bmod 7$.

$15 = 1 \pmod 7$. the solution to this system those x such that. (previous exercise of inverse $a_1 = 2$, $a_2 = 3$, $a_3 = 2$)

$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3$

$\equiv 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1$

$= 233 \equiv 23 \pmod{105}$

Example No 6:-

use the method of back substitution to find all integer x such that $x \equiv 1 \pmod 5$, $x = 2 \pmod 6$ and $x \equiv 3 \pmod 7$.

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Solution:-

We know that first congurence theorem.

$x = 5t + 1$.

wher x second congurence tell. were t is integer.

$5t+1 = 2 \pmod 6$

$t = 5 \pmod 6$

$t = 6u + 5$

$x = 5t+1$

$x \equiv 5(6u+5)+1$

$x \equiv 30u+26$

$30u+26 = 3 \pmod 7$

$u = 6 \pmod 7$

$u = 7v+6$.

$x = 30u+26$

$x = 30(7v+6)+26$

$= 210v + 206$.

$x = 206 \pmod{210}$.

Theorem 3:-

Fermat's Little theorem:-

If p is prime and a is an integer not divisible by p, then:

$a^{p-1} \equiv 1 \pmod p$

Furthur more, For every integer a we have

$a^p \equiv a \pmod p$

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Fermat's little theorem tell us that if $a \in Z_p$ then $a^{p-1} = 1$ in Z_p.

Example No 9:-

Find $7^{222} \bmod 11$.

We use fermats little theorm

$a^p = a \pmod p$

$7^{10} \equiv 1 \pmod{11}$

$(7^{10})^k = 1 \pmod{11}$ in integer 11. k.

$222 = 22 * 10 + 2$.

$7^{222} = (7^{10})^{22} \cdot 7^2$

$= (1)^{22} \cdot 7^2 \pmod{11}$

$= (1)^{22} \cdot 49 = 5 \pmod{11}$

It follow $7^{222} \bmod 11 = 5$.

Pseudoprime:-

Let b be a positive integer. If n is a composite positive integer, and $b^{n-1} \equiv 1 \pmod n$; then n is called a Pseduoprime to the base b.

Example 11:-

The integer 561.

$561 = 3 \cdot 11 \cdot 17$.

$\gcd(b, 561) = 1$, $\gcd(b, 3) = 1$, $\gcd(b, 11) = 1$

$\gcd(b, 17) = 1$.

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Using fermat's little theorem.

$b^2 = 1 \pmod 3$ $\quad b^{10} = 1 \pmod{11}$ $\quad b^{16} = 1 \pmod{17}$

$b^{560} = (b^2)^{280} = 1 \pmod 3$

$b^{560} = (b^{10})^{56} = 1 \pmod{11}$

$b^{560} = (b^{16})^{35} = 1 \pmod{17}$

Question No 15:-

Show that if m is an integer greater than 1 and $ac \equiv bc \pmod m$, then $a \equiv b \pmod{m/gcd(c,m)}$

Solution:-

By using given condition as

$ac \equiv bc \pmod m$

it means that a divides $ac - bc$. as

$m \mid ac - bc$

We can write it as

$ac - bc = md \longrightarrow (i)$

Now we use that,,

Let $f = gcd(c,m)$. According to defination of gcd it divided both m and c as

$f \mid m$

$f \mid c$

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Dividing 'f' an b/s of eq (i)

$$\frac{ac-bc}{f} = \frac{md}{f}$$

$$(a-b)\frac{c}{f} = \frac{m}{f}d$$

Let $\frac{m}{f} = m'$

$$(a-b)\frac{c}{f} = m'd$$

As c/f is an integers we can write it as

$m' \mid a-b$ which can be Written as $a \equiv b \pmod{m'}$

So, proved.

Question No 17:-

Show that if p is prime the only Solution of $x^2 \equiv 1 \pmod p$ are integers x such that $x \equiv 1 \pmod p$ or $x \equiv -1 \pmod p$

Solution:

We have to find that p is prime and $x = -1 \pmod p$ or $x = 1 \pmod p$

We have the following condition.

$x^2 \equiv 1 \pmod p$.

Which can be written as, i

$x^2 - 1$ is divide by p which can be explain as.

$p \mid x^2 - 1$.

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By factorization of x^2-1

$P|(x+1)(x-1)$

We can say that.

$P|(x+1)$ and $P|(x-1)$

$P|(x+1)$ can be written as $x=-1 \pmod p$

and $P|(x-1)$ can be written as $x=+1 \pmod p$

Which is proved.

Question No 20:-

Use construction in proof of chinese remainder theorem is to find all to solution to system of conqurence.

$x \equiv 2 \pmod 3$ $\quad x \equiv 3 \pmod 5$

$x \equiv 1 \pmod 4$

Solution:-

We have a formula for chines remainder theorem.

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \pmod M \quad \text{--(i)}$$

Now we have.

$a_1 = 2$, $a_2 = 1$ $\quad a_3 = 3$.

$m_1 = 3$ $\quad m_2 = 5$ $\quad m_3 = 4$.

For 'm'

$M = m_1 \cdot m_2 \cdot m_3 = 3 \cdot 5 \cdot 4 = 60$

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For 'y_1'

$M_1 y_1 = 1 \pmod{m_1}$

$20 y_1 = 1 \pmod 3$

$2 y_1 = 1 \mod 3$

$2 \times 5 y_1 = 5 \pmod 3$

$y_1 = 2 \pmod 3$

$\boxed{y_1 = 2}$

$M_1 = \frac{M}{m_1} = \frac{60}{3} = 20$

$M_2 = \frac{M}{m_2} = 60/5 = 12$

$M_3 = \frac{M}{m_3} = 60/4 = 15$

For 'y_2'

$M_2 y_2 = 1 \pmod{m_2}$

$15 y_2 = 1 \pmod 4$

$3 y_2 = 1 \pmod 4$

$3 \cdot 3 y_2 = 3 \pmod 4$

$y_2 = 3 \pmod 4$

$\boxed{y_2 = 3}$

For 'y_3'

$M_3 y_3 = 1 \pmod{m_3}$

$12 y_3 = 1 \pmod 5$

$2 y_3 = 1 \pmod 5$

$2 \cdot 8 y_3 = 8 \pmod 5$

$y_3 = 8 \pmod 5$

$\boxed{y_3 = 8}$

So, By putting value.

$x = [(2)(20)(2) + (1)(15)(3) + (3)(12)(3)] \pmod{60}$

$x = (80 + 45 + 108) \bmod 60$

$x = 233 \pmod{60}$

$x = 53 \pmod{60}$

$\boxed{x = 53}$

Question No 23:-

Solve the system of conqurence in Exercise 20 using method of back substitution:

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$x = 2 \pmod 3 \longrightarrow (i)$

$x = 1 \pmod 5 \longrightarrow (iii)$

$x = 1 \pmod 4 \longrightarrow (iv)$

We first use first congurence

$x = 2 \pmod 3$

Which cant be written

$3t + 2 = x$

$x = 3t + 2$ where t is integer

We just put $x = 3t + 2$ in second

$3t + 2 = 1 \pmod 4$

$3t = -1 \pmod 4$

$9t = -3 \pmod 4$

$t = 1 \pmod 4$

this can be express as

$t = 4u + 1$. $\because u$ is integer

Now we have an equation

$3t + 2 = x$

$x = 3(4u + 1) + 2$

$x = 12u + 5$

Put x in third congurence which is give as.

$x = 3 \pmod 5$

$12u + 5 = 3 \pmod 5$

$12u = -2 \pmod 5$

$2u = -2 \pmod 5$

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$16u = -16 \pmod 5$

$u = -16 \pmod 5$

$u = 4 \pmod 5$

$u = 5v + 4$

$x = 12(5v + 4) + 5$

$x = 60v + 53$.

So, $\boxed{x = 53}$

Q.No 31:-

which integer leave a remainder of 1 which divided by 2 and also leave remainder of 1 when divided by 3?

Solution:-

$x = 1 \pmod 2$

$x = 1 \pmod 3$

We are asked to solve simultaneous congurence $x \equiv 1 \pmod 2$ and $x = 1 \pmod 3$

the solution will be unique modulo $2 \cdot 3 = 6$

By insepection we see that the answer simply that

$x \equiv 1 \pmod 6$

the solution set is

$\{\ldots, -11, -5, 1, 7, 13, \ldots\}$

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Q.No 33:-

use fermat's little theorem to find 7^{222} mod 13.

Solution:

Farmat's little theorem is tell us that.

$$7^{12} = 1 \pmod{13}$$

Note that

$$121 = 10 \cdot 12 + 1.$$

therefore,

$$7^{121} = 7^{12 \cdot 10 + 1} \Rightarrow (7)^{12\,10} \cdot 7.$$

$$= (1)^{10} \cdot 7 \Rightarrow 7 \pmod{13}$$

$$7^{121} = 7 \pmod{13}$$

Q. NO 34:-

Use fermat's theorem to find 23^{1002} mod 41

Solution:-

We know that fermat's theorem tells that $(23)^{40} = 1 \pmod{41}$

Note that

$$1002 = 40 \cdot 25 + 2.$$

$$(23)^{1002} = (23)^{40 \cdot 25 + 2}$$

$$= (23)^{40^{25}} \cdot (23)^2 \cdot \pmod{41}$$

$$(1)^{25} \cdot 529 \pmod{41}$$

$$(23)^{1002} = 529 \pmod{41}$$

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M T W T F S S

Q.No 35:-

use fermat's little theorem to show that if p is prime and $p \nmid a$, then a^{p-2} is an inverse of a modulo p.

Solution:-

Fermas little theorem tell us that under given condition.

$$a^{p-1} = 1 \pmod{p}$$

Now we take a^{p-2}.

$$a^{p-2} = a^{p-2} \cdot a = a \cdot a^{p-1} = a^{p-1} = 1 \pmod{p}$$

As a^{p-2} is multiple of a so, it is inverse of a mod p.

Question: No 36:-

use exercise 35 to find an inverse of 5 modulo 41.

Here, $p - 41$

$a = 5$

By using above relation which is discussed in question no 35.

a^{p-2} is inverse of a mod p.

5^{41-2} is inverse of 5 mod 41.

5^{39} is inverse of 5 mod 41.

Question No 37:-

Show that $2^{340} = 1 \pmod{11}$ by fermat's little theorem and nothing that $2^{340} = (2^{10})^{34}$.

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Solution:-

A fermat's little theorem tells as that as

$2^{10} = 1 \pmod{11}$

Now as the question.

(a) $2^{340} = (2^{10})^{34} (1)^{34} = 1 \pmod{11}$

(b) $2^{340} = (25)^{68} = (32)^{68} = (1)^{68} = 1 \pmod{341}$

(c) Calculate. from part (a) and (b) that $2^{340} = 1 \pmod{341}$

As we know that from the a and b) that,

11 and 31 are relatively prime.

and $11 \cdot 31 = 341$. So.

$2^{340} = 1 \pmod{341}$

Question No 39:-

(a) Use fermat's little theorem to compute $5^{2003} \bmod 7 \cdot 5^{2003} \bmod 11$, $5^{2003} \bmod 13$.

Solution:-

By using fermat's theorem first value as:

$5^{2003} \bmod 7$ According fermat's theorm.

$5^6 = 1 \pmod 7$

$2003 = 6 \cdot 333 + 5$

$= (5)^{6 \cdot 333} (5)^5 \pmod 7$

$= (1)^{333} \cdot (5)^5 \pmod 7$

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$5^{2003} = 3 \pmod 7$

Now we use fermat's theorem on second value.

$5^{2003} \bmod 11.$

$5^{10} = 1 \pmod{11}$

$5^{2003} = (5)^{10 \cdot 200} \cdot 5^3 \pmod{11}$

$= (5^{10})^{200} 5^2 \pmod{11}$

$= (1)^{200} \cdot 5^3 \pmod{11}$

$5^{2003} = 4 \pmod{11}$

For third value.

$5^{2003} = 1 \pmod{13}$

$5^{12} = 1 \pmod{13}$

$5^{2003} = 5^{12 \cdot 166} \cdot 5^{11} \pmod{13}$

$= (1)^{166} \cdot 5^{11} \pmod{13}$

$5^{2003} = 8 \pmod{13}$

Question No 43:-

Determine whethe $M_{11} = 2^{11} - 1 = 2047$ and $M_{17} = 2^{17} - 1 = 131071$ are prime.

Solution:-

To decide whether $2^{11} - 1 = 2047$ is prime we need only look for a prime factor not exceeding $\sqrt{2047} \approx 45$ By exercise 41 every such prime divisor of the sum form $22k+1$.

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The only candidate is therefore 23. In fact,

$2047 = 23 \cdot 89$ So, we conclude that 2047 is not prime. And now the second is a prime by procedure of the above number as same.

Q. No 46:-

Show that 1729 is a carmichead number.

To solve this we first check it whether it is composite or not.

$1729 = 7 \cdot 13 \cdot 19$.

So, we must show that this number meets the defination of carmichead number namely that.

$= 1 \pmod{1729}$ For

b relatively prime to 1729.

Note that if.

gcd $(b, 1729) = 1$, then gcd $(b,13) = 1$.

gcd $(b,19) = 1$, gcd $(b,7) = 1$,.

By using fermats little theorem we find that.

$b^{1728} = 1 \pmod{13}$, $b^{180} = 1 \pmod{19}$

$b^{6} = 1 \pmod{11}$.

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It follows that:

$b^{1728} = (b^{12})^{144} = (1)^{144} \pmod{13}$

$b^{1728} = 1 \pmod{13} \rightarrow (i)$

For b^{6}

$(b)^{1728} = (b^{6})^{288} \pmod{7}$

$(b)^{1728} = 1 \pmod{7} \rightarrow (ii)$

For b^{18}

$(b)^{1728} = (b^{18})^{96} \pmod{19}$

$b^{1728} = 1 \pmod{19} \rightarrow (iii)$

By comparing eq (i), (ii) and (iii) we conclude that.

$b^{1728} = 1 \pmod{(7 \cdot 13 \cdot 19)}$

$b^{1728} = 1 \pmod{1729}$.

which is required.

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Chapter No 6:-

Counting:-

Exercise No 6.1:-

Q.No 1:-

there are 18 mathematics majors and 325 computer science major at a college.

(a) In how many ways can two representative be picked so that one is mathematical major and other is a computer Science majors?

As We have to pick two representation. first from 18 mathematics majors and other from 325 computer science majors. there are 18 ways to select mathematics major and there are 325 ways to select computer science major.

So, By using product rule:-

$18 \times 325 = 5850$ ways.

to select two representative.

(b) In how many ways can one representative be picked who is either a mathematics major or a computer science major?

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Solution:-

In this part we have to choose only one representative either from mathematics major are from computer science majors, there are 18 ways to choose a representative from mathematics major and there are 325 ways to choose the same representative from computer science major, So, they are mutually exclusive. So, we use sum rule which is given as

$18 + 325 = 343$

they are 343 ways to pick one representative either from mathematics major. or form computer science majors.

Q.No 2:-

An office building contain 27 floors and has 37 office one each floor. How many office are in building:-

Solution:-

We have to find the number of office in the building. Now the number of office in one floor is given as 27. and no of floors in the building So, we use product rule as.

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$37 \times 27 = 999$.

So, 999 office are there in the building.

Question No 3:-

A multiple choice test contain 10 questions. there are four possible answer for question.

(a) In how many ways on a student answers the question on the test if student answer every.

Solution:-

Since student will perform each of 10 tasks, one after other. there are 4 ways to do each task. therefore we use product rule as

$4 \cdot 4 \cdots 4^n = 4^{10} = 1048576$.

So,

therefore 1048576 ways to answers the question on test.

b)

In how many ways can a student answers the questions on the test if student can leaves answer blank.

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Solution:-

AS in this part student have five 5 to solve the question as 4 ways to do the all question and 1. way to leave question blank. So, student have 5 ways to solve the question. Now we use product rule.

$5^{10} = 9765625$.

Question No 04:-

A particals brand of shirt comes in 12 colors, has a male version and a female version, and comes in 3 size for each sex. How many different types of this shirt are made?

Solution:-

the colour of the shirt has nothing to do thing with size or sex. Simularly, this applies to other case as well. So, the product rule will be applied to get.

$12 \times 2 \times 3 = 72$

therefore types of shirts most be made to satisfy demand.

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Q.No 5:-

Six different airlines fly from New York to Denver and seven fly from Denver to san fancisco. How many different pairs of airlines can you choose on which to back a trip from New York to san Francisco via Denver.

[redacted] When you pick an online for flight to Denver and an airline for continuation flight to San Francisco?

Solution:-

Since a flight is determined by choosing an airline for flight from New york to Denver which an be done in 6 ways and then choosing an cirline for flight from Denver to San Francisco. (which can be done in 7 ways)

therefore there are

$6 \cdot 7 = 42$ different possibilities for entire flight.

Question No 6:-

there are four major auto routes from Boston to Detroit and six from detriot to Los Angels

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How many major auto routes are there from Boston to Los Angles via Detroit?

Solution:-

the number of routes from Boston to Detroit = 4.

the number of routes from Detroit to Los Angeles = 6

So, the number of routes from Boston to los Angeles via Detroit is given

$6 \cdot 4 = 24$. major auto routes are there from Boston to los Angeles via Detroit.

Q.No 7:-

How many different three-letter-intial can are people have?

Solution:-

three-letters intials are determined by specifying the first intial (26 ways) therefore by product rule there are,

$26 \cdot 26 \cdot 26 = 26^3 = 17576$.

possible three-letter intials

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Q.NO 8:-

How many different three-Letter initial with none of the Letters repeated can people have?

Solution:-

We know that there are 26 letters in alphabats first intial = 26

Second intial = 25 (as the first letter can be choosen)

third intial = 24 (as first and second letter cannot be chosen)

So we use product rule:-

$$26 \cdot 25 \cdot 24 = 15600$$

So, 15600 different three-Letter intial with none of letter repeated.

Q.NO 9:-

How many different three Letter intials are thre that begin with on A?

Solution

there is only one way to specify first intials, but as in Exercise 7. there are are 26 ways to specify each of the other intial.

there fore there are by product rule,

$$1 \cdot 26 \cdot 26 = 26^2 = 676.$$

possible three-Little intials begining with A.

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Question No 10:-

How many bit string are there of Length eight?

Solution:-

We know that there are two bits strings Say 0 and 1.

As give in question we have to finds two of bits string that of Length. So, use product rule.

$$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^8 = 256$$

So thre are 256 bit strings of Length eight.

Question No 11:-

How many bit string of Length ten both and end with an A?

Solution:-

thre are two bits string 0 and 1

As given in the question that string begin and end with 1.

So, first bit string = 1

Second = = 2

3rd .... = 2

⋮

9th .... = 2

10th .... = 2 bit string = 1

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=1 because there is only are, zero does not not exist. So, we use product rule.

$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^8 = 256.$

Q.NO 12:-

How many bit string are thre of length six or less, not counting the empty string?

Solution:-

We have to find number of bits string that are of length six or less, but they should not be empty.

For 1st bit string $= 2^6$

2nd ... $= 2^5$	5th .... $= 2^2$
3rd ... $= 2^4$	6th .... $= 2^1$
4th .... $= 2^3$	

So by using sum rule.

the number of bit string

$= 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1$

$= 64 + 32 + 16 + 8 + 4 + 2$

$= 126$

So, number of bit strings are 126.

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Q.NO 14:-

How many bit string of length n, where n is positive integer, start and end with 1S?

Solution:-

there are two possible bits 0 and 1. First bit contain, = 1 (as start with 1S)

For second bit = 2.

3rd bit = 2.

⋮

For (n-1) bit = 2.

For n bit = 1.

So, there we use product rule as

$1 \cdot 2 \cdot 2 \cdot \dots \cdot 2 \cdot 1 = 2^{n-2}$

So, there are 2^{n-2} bit string when $n \geq 2$ and 1 bit string when $n = 1$.

Q. No 15:-

How many string are there of four lowerance Letter of length four or less, not containg the empty string?

Solution:-

As we have number of lowerecase letter are = 26.

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For 4th length string bits contain
$= 26 \cdot 26 \cdot 26 \cdot 26$
$= 26^4$

For 3rd length string bits contain
$= 26 \cdot 26 \cdot 26$
$= 26^3$

For 2nd length string $= 26 \cdot 26 = 26^2$
For 1st length string $= 26$

Q.No 16:-

How many string are there of four lowercase Letters that have the Letter x in them?

We know that these are 26 Letters. So, for four length string
$26 \cdot 26 \cdot 26 \cdot 26 = 26^4 = 456976$

For 4 length string which have no letter x as $26 - 1 = 25$
$25 \cdot 25 \cdot 25 \cdot 25 = 25^4 = 390625.$

So, we have to find the bit String of length 4 with at least one x which is gives as,
bit string for all Letters − bit string for non−x.
Letter
$456976 - 390625 = 66351.$

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Q.No 20:-

How many positive integers blw 5 and 31? (c) Are divisble by 3 and by 4? which integers are these?

Solution:-

Let A be positive integers blw 5 and 31 (thus from 6 to 30)
We have such integers are 25 as
$|A| = 25$

which the integers which are both divisible by 3 and 4 are as which are divisible by
$3 \cdot 4 = 12$
$d = 12$
$\frac{|A|}{d} = 25/12 = 2.883$

So, there are 2 integer blw 5-31 which are divisible by both 3 and 4 12 and 24.

(b) Are divisble by 4? which integers are these?

Solution:-

we have total number that are blw 5-31 are 25
$|A| = 25.$

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which are divisible by 4 d=4

$|A| = 25/4 = 6.25$

thus there are 6 or 7 numbers that are divisible by 4.

8, 12, 16, 20, 24, 28.

we not that there are 6 integers

Q. No 24:-

How many positive integers between 1000 and 9999 inclusive.

(a) Are divisble by 9?

Let A be the positive integers blw 1000 and 9999 inclusive.

We know that are 9000 such integer

$|A| = 9000$

the integers that are divisible by 9 are as d=9.

$\frac{|A|}{d} = \frac{9000}{9} = 1000.$

(c) have distinct digits?

there are in total 10 digits.

First digits 9 (ways since first digits cannot be zero)

Second digits : 9 ways

third digits 8 ways

fourth digits 7 ways

So, by product rule.

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$9 \cdot 9 \cdot 8 \cdot 7 = 4536$

thus 4536 integers have distint digits.

Q. No 20:-

How many integers between 5 and 31.

(a) which are divisible by 3?

We have integer that blw 5 and 31 are these which are from 6 to 30.

So, there are 25 integers blw

5-30 = 25

$|A| = 25$

we have to find such integers that are divisible by 3. d=3

$\frac{|A|}{d} = 25/3 = 8.33$

So, there are 8 or 9 integers blw 5 to 31.

We have to integers.

6, 9, 12, 15, 18, 21, 24, 27, 30

So, there are 9 integer. blw 5-31.

Q. No 25:-

How many strings of three decimals digits.

(a) Do not contain the same digits three times?

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No. of string with third decimal, $10 \cdot 10 \cdot 10 = 1000$

there are 10 possible digits from 0 to 9.

No. of string with 3 decimal digits first digit : 10 ways.

2nd digits = 1 ways (since it has to be equal to first digits)

3rd digits = 1 ways (since it has to be equal to 1st and 2nd)

So, their product.

$10 \cdot 1 \cdot 1 = 10$

So, No. of string that are not same three digits are

$1000 - 10 = \boxed{990}$

(b) began with an odd digits?

there are 10 possible digits from 0 to 9.

1st digits = 5 way (since there are 5 odd digits)

2nd digits = 10 ways

3rd digits = 10 ways

we product rule

$5 \cdot 10 \cdot 10 = 500$

have exactly two digits that are 4s?

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there are 10 possible digits from 0 to 9.

First four : 1 way

Second four : 1 way

other digits : 9 way

positive 4s = 3 way

use product rule.

$3 \cdot 9 \cdot 1 \cdot 1 = \boxed{27}$

Q.No 27:-

A commitee is formed consisting of one representative for each of 50 state in the united state, where the representative from a state is either the governor or one of two two senators from that State. How many ways are there to form this committee?

Solution:-

the repesentative of state is either to governor. or of two snators from the state, thus per state there are 3 ways to select representative per state. State = 50

Representative = 3 ways.

using product rule

$3 \cdot 3 \cdot 3 \ldots 3^{50} = 3^{50} = 717,897,987,691,852$

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=588,770,249 ways

Question No 28:-

How many license plates can be made using either three digits followed by three uppercase English letters or three uppercase English letter followed by three digits?

We know that there are 26 possible letters and 10 possible digits.

1st letter = 26 way	1st digits = 10 ways
2nd letter = 26 way	2nd digits = 10 ways
3rd letter = 26 way	3rd digits = 10 ways.

order = 2 ways (Becaus letter are before digits or digits are before letter)

Product rule.

$26 \cdot 26 \cdot 26 \cdot 2 \cdot 10 \cdot 10 \cdot 10 = 26^3 10^3 \cdot 2$

$= 35152000$ ways

Question No 30:-

How many license plate can be made using either three uppercase English letters followed by ~~four~~ three digits or ~~two~~ four digits followed uppercase English letters followed two digits?

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Solution:-

there are 26 possible letter and 10 possible digits.

(i) three letter followed by three digits?

1st letter = 26 ways	1st digits = 10 ways
2nd = 26 ways	2nd = 10 ways
3rd = 26 ways	3rd = 10 ways.

use product rule

$26^3 \cdot 10^3 = 17576000$

(ii) Four letter followed by two digits

1st letter = 26 ways	1st digits = 10 ways
2nd ... = 26 ways	2nd digits = 10 ways.
3rd ... = 26 ways	
4th .... = 26 ways	

use product rule.

$26^4 \cdot 10^2 = 45697600$

there letter followed by three digits or four letters followed by two digits.

use sum rule.

$17576000 + 45697600 = 63273600$ ways

Question No 34:-

How many different function are there from a set with 10 element to sets with following number of element

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(a) 2

In this part their are two(2) possible images (since the set that contains range has 2 element)

1st element : 2 way	6th element = 2 ways
2nd .... = 2 way	7th ... = 2 ways
3rd ... = 2 way	8th ... = 2 ways
4th ..... = 2 way	9th .... = 2 ways
5th ..... = 2 way	10th ....= 2 Ways.

So by using Product rule.

$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

$= 1024$

Q.No 35 :-

How many one-to-one function are there from a set with five element to sets with the following number of element?

(a) 4

A function is one-to-one, if each element has a unique image since there are five 5 element in domain and 4 element in range (images) it is not possible for function to one-to-one.

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(c) 6

Since Domain = 5 and Range = 6

1st element = 6 ways	use product rule.
2nd ... = 5 ways	$6 \cdot 5 \cdot 4 \cdot 3 \cdot 2$
3rd ... = 4 ways	$= 720$
4th .... = 3 ways	
5th ... = 2 ways.	

Question No 44:-

How many ways to there to seat four of a grop of ten people around a circular table where two setting are (non) considered the same when everyone has the same immediate left and immediate right neighbor?

Solution :-

Let A be groups of four people selected from 10 people.

1st person = 10 ways, 2nd person = 9 Ways

3rd person = 8 ways, 4th person = 7 Ways.

use product rule.

$|A| = 10 \cdot 9 \cdot 8 \cdot 7 = 5040$

there are four people in each possible group of A.

$d = 4$

$n = |A|/d = \frac{5040}{4} = 1600$

$= 1260$

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Q.No 45:-

How many way are there to set six people around a circular table where two seating are considered the same when everyone has the same two neighbors without regard to whether they are right or left neighbors?

Solution:-

As we know that there are 6 people.

$= 6$ Ways (to sit around a circle)

For 2nd person

$= 5$ Ways (since seat have been assigned to first person)

Similarly,

3rd person $= 4$ ways, 4th person $= 3$ ways

5th person $= 2$ ways, 6th person $= 1$ ways

$$6 \cdot 2 \cdot 5 \cdot 4 \cdot 3 \cdot 1 = 720$$

As there are total 6 person.

$$d = 6$$

thus $\frac{|A|}{d} = 720/6 = 120$

So, there are 120 ways to sit 6 persons But here a point is hidden which

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tell us that two seat are considere same. So, we have to divide 120 by two. As

$$\frac{120}{2} = \boxed{60 \text{ Ways}}$$

Q.NO 46:-

In how many ways can a photographer at a wedding arange 6 people in a row from a group of 10 people. where the bride and the the groom are among these 10 people, if

(a) the bride must be in the picture?

We pick 6 people from 10 people

1st person $= 1$ way (bride)

2nd ... $= 9$ way

3rd ... $= 8$ way

4th ... $= 7$ Way

5th ... $= 6$ Way

6th ... $= 5$ Way

use product rule

$$1 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 15120$$

Since bride can be placed in 6 position along row, with 15120 possible graph each.

$$6 \cdot 15120 = 90720$$

(b) the bride is not next to the groom?

We pick 6 people from 10 people.

1st person $= 1$ ways (bride) 2nd .... person $= 1$ way (groom)

3rd person $= 8$ ways 4th ... person $= 7$ way

5th person $= 6$ ways 6th ... person $= 5$ Way

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use product rule.

$1 \cdot 1 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 1680$

Since bride can be placed in six position along row. after which groom can be placed in 5 person. and there are 1680 possible groups each.

$6 \cdot 5 \cdot 1680 = 50400$

Q.No 48:-

How many bit string of length seven either begin with two 0s or end with three 1s?

Let A be the bit string of length Seven begining with two Zeros. then

$|A| = 1 \cdot 1 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^5 = 32$

Let set A_2 be bit string of length seven ending with three ones. then

$|A_2| = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 1 \cdot 1 \cdot 1 = 2^4 = 16$

Let $A_1 \cap A_2$ be the set of bit string of length seven begining with two zero and ending with three ones. then,

$(A_1 \cap A_2) = 1 \cdot 1 \cdot 2 \cdot 2 \cdot 1 \cdot 1 \cdot 1 = 2^2 = 4$

$(A_1 \cup A_2) = |A_2| + |A_1| - |A_1 \cap A_2| = 32 + 16 - 4$

$= 44$

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Exercise No 6.2:-

the Pigeonhole Principle:

If K is a positive integer and K+1 or more objects are placed into K boxes. then there is at Least one box containing two or more of the objects.

theorem 2:-

the Generalized Pigeonhole Principle:-

If N objective are placed into K boxes, then there is at least one box containing at at Least $\lceil N/K \rceil$ objects.

Proof:-

We will use a proof by contraposition. Suppose that none of boxes contain more than $\lceil N/K \rceil - 1$ objects. the total number of objects is at least. Most.

$$K\left(\left\lceil \frac{N}{K} \right\rceil - 1\right) < K\left(\left(\frac{N}{K} + 1\right) - 1\right) = N$$

Where the inequality $\lceil N/K \rceil < (N/K) + 1$ has been used. this is a contradiction because there are a total of N objects.

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Question No 1:-

Show that in any set of six classes, each meeting regularly once a week on a particular day of the week, there must be two that meet on the same day, assuming that no classes are held on weekends.

holes = days of week (not including weekend).

object = k+5

= 6 = 5+1 = k+1

So, Eeach class meets only one day a week. So, by pigeonhole principle. the must be atleast 2 classes that meet on same day.

Question No 2:-

Show that if there are 30 students in a class, then at least two have last name that begin with the same letter.

there are 30 students in class. Assume that each student has at least a first name and a last name. Each last name has to begin with a letter of alphabet.

objects = first letter of each last name

=N = 30

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Holes letter of alphabets = k=26

$\left[\frac{30}{26}\right] = 2$

So, by pigonhole principle, theorem at least two student who have a last name that begin with same letters.

Question No 3:-

A drawer contain a dozen brown socks and a dozen black socks, all unmatched. A man takes socks out at random in the dark.

(a) How many socks must be take out to be sure that the has at least two socks of the same colour.

there are 12 black socks and 12 brown sock. the total socks are 24.

Hence.

holes = k = colour = 2

object = N = ?

we know that $\lceil N/2 \rceil$ must be equal to 2.

the least positive integer that satisfy the above condition is 3. As

$\lceil 3/2 \rceil = \lceil 1.5 \rceil \approx 2$

So, the positive number is = 3.

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(b) How many socks must he take out to be sure that he has at least balls two black socks?

the first 12 choices may all be brown, so the person needs to choose at least 14 socks be be sure that least 2 of then are black.

So, Answer is 14.

Question No 5:-

Show that among any group of five (not necessarily consecutive) integer, there are two with the same remainder when divided by 4.

there are four possible remainder when an integer is divided by 4.

0, 1, 2, 3

holes = remainde when divided by 4 = k = 4

the number of integer = N = ?

By pigeonhole principle at least.

$$\left[\frac{N}{k}\right] = 2$$

$$\left[\frac{N}{4}\right] = 2$$

only the positive number that satisfy the above condition is 5.

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As.

$$\left[\frac{5}{4}\right] = [18] = 2$$

So, Answer is 5. so it is proved.

Question No 6:-

Let d be a position integer. Show that among any group of d+1 (not necessarily consecutive) integers there are two with exactly the same remainder when they are divided by d.

Let A be the set of integers $\{a, b, c \cdots n\}$. $|A| = |\{a, b, c \cdots n\}| = d+1$

When dividing by d, the possible

Remainder of d are $\{0, 1, 2, \cdots d-1\}$.

the possible number of remainder is d. or

$$|\{0, 1, 2, \cdots d-1\}| = d$$

objects = number of integer = d+1

holes = number of remainder = d

$$\left[\frac{d+1}{d}\right] = 2$$

So, By pigeonhole principle, among any group of d+1. integer there are d+1.

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Question No 9:-

what is the minimum number of students, each of whom comes from one of the 50 states, who must be enrolled in a university to guarantee that there are at least 100 who come from the same state?

Pigeon hole = k = state = 50

object = N = Number of student = ?

As given in quest that by pigeonhole principle.

$$\left\lceil \frac{N}{k} \right\rceil = \left\lceil \frac{N}{50} \right\rceil = 100$$

the above condition is satisfied when we put N = 4951. As

$$\left\lceil \frac{4951}{50} \right\rceil = 99.02 \not\geq 100$$

So, our Ansaser is 4951.

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Exercise No 6.3:-

Q.1:-

List all permutation of {a,b,c}

As we know that permutation are ordered carrangements. thus we need to list all ordered arrangement of all 3 of these letters.

there are following orders.

(a,b,c), (a,c,b), (c,a,b), (b,a,c), (b,c,a) (c,a,b) and (c,b,a).

We conclude that that are 6 permutation of {a,b,c}.

Q.2:- How many different permulation are there of set {a,b,c,d,e,f,g}?

there are given

$n=7$

$r=7$

By using permutation formula.

$$P(7,7) = \frac{7!}{(7-7)!} = \frac{7!}{0!}$$

$$= 7!$$

$$= 7\times6\times5\times4\times3\times2\times1$$

$$= 5040$$

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Q.3:- How many permutation of {a,b,c,d,e,f,g} end with a?

Here we have; As we forget 'a'

$r = 6$

$n = 6$.

By using formula.

$P(6,6) = \frac{6!}{(6-6)!} = \frac{6!}{0!}$

$= 6!$ $= 720$

Q.4:- Let $S = \{1,2,3,4,5\}$

(a) List all three-permutation of 5

For three-permutation.

$r = 3$

Total number $n = 5$

By using formula.

$P(5,3) = \frac{5!}{(5-3)!} = \frac{5!}{2!} = \frac{5 \times 3 \times 4 \times 2!}{2!}$

$= 60$

(b) List all 3-combination of 5.

For all-combination

$r = 3 \quad n = 5$

$C\binom{5}{3} = \frac{5!}{(5-3)!\,3!} = \frac{5!}{2!\,3!} = \frac{3!\,4^2 \times 5}{3! \cdot 2}$

$= 10$

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Q.5:- Find value of each of these qualities:-

(a) $P(6,3)$

$n = 6$

$r = 3$

$P(6,3) = \frac{6!}{(6-3)!}$

$= \frac{6!}{3!}$

$= \frac{3! \cdot 4 \times 5 \times 6}{3!}$

$= 120$

(b) $P(6,5)$

$n = 6$

$r = 3$

$P(6,5) = \frac{6!}{(6-5)!}$

$= \frac{6!}{1!}$

$= 720$

Q.07: Find number of 5-permutation of a set with nine element.

$r = 5$

$n = 9$

$P(9,5) = \frac{9!}{(9-5)!} = \frac{9!}{4!}$

$= \frac{4! \times 5 \times 6 \times 7 \times 8 \times 9}{4!}$

$= 15120$

Q. 08:-

In how many different order can five runners finish a race if no ties are allowed?

We select 5 runner. For 1st number there is only 5 chance. For 1st runner

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there are 4 change and for 3rd, 4th, and 5th there are 3, 2 and 1 ways respectly.

So, For selection of 5 runners.

$P(5,5) = 5.3.4.2.1.$

$= 120$ way

Q.09:-

How many possibilities are there for the win, place, and show (first, second, and third) position in a horse race with 12 horses if all order of finish are possible?

Total horse $= n = 12$

the horse getting positio $= r = 3$

$P(12,3) = \frac{12!}{(12-3)!}$

$= \frac{9! \times 10 \times 11 \times 12}{9!}$

$= 1320$

Q.20:-

the are six different candidates for governor of a state. In how many different orders can the name of the candidate be printed on a ballot.

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Total candidate $= 6$

For $1^{st} = 6$ ways, $3^{rd} = 4$ ways

$2^{nd} = 5$ ways, $4^{th} = 3$ ways

$5^{th} = 2$ ways, $6^{th} = 1$ way

So By using product rule.

$P(6,6) = 6.5.4.3.2.1.$

$= 720$

Q.21

How many permutation of letters ABCDEFG contain

(a) the string BCD?

Let we suppose that BCD are in some block. then we have to find the of 5. Letters as of A, E, F, G.

and block of BCD.

$P(5,5) = \frac{5!}{(5-5)!} = 5!$ $= 120$

(C) the strings BA and GF?

Let we put BA one letter and GF as one letter. thus we have to find. permutation of 5 letter namely: (BA), C, D, E and (GF)

$P(5,5) = \frac{5!}{(5-5)!} = 5!$ $= 120$

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(f) the strings CBA and BED?

there are no permutation with both of these substring, since B cannot be followed by both A and E at some time.

Q.11:- How many bit string of length 10:-

(a) contain exactly four 1s?

Here order does not matter:-

So, we use combination

Here $n = 10$

$r = 4$ (exactly)

By using formula:

$$C(n,r) = \binom{n}{r} \Rightarrow \binom{10}{4} = \frac{10!}{(10-4)!\,4!}$$

$$= \frac{10!}{6!\,4!} \Rightarrow \frac{6!7\times 8\times 9\times 10}{4\times 3\times 2\times 6!}$$

$= 210$

(b) at most four 1s:-

So, we have to find combination for almost 4 1s.

For one 1s;

$$C(10,1) = \binom{10}{1} = \frac{(10!)}{(10-1)!\,1!} = 10$$

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For two 1s;

$$C(10,2) = \binom{10}{2} = \frac{10!}{(10-2)!\,2!} = 45$$

For three 1s:

$$C(10,3) = \binom{10}{3} = \frac{10!}{(10-3)!\,3!} = 120$$

For 4 1s;

$$C(10,4) = \binom{10}{4} = \frac{10!}{(10-4)!\,4!} = 210$$

For at most four 1s;

$$C(10,1) + C(10+2) + C(10+3) + C(10+4)$$

$$= 1 + 45 + 120 + 210 = 386$$

(C) at least four 1s?

For at least four 1s;

$$C(10,4) + C(10,5) + C(10+6) + C(10,7) + C(10,8)$$

$$C(10,9), C(10,10)$$

$$= 210 + 252 + 210 + 120 + 45 + 10 + 1.$$

$= 848$

d) An equal number of 0s and 1s?

For equal number of 0s and 1s. it means that there are 5 0s and 1s;

So, $n = 10$ For (1s)

$r = 5$

$$C(10,5) = \binom{10}{5} = \frac{10!}{(10-5)!\,5!} = \frac{10!}{5!\,5!} = 252$$

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Q.13:-

A group contains n men n women. How many ways are there to arrange these People in a row if the men and women alternate?

Here order is important, So, We Use permutation. for men:

$$P(n,n) = \frac{n!}{(n-n)!\,n!} = \frac{n!}{0!} = \boxed{n!}$$

Now, We use total.

(i) When men is first ant then women respectively, using product rules:

$$n!\,n! = (n!)^2$$

(ii) When Women is first and then man respectively, using Product rule:

$$(n!)(n!) = (n!)^2$$

For both using Sum $= (n!)^2 + (n!)^2$

$$= 2(n!)^2$$

Q.15:- In how many ways can a set of five letter be selected from the English alphabet?

For integer order is not important So, We use deffination of combination

$n = 99$ (as less than 100);

$r = 2$

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By using formula:

$$C(n,c) = \binom{n}{r} = \binom{99}{2} = \frac{99!}{(99-2)!\,2!}$$

$$= \boxed{4851}$$

Q.NO16:-

How many subsets with an odd number of elements does a set with 10 element have?

We find all odd number that are less than 10 are.

1, 3, 5, 7, 9

Hence order is not important So, We use combination formula;

$n = 10$

For all odd number.

$$C(10,1) + C(10,3) + C(10,5) + C(10,7) + C(10,9)$$

$$= 10 + 120 + 252 + 120 + 10$$

$$= \boxed{512}$$

Q:19:- A coin 1st flipped 10 time...

(a) are thre in total?

For 1st flipp of coin there are two passibities (had or tail)

∴ For total at 10 flippes

$2^{10} = \boxed{1024}$ outcome.

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(b) Contain exactly two heads.

Here order is not mattered

So, We use combination formula

$n = 10$, $r = 2$

$C(10,2) = \binom{10}{2} = \frac{10!}{(10-2)!\,2!}$ = 45

(d) Contain same number of heads and tails.

For same heads and tails there are 5 heads and 5 tails.

$n = 10$

$r = 5$

By combination formula:

$C(n,r) = \binom{n}{r} = \binom{10}{5}$ = 252

Q.26:- thirteen people on a softball team show up for a game.

(a) How many ways are there to choose 10 player to take the field?

Now, total people $= n = 13$

selected players $= r = 10$

here order is not important

So, we use combination formula.

$C(n,r) = C(13,10)$ = 286

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(b) How many way are there to assign the 10 positions by selecting Player from the 13 people who show up?

Here order is important So, use use permutation formula;

$n = 13$

$r = 10$

$P(13,10) = \frac{13!}{(13-10)!}$ = 1,037,836,800

(c) of the 13 people who show up, three are women. How many ways are there to choose 10 players to take the field if at least one of these players must be a woman?

First We find ways for which woman are not include:-

$13 - 3 = 10$

$n = 10$

$r = 10$

$C(10,10) = \binom{10}{0} = 1$

We know that for total selection of 10 choose are 286 ways

Number of Ways with atleast are women

= Total ways - No. of way without Women = 286 - 1 = 285

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Question No 28:-

A Professor writes 40 discrete math true/false question. of the statement in these question. 17 are true. If the question can be positioned in any order, how many different answer Key are possible?

the order of question does not matter. So, We use combination formula:

$n = 40$

$r = 17.$

$C(40,17) = \binom{40}{17} = \frac{40!}{(40-17)!\,17!} = 8873 \times 10^{10}$

Question No 29:-

Seven women and nine men are on the faculty in the mathematics department at a school.

(a) How many ways are there to select a committee of five member

We select 5 number from.

$9+7 = 16$ people

$n = 16$

$r = 5$

the order does not matter use combination.

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$C(16,5) = 4368.$

Now, we find the committee for only men:

$n = 9$

$r = 5$

$C(n,r) = C(9,5) = 125$

No, to find committe with atleast one women:-

$4368 - 128 = 4242$

Question No 33:-

Suppose that a department contains 10 men and 15 women. How many ways are there to form a committee with six member if it must have the same number of men and women?

Total men = 10

total Women = 15

selected = six(6)

As there are equal 10 number

So, 3 men and 3 Women.

For 3 men

$n = 10$

$r = 3$

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$C(10,3) = \binom{10}{3} \frac{10!}{(10-3)!\,3!}$ $= 120$

(ii) For 3 women.

$n = 15$

$r = 3$

$C(15,3) = \frac{15!}{(15-3)!\,3!}$ $= 455$

for Both men and women we use product rule:-

$C(10,3) \cdot C(15,3) = (120)(455)$

$= 254600$

Question No 37:-

How many bit string of length 10 contain at least three 1s and at least three 0s?

Total number of String of length 10.

$2^{10} = 1024$

(i) string with no ones:

$C(10,0) = 1$

Similarly String with no zeros is $C(10,0) = 1$.

(ii) string with 1 ones.

$C(10,1) = 10$.

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Similarly string with 1 zero is

$C(10,1) = 10$

String with 2 ones

$C(10,2) = 45$

Similarly String with 2 zero is

$C(10,2) = 45$

String with atmost 3 zeros and atmost 3 ones.

$1+1+10+10+45+45$ $= 112$

We have to find string with atleast 3 ones and 3 zeros.

= Total - atmost 3 one and 3 zero

$1024 - 112$ $= 912$

Question No 38:-

How many ways are there to select 12 countries in the united Nation to serve on a council if 3 are selected from a block of 45, 4 are selected from a block of 57, and the others are selected from the remaining 69 Countries?

the order is not method in this question So, we use combination:-

As given in the question:-

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(i) 3 are selected from black of 45

$C(45,3) = \frac{45!}{3!(45-3)!} = 14190$

(ii) 4 are selected from black of 57

$C(57,4) = \frac{57!}{(57-4)!4!} = 395010$

12 = 4+3+5 are selected from block of 69.

$C(69,5) = 11238513$

Use product rule:-

$14190 \cdot 395010 \cdot 11238513$

$= 62,994,022,035,644,700$

Question No 39:-

How many license plates consisting of three letters followed by there digits contain no letter or digits twice?

For letter (order matter)

$n = 26$

$r = 3$

$P(26,3) = \frac{26!}{(26-3)!} = 15600$

For "digits"

$n = 10$

$r = 3$

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$P(10,3) = \frac{10!}{(10-3)!} = 720$

use product rule:-

$15,600 \cdot 720 = 11,232,000$

Question NO 40:-

Find a ~~formula~~ ~~formula~~ number of circuler 3-permutations of 5 people.

Here

$n = 5$

$r = 3$

By using formula.

$P(5,3) = \frac{5!}{(5-3)!} = 60$

Here let.

$P(5,3) = 60$

We divide it by number of permutation using quotient rule.

$|A| = 60/3 = 20$

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Exercise No 6.4:-

the binomial theorem:-

Let x and y be variable, and let n be a nonnegative integer.

then,

$$(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \cdots +$$

$$\binom{n}{n-1} xy^{n-1} + \binom{n}{n} y^n.$$

Example No 2:-

What the expension of $(x+y)^4$?

By using binomial theorem.

$$(x+y)^4 = \sum_{j=0}^{4} \binom{4}{j} x^{4-j} y^j$$

$$= \binom{4}{0} x^4 + \binom{4}{1} x^3 y + \binom{4}{2} x^2 y^2 + \binom{4}{3} xy^3 + \binom{4}{4} y^4.$$

$$= x^4 + 4x^3y + 6x^2y^2 + 4xy^2 + y^4.$$

Example No 3:-

What is the coeffeicent of $x^{12}y^{13}$ in the expansion of $(x+y)^{25}$?

From the binomial theorem it follows that this coeffecient is

$$\binom{25}{13} = \frac{25!}{13!\,12!} = 5200,300$$

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First, note that this expression equal $(2x + (-3y))^{25}$. By the binomial theorem, we have.

$$(2x + (-3y))^{25} = \sum_{j=0}^{25} \binom{25}{j} (2x)^{25-j} (-3y)^j.$$

the coefficient of $x^{12}y^{13}$ in the expension is obtain when $j=13$, namely

$$\binom{25}{13} 2^{12} \cdot (-3)^{13} = -\frac{25!}{13!\,12!} 2^{12} 3^{13}.$$

Corollary 1:-

Let n be a nonnegative integer. then. then.

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

Proof:-

Using the binomial theorem with $x=1$ and $y=1$, we see that.

$$2^n = (1+1)^n = \sum_{k=0}^{n} \binom{n}{k} 1^k 1^{n-k} = \sum_{k=0}^{n} \binom{n}{k}$$

there is also a nice combinational proof of collory.

$$\sum_{k=0}^{n} \binom{n}{k}$$

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

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Collary 2:-

Let n be a positive integer:

then

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$$

Proof:-

When we use the binomial theorem with $x = -1$ and $y = 1$, we see that.

$$0 = (0)^n = ((-1)+1)^n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k 1^{n-k}$$

$$= \sum_{k=0}^{n} \binom{n}{k} (-1)^k$$

this prove corollary.

Remark:-

Corollary 2 implies that:

$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots$$

Corollary 3:-

Let n be a nonnegative integer then.

$$\sum_{k=0}^{n} 2^k \binom{n}{k} = 3^n.$$

Proof:-

Left-hand side of this formula is the exponent of $(1+2)^n$ provided by the binomial theorem.

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there, by binomial theorem.

We see that.

$$(1+2)^n = \sum_{k=0}^{n} \binom{n}{k} 1^{n-k} 2^k = \sum_{k=0}^{n} \binom{n}{k} 2^k.$$

Hence,

$$\sum_{k=0}^{n} 2^k \binom{n}{k} = 3^n.$$

Pascal's Identity:-

Let n and k be positive integers with $n \geq k$. then.

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

Remark:

$\binom{0}{0}$

$\binom{1}{0}$ $\binom{1}{1}$

$\binom{2}{0}$ $\binom{2}{1}$ $\binom{2}{2}$

$\binom{3}{0}$ $\binom{3}{1}$ $\binom{3}{2}$ $\binom{3}{3}$

$\binom{4}{0}$ $\binom{4}{1}$ $\binom{4}{2}$ $\binom{4}{3}$ $\binom{4}{4}$

$\binom{5}{0}$ $\binom{5}{1}$ $\binom{5}{2}$ $\binom{5}{3}$ $\binom{5}{4}$ $\binom{5}{5}$

$\binom{6}{0}$ $\binom{6}{1}$ $\binom{6}{2}$ $\binom{6}{3}$ $\binom{6}{4}$ $\binom{6}{5}$ $\binom{6}{6}$

$\binom{7}{0}$ $\binom{7}{1}$ $\binom{7}{2}$ $\binom{7}{3}$ $\binom{7}{4}$ $\binom{7}{5}$ $\binom{7}{6}$ $\binom{7}{7}$

$\binom{8}{0}$ $\binom{8}{1}$ $\binom{8}{2}$ $\binom{8}{3}$ $\binom{8}{4}$ $\binom{8}{5}$ $\binom{8}{6}$ $\binom{8}{7}$ $\binom{8}{8}$.

By pascal Identity

$$\binom{6}{4} + \binom{6}{5} = \binom{7}{5}$$

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1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

1 6 15 20 15 6 1

1 7 21 35 35 21 7 1

1 8 28 56 70 56 28 8 1.

(2) :-

Find the expansion of $(x+y)^4$.

(a) using combinatorial reasoning, as in example 1.

$= (x+y)^4$.

$(x+y)^4 = (x+y)(x+y)(x+y)(x+y)$

1st combination

$= \binom{4}{0} x^4 = x^4$

2nd combination

$= \binom{4}{1} x^3 y = 4x^3y$.

3rd combination.

$= \binom{4}{2} x^2y^2 = 6x^2y^2$

4th combination

$\binom{4}{3} xy^3 = 4xy^3$

5th combination

$= \binom{4}{4} y^4 = y^4$.

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consequently, the product.

$x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$.

(b) using the binomial theorem

$= (x+y)^4$

By using binomial theorem.

$(x+y)^n = \binom{n}{0} x^n + \binom{n}{1} x^{n-1}y + \binom{n}{2} x^{n-2}y^2 + \binom{n}{3} x^{n-3}y^3 + \binom{n}{4} x^{n-4}y^4$.

$(x+y)^4 = \binom{4}{0} x^4 + \binom{4}{1} x^{4-1}y + \binom{4}{2} x^{4-2}y^2 + \binom{4}{3} x^{4-3}y^3 + \binom{4}{4} x^{4-4}y^3$.

$\boxed{(x+y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^3}$

(3) Find the expension $(x+y)^6$

$= (x+y)^6$.

By using binomial theorem.

$(x+y)^n = \binom{n}{0} x^n + \binom{n}{1} x^{n-1}y + \binom{n}{2} x^{n-1}y^2 + \cdots + \binom{n}{n-5} x^{n-5}y^5 + \binom{n}{n} x^{n-n}y^6$

$(x+y)^6 = \binom{6}{0} x^6 + \binom{6}{1} x^5y + \binom{6}{2} x^4y^2 + \binom{6}{3} x^3y^3 + \binom{6}{4} x^2y^4 + \binom{6}{5} xy^5 + \binom{6}{6} y^6$.

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$$(x+y)^6 = x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6.$$

Question No 4:-

Find the coefficient of x^5y^8 in $(x+y)^{13}$.

Coefficient of x^5y^8 in $(x+y)^{13}$.

From the binomial theorem it follows that this coefficient is.

$$\binom{13}{8} = \frac{13!}{(13-8)!\,8!}$$

$$= \frac{13!}{8!\,5!} = 1287$$

First, note that this expression equal $(x+y)^{13}$. By the binomial theorem,

$$(x+y)^{13} = \sum_{j=0}^{13} \binom{13}{j} x^{13-j} y^j$$

obtained $y=8$, when $j=8$.

$$(x+y)^{13} = \binom{13}{8} x^{13-8} y^8$$

$$(x+y)^{13} = \frac{13!}{8!\,5!} x^5y^8$$

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Question No 5:-

How many term are there in the expension of $(x+y)^{100}$ after like terms are collected?

In expension of $(x+y)^n$, there are $n+1$ terms.

As there are $n=100$ We get.

$100+1 = 101$

Question No 7:-

What is the coefficient of x^9 in $(2-x)^{19}$?

By using binomial theorem the term involving x^9 in the expansion.

$$= (2+(-x))^{19}$$

$$= \binom{19}{9} 2^{10} (-x)^9$$

therefore coefficient is

$$= +2^{10}\binom{19}{9}(-1)^9$$

$$= -2^{10}\binom{19}{9} = -94595.072.$$

Question No 8:-

what is the coefficient of x^8y^9 in the expension of $(3x+2y)^{17}$?

We know the binomial theorem:

$$(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j$$

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We first write j=0 question in term of binomial expression :-

Let $x-3x$; $y=2y$

$(3x+2y)^{17} = (x+y)^{17}$.

$n = 17$

$j = 9$

By using binomial theorem.

$(x+y)^{17} = \binom{17}{9} x^{17-9} y^{9}$

$= \binom{17}{9} (x^{8})(y^{9})$: $x = 3x$

$= \binom{17}{9} (3x)^{8} (2y)^{9}$. $y = 2y$.

As we only find coefficient.

$= 8.1662 \times 10^{10}$

Question No 12:-

$\binom{10}{k}$, $0 \le k \le 10$ By using pascals identity :-

We know the pascals identity as $\binom{n+1}{k} = \binom{n}{k+1} + \binom{n}{k}$

As $n = 10$.

By using parcals's identity

$\binom{11}{0} = 1$

$\binom{11}{1} = \binom{10}{0} + \binom{10}{1} = 1 + 10 = 11$

$\binom{11}{2} = \binom{10}{1} + \binom{10}{2} = 10 + 45 = 55$

$\binom{11}{3} = \binom{10}{2} + \binom{10}{3} = 45 + 120 = 165$

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$\binom{11}{4} = \binom{10}{3} + \binom{10}{4} = 120 + 210 = 330$

$\binom{11}{5} = \binom{10}{4} + \binom{10}{5} = 210 + 252 = 462$

$\binom{11}{6} = \binom{10}{5} + \binom{10}{6} = 252 + 210 = 462$

$\binom{11}{7} = \binom{10}{6} + \binom{10}{7} = 210 + 120 = 330$

$\binom{11}{8} = \binom{10}{7} + \binom{10}{8} = 120 + 45 = 165$

$\binom{11}{9} = \binom{10}{8} + \binom{10}{9} = 45 + 10 = 55$

$\binom{11}{10} = \binom{10}{9} + \binom{10}{10} = 10 + 1 = 11$

$\binom{11}{11} = 1$.

thus the row $\binom{11}{k}$ is then

1, 11, 55, 165, 330, 462 + 462 + 330 + 165 + 55 + 11 + 1

Question No 13:-

$\binom{9}{k}$, $0 \le k \le 9$?

the row of $\binom{9}{k}$ is simply coefficient K.

$\binom{9}{0} = \frac{9!}{0!\,(9-0)!} = 1$.

$\binom{9}{1} = 9$

$\binom{9}{2} = 36$

$\binom{9}{3} = 84$

$\binom{9}{4} = 126$

$\binom{9}{5} = 126$

$\binom{9}{6} = 84$

$\binom{9}{7} = 36$

$\binom{9}{8} = 9$

$\binom{9}{9} = 1$

So, Row is 1, 9, 36, 84, 126, 126, 84, 36, 9, 1.

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Q.15:-

Show that $\binom{n}{k} \leq 2^n$ for all positive integers n and all integers k with $0 \leq k \leq n$.

As we know from corollary 1.

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

It means that sum of all $\binom{n}{k}$ subset is equal to 2^n.

So, From above discussion it is obviously true that the only one subset.

$$\binom{n}{k} < 2^n.$$

Moreover; a subset can never be greater than a set.

Question No 18:-

Suppose that b is an integer with $b \geq 7$. use the binomial theorem and the appropriate row of pascal's triangle to find the base-b expansion of $(11)_b^4$ [that is, the fourth power of the number $(11)_b$ in base-b notation].

We know that.

$(11)_b$ is base b notation means

$$1.b + 1.1 = b+1.$$

So,

$$(11)_b)^4 = (b+1)^4.$$

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By using Binomial theorem;

$$(1+b)^4 = (b+1)^4 = \sum_{i=0}^{4} \binom{4}{i} . b^i$$

$$= \binom{4}{0} . (1)^4 (b)^0 + \binom{4}{1} (1)^3 (b) + \binom{4}{2} (1)^2 (b)^2 + \binom{4}{3} (1)^1 (b)^3 + \binom{4}{4} (1)^0 . (b)^4$$

$$= 1.1 + 4b + 6.b^2 + 4.b^3 + 1.b^4$$

$$= (1\,4\,6\,4\,1)_b$$

Question No 19:-

Prove pascal's identity. using the formula for $\binom{n}{r}$.

We know formula of pascal's identity as

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

Proof:-

Let n be a positive integer and k an integer with $0 \leq k \leq n$.

L.H.S =

$$\binom{n}{k-1} + \binom{n}{k} = \frac{n!}{(k-1)!\,(n-(k-1))!} + \frac{n!}{k!\,(n-k)!}$$

$$= \frac{n!\,k}{(k-1)!\,(n-k+1)!\,.k} + \frac{n!\,(n-k+1)!}{k!\,(n-k)!\,(n-k+1)!}$$

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$$= \frac{n!\cdot k}{k!\,(n-k+1)!} + \frac{n!\,(n-k+1)!}{k!\,(n-k+1)!}$$

$$= \frac{n!\,(k+(n-k+1))}{k!\,(n-k+1)!}$$

$$= \frac{n!\,(n+1)}{k!\,(n+1-k)!} \Rightarrow \frac{(n+1)!}{k!\,(n+1-k)!}$$

$$\binom{n+1}{k} = L.H.S$$

So, it is proved:-

Question No 23:-

Show

$$\binom{n+1}{k} = (n+1)\binom{n}{k-1}\Big/k$$

Let us Take R.H.S.

$$R.H.S = \frac{(n+1)\binom{n}{k-1}}{k}$$

$$= \frac{1}{k}(n+1)\frac{n!}{(k-1)!\,(n-k+1)!}$$

$$= \frac{(n+1)\,n!}{k\,(k-1)!\,(n-k+1)!}$$

$$= \frac{(n+1)!}{k!\,(n-k+1)!}$$

$$= \frac{(n+1)!}{k!\,(n+1-k)!}$$

$$\binom{n+1}{k} = L.H.S$$

Question No 24:-

Show that if p is a prime and k is an integer such that $1 \le k \le p-1$, then p divides $\binom{p}{k}$

We know the formula of combination as

$${}^{p}C_{r} = \frac{p!}{(p-k)!\,k!}$$

Each term in the product in denominator $< p$ and so, none of terms divide the 'p' in numerator So, the resulting number must be divisible by p.

Q:25

Let n be a positive integers Show that.

$$\binom{2n}{n+1} + \binom{2n}{n} = \binom{2n+2}{n+1}\Big/2$$

So, Now We first use.

$$L.H.S = \binom{2n}{n+1} + \binom{2n}{n}$$

By using pascal's identity

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$\binom{2n+1}{n+1} = \binom{2n}{n+1} + \binom{2n}{n}$ using pascals identity

By multiply and divide 2.

$= \frac{1}{2}\left[2\binom{2n+1}{n+1}\right]$

$= \frac{1}{2}\left[\binom{2n+1}{n+1} + \binom{2n+1}{n+1}\right]$

Again using pascals identity.

$= 1/2\left[\binom{2n+1}{n+1} + \binom{2n+1}{n}\right]$

$= 1/2\left[\binom{2n+1+1}{n+1+1}\right]$

$= 1/2\binom{2n+2}{n+2} = R.H.S$

Q:22 (b) Show that:-

$$\binom{n}{r}\binom{r}{k} = \binom{n}{k}\binom{n-k}{r-k}$$

$L.H.S = \binom{n}{r}\binom{r}{k}$

$= \frac{n!}{r!(n-r)!} \times \frac{r!}{k!(r-k)!}$

$= \frac{n!}{k!(n-r)!(r-k)!} \rightarrow (1)$

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$R.H.S = \binom{n}{k}\binom{n-k}{r-k}$

$= \frac{n!}{k!(n-k)!} \times \frac{(n-k)!}{(r-k)!(n-k-r+k)!}$

$= \frac{n!}{k!(r-k)!(n-r)!} \rightarrow (2)$

From eq (i) and (2) it is proved that $L.H.S = R.H.S$.

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Chapter No 9:

Relations and their properties:-

Definition:-

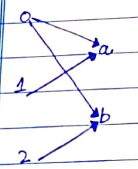
Let A and B be sets. A binary relation from A to B is a subset of AXB.

Example No 3.

Let A = {0, 1, 2}

B = {a, b}

AXB = (0, a) (1, a) (2, b), (0, b)



R	a	b
0	X	X
1	X	
2		X

Relation on a set:-

Relation from a set A to itself are of special interest.

Relation. Defination:

A relations on a set A is a relation from A to A.

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a relation on a set A is a subset of AXA.

Example No 4:-

Let A be the set {1, 2, 3, 4}. Which ordered pair are in the relation R = {(a, b) | a divide b}

Solution

Because (a, b) is in R if and only if a and are positive integers not exceeding 4 such that a divides b.

R = {(1,1), (1,2), (1,3), (1,4), (2,2), (3,3), (4,4)}

Example No 5:-

Consider these relation on the set of integers.

$R_1 = \{(a,b) \mid a \leq b\}$ (1,1)

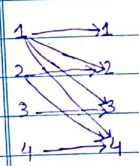
$R_2 = \{(a,b) \mid a > b\}$ (2,1)

$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\}$ (1 = -1)

$R_4 = \{(a,b) \mid a = b\}$ (2,2)

$R_5 = \{(a,b) \mid a = b+1\}$ (2,3)

$R_6 = \{(a,b) \mid a+b \leq 3\}$ (3,3)



R	1	2	3	4
1	X	X	X	X
2		X		X
3			X	
4				

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Definition:-

A Relation R on a set A is called Reflexive if $(a,a) \in R$ for every element $a \in A$.

using quantifier relation R on the set A is reflexive if $\forall a((a,a) \in R)$.

Example No 7:-

consider following relation on $\{1,2,3,4\}$

$R_1 = \{(1,1),(1,2),(2,1),(2,2),(3,4)(1,4),(4,4)\}$

$R_2 = \{(1,1),(1,2),(2,1)\}$

$R_3 = \{(1,1),(1,2),(1,4),(2,1),(2,2),(3,3),(4,1),(4,4)\}$

$R_4 = \{(2,1),(3,1),(3,2),(4,2),(4,2),(4,3)\}$

$R_5 = \{(1,1)(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,3),(3,4)(4,4)\}$

$R_6 = \{(3,4)\}$

R_3 and R_5 Reflexive. (1,1),(2,2),(3,3),(4,4)

Definition:-

A relation R on a set A is called transitive if whenever $(a,b) \in R$ and $(b,c) \in R$, than $(a,c) \in R$, for all $a,b,c \in A$.

Example No 15:-

Is the divides relation on the set of positive integers transitive?

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Suppose that a divide b & b divide c. then there are positive integers k and l such that $b = ak$ and $c = bl$.

Hence

$c = a(kl)$

So,

$a = c$ F_1 a divides c.

Transitive relation.

Combination Relations:-

Because relation from A to B are subset of AXB, two relation from A to B. can be combine in way two sets.

Example No 17:-

Let $A = \{1,2,3\}$ and $B = \{1,3,4,2\}$

Relation $R_1 = (1,1),(2,2),(3,3)$

$R_2 = (1,1)(1,2),(1,3),(1,4)$

$R_1 \cup R_2 = \{(1,1),(1,2),(1,3)(1,4),(2,2),(3,3)\}$

$R_1 \cap R_2 = \{1,1\}$

$R_1 - R_2 = \{(2,2)(3,3)\}$

$R_2 - R_1 = \{(1,2),(1,3),(1,4)\}$

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Question No 1:

List the ordered pairs in the Relation R from $A=\{0,1,2,3,4\}$ to $B=\{0,1,2,3\}$ where $(a,b) \in R$.
if and only if

a) $a=b$.
(0,0), (1,1), (2,2), (3,3)

(b) $a+b=4$
(1,3), (2,2), (3,1), (4,0), (0,4) Ans.

(c) $a>b$
So, order pairs are
(1,0), (2,0), (2,1), (3,0), (3,1), (3,2)
(4,0), (4,1), (4,2), (4,3)

(d) $a|b$ (a divide b)
It means that we chose b in the form of $b=ak$ ∵ K is an integer.
{(1,0), (1,1), (1,3), (2,4), (2,0), (2,2), (3,0) (3,3), (4,0)}.

(e) gcd $(a,b)=1$
(It means a and b are relatively prime)
{(0,1), (1,0), (1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (4,1), (4,3)}

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(f) = lcm $(a,b)=2$.
(2,2), (2,1), (2,2)

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Question No 3:-

Set = {1, 2, 3, 4}

(a) {(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)}

this is transitive relative because here (a,b), (b,c), and (a,c) exist in the relation for eg.

(3,2) (2,4) (3,4) are exit.

a b b c a c

(b) {(1,1), (1,2) (2,1), (2,2) (3,3), (4,4)}

this relation is reflexive, symetric and transitive.

(c) {(2,4) ,(4,2)}

this symmetric relation.

(d) {(1,2), (2,3) , (3,4)}

this is antysymmetric relations.

(e) {(1,1), (2,2), (3,3), (4,4)}

this is clearly reflex, symmetric antisymmetric and transitive relation

(f) {(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)}

It is neither retlexive relative Symmetric nor antisymmetric transitive

Question NO 4:-

(a,b) ∈ R if and only if

(a) a is taller than b.

- It is not reflexive, because a Person cannot be taller than himself/herself.
- It is not symmetric, because if person A is taller than person B, then person B is not tallar than A.
- It is antisymmetric because (a,b) ∈ R & (b,a) ∈ R cannot occure at some time.
- It is transitive, because if person A is taller than B and B is taller than C. then person A need to be tallar than C.

(b) a and were born on some day.

So, it is reflex, Symmetic and transitive but not antisymmentric relative.

(c) a has some first name as b.

It is reflexive, Symetric and transtive but not antisymmetric relation.

(d) a and b have common grandparent.

It is reflexive, Symmetric but neither antisymmetric nor transitive

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Question No 6:-

Determine whether the relation R on the set of all real number is reflexive, symmetric, antisymmetric, and/or transitive, where $(x,y) \in R$ if and only if.

(a) $x+y=0$.

- It is not reflexive, because $x+x=0$ is only true when $x=0$ but not for all real number.
- It is symmetric because $x+y=0$ $y+x=0$
- It is not antisymmetric, because $1+(-1)=0$ and $(-1)+(1)=0$ while $1 \neq -1$
- It is not transitive, because $1+(-1)=0$ and $(-1)+1=0$ while $1+1 \neq 0$

(b) $x=\pm y$

- It is reflexive, because $x=x$ is always true.
- It is is symmetric because, if $x=\pm y$ then $y=\pm x$
- It is not antisymmetric, because $1=(1)$ and $(-1)=-1$ while $1 \neq -1$
- It is transitive because if $x=\pm y$ and if $y=\pm z$ then $x=\pm z$.

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(c) $x-y$ is a rational numbers.

- Reflexive: $x-x=0$ which is rational.
- Symmetric: $x-y$ is rational and $y-x=-(x-y)$ rational
- Transitive: if $x-y$ is rational and if $y-z$ is rational then $x-z$ is rational.
- not antisymmetric $1-2$ and $2-1$ are both rational but $1 \neq 2$.

(d) $x=2y$

- Not Reflexive: $\because x=2x$ is only true when $x=0$ and thus is not true for all real numbers.
- Not symmetric: $\because x=2y$ but $y \neq 2x$.
- Antisymmetric: $\because$ As it is not symmetric
- Not transitive:

Because if $x=2y$ and $y=2z$ then $x=2y=2(2z)=4z$ which is not equal to $x=2z$ when x,y,z are positive.

(e) $xy \geq 0$

Reflexive: $\because x \cdot x = x^2 \geq 0$ is always true.

Symmetric: $\because$ if $xy \geq 0$, then $yx \geq 0$.

Not antisymmetric: $\because$ if $xy \geq 0$ and $yx \geq$ but $x \neq y$.

Not transitive:- $\because$ if $x=-1$, $y=0$ and $z=1$ then $xy \geq 0$, $yz \geq 0$ and $xz < 0$.

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(f) $xy = 0$.

- Not reflexive:- $\because x \cdot x = x^2 = 0$ only for $x = 0$ not for all real no.
- Symmetric:- $\because xy = 0$ then $yx = 0$ $\therefore$
- Not antisymmetric:- $\because (1)(0) = 0$ and $(0)(1) = 0$ while $0 \neq 1$.
- No transitive:- If $x = -1$, $y = 0$ and $z = 1$ than $xy = 0$, $yz = 0$ and $xz \neq 0$

(g) $x = 1$

- Not reflexive:- $\because (0,0)$ not include in it
- Not symmetric:- $\because$ if $x = 1$, $y = 0$ then $(x,y) \in R$ while $(y,x) \in R$.
- Antisymmetric:- $\because$ As given as above
- Transitive:- $\because (x,y) \in R$ & $(y,z) \in R$ then $x = 1$ and thus $(x,z) \in R$ is also true

(h) $x = 1$ or $y = 1$

- It is only symmetric :- (x,y) & R and (y,x) & R exist.
- Not reflexive :- as $(0,0)$ not include in it.
- Not antisymmetric because if $x = 1$ and $y = 0$ the $(x,y) \in R$ and $(y,x) \in R$ while $x \neq y$.

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- Not transitive $\because$ if $x = 0$ and $y = 1$ and $z = 2$, then, $(x,y) \in R$ and $(y,z) \in R$ while $(x,z) \notin R$.

Question No 7:-

(a) $x \neq y$

- this relation is not reflexive; $\because 1 \neq 1$ or $x \neq y$.
- It is symmetric; $\because x \neq y$ then $y \neq x$ exist.
- It is not antisymmetric; $\because 1 \neq 2$ and $2 \neq 1$
- It is not transitive; $\because 1 \neq 2$ and $2 \neq 1$ but $1 \neq 1$.

(b) $xy \geq 1$

- Not reflexive:- $\because (0,0)$ not include
- Symmetric; $\because xy = yx$; So one of these quantity is greater than or equal to 1.
- Not antisymmetric; $\because (2,3)$ and $(3,2)$ included but they are not equal.
- Transitive; $\because x \geq 1/y$, $y \geq 1/z$, $x \geq 1/z$

(c) $x = y + 1$ or $x = y - 1$

- Not reflexive; $\because (1,1)$ not included.
- Symmetry, $\because x = y - 1$ is equivalent to equation $y = x + 1$, which is same as $x = y + 1$ with rule of reversed x and y.

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- Not antisymmetrics ∵ (1,2) and (2,1) are relation.
- Not transitives- ∵ (1,2) and (2,1) are in relation but (1,1) is not.

(d) $x \equiv y \pmod 7$

Recall that $x \equiv y \pmod 7$ means that $x-y$ is a multiple of 7. i.e $x-y=7t$ for some integers.

- Reflexive ∵ $x-x=7$ for all x.
- Symmetric ∵ if $x \equiv y \pmod 7$ than $x-y = 7t$. for some

therefore $y-x = 7(-t)$, so, $y \equiv x \pmod 7$

- Not antisymmetrics ∵ eg $2 \equiv 9$ and $9 \equiv 2 \pmod 7$
- Transitives ∵ if $x \equiv y \pmod 7$, if $y \equiv z \pmod 7$ then $x \equiv z \pmod 7$ include.

(e) x is multiple of y.

- Reflexive ∵ Every number is a multiple of its
- Not Symmetric ∵ for eg 6 is multiple of 2 but 2 is not multiple of 6.
- Not antisymmetric ∵ 2 is multiple of (-2) and (-2) multiple of 2 but $-2 \neq 2$.
- Transitive ∵ $x = yt$ $y = 5z$ then $x = 5z(t) = 5(zt)$

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(f) x and y are both negative or non negative :-

- Reflexive ∵ a and a are either both (1) or (-1)
- Symmetric ∵ e.g :- (x,y) and (y,x) exist
- Not symmetric ∵ 5 is related to 6 and 6 is related to 5 but $5 \neq 6$.
- Transitive ∵ if related to b and related then a related to c.

(g) $x = y^2$

- Not reflexive ∵ $17 \neq 17^2$
- Not symmetric :- $289 = 17^2$ but it is not the case that $289^2 = 17$.
- Antisymmetric - Due to above case.
- Not transitive For e.g :- $16 = 4^2$ and $4 = 2^2$ but $16 \neq 2^2$.

(h) $x \geq y^2$

- Not reflexive ∵ $17 \not\geq 17^2$
- Not symmetric ∵ $289 \geq 17^2$ but it is not the case that $17 \geq 289^2$
- Antysymmetric ∵ Due to above case
- Transitive ∵ For e.g $4^2 = 16$ and $2^2 = 4$ but $4^2 \neq 16$.

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Question No 11:-

which is relation in exercise 3 are irreflexive?

the relation in parts (a), (b) and (e) all have at least one pair of the form (x,x) in them. So they are not irreflexive. the relation in part (c), (d) and (f) do not, so they are irreflexive.

Question No 15:-

can relation on a set be neither reflexive nor irreflexive?

the relation in Exercise (3) (a) is neither reflexive nor irreflexive it contain some of pairs (a,a) but not all of them.

$\{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$.

Question No 17:-

Give an example of an irreflexive relation on the set of all people there are many such example. the empty relation is always irreflexive (x is never related to y).

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An example would be $(a,b) \in R$ if and only if a is taller than b. since nobody is taller than hirself or herself.

We always have $(a,a) \notin R$.

Question No 19:-

Which relation in Exercise (4). are asymmetric?

- the relation in part (a) is asymmetric since if a is taller than b, then b cannot be taller than a.
- In part (b). it is not asymmetric since there many instances of a and b born on the some day, and in all such cases, it is also case that b and a were born on some day.
- the relation in part (c) and (d) are also not asymmetric.

Question No 21:-

which relation in Exercise (6) are asymmetric?

there are all not asymmetric in this question.

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Question No 23:-

use quantifier to express what it means for a relation to be asymmetric.

A relation is asymmetric relation in which $(a,b) \in R$ and $(b,a) \in R$ can never hold simultaneously in symbol.

$\forall a \forall b \ ((a,b) \in R \rightarrow (b,a) \notin R)$

Question No 25:-

How many different relation are there from a set with m element to a set with n elements.

there are mn element of the set $A \times B$, if A is a set with m element and B is a set with n element. A relation from A to B is subset of $A \times B$.

So,

By product rule.

2^{mn} are number of subset.

Question No 20:-

Given example of relation on a set.

(a) both symmetric and antisymmetric.

(i) $R = \{1,1\}$

(ii) $R = \{(2,2), (3,3), (4,4)\}$

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(b) neither symmetric nor antisymmetric

(i) $R = \{(0,1), (1,2), (2,1)\}$

$R = \{(0,0), (1,5), (2,10), (3,11)\}$

Question 24:-

Give an example of an antisymmetric relation on the set of all people.

Let R be a relation on the set of all people such that $(a,b) \in R$ if and only if a has one inch longer hair than b.

Question No 30:-

$R_1 = \{(1,2), (2,3), (3,4)\}$, $R = \{(1,1), (1,2), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3), (3,4)\}$ Find.

(a) $R_1 \cup R_2$

$R_1 \cup R_2 = \{(1,1), (1,2), (2,3), (3,4), (2,1), (2,2), (3,1), (3,2), (3,3)\}$.

(b) $R_1 \cap R_2$

$R_1 \cap R_2 = \{(1,2), (2,3), (3,4)\}$

(c) $R_1 - R_2$

$R_1 - R_2 = \{\phi\}$

(d) $R_2 - R_1$

$R_2 - R_1 = \{(1,1), (2,1), (2,2), (3,1), (3,2), (3,3)\}$

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Question No 32:-

Let R be the relation $\{(2,2),(1,3),(2,3)(2,4),(3,1),\}$ and let S be the the relation $\{(2,1),(3,1),(3,2),(4,2)\}$ find SoR.

$R = \{(1,2),(1,3),(2,3),(2,4),(3,1)\}$

$S = \{(2,1),(3,1),(3,2),(4,2)\}$

$\because S = a,b \quad \because R = b,c \quad , \because SoR = a,c$

Find So R.

$SoR = \{(2,2)(2,3)(3,2),(3,3),(3,4),(4,3)(4,4)\}$

Question No 33:-

$R = (a,b)$: a is present of b.

$S = (a,b)$ a and b are sibling (brother or sister)

Find $(SoR) = ?$ and $(RoS) = ?$

Solution:-

$SoR = \{(a,b) \mid a$ is a parent of b but and b has a sibling$\}$.

$RoS = \{(a,b) \mid a$ is an auntor uncle $b\}$

Question No 48:-

How many trasitive relations are there on a set with n element if;

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(a) $n = 1$.

Let us assume that a is only element in A. then there is only are ordered pair on set A.

(a,a).

it also have ϕ set.

So, All posible relation are then;

ϕ Transitive

$\{(a,a)\}$ Transitive

So,

there are 2 transitive element.

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